FIELDCOMMAND

Incident Management System v1.0

COMPLETE USER MANUAL

v1.0

ICS/NIMS All-Hazards · Amateur Radio EMCOMM · Public Safety

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1 Introduction & System Overview

CHANGED **Reworded intro: the communications + ICS "marriage" and the vital role of amateur radio; added the tornado-outbreak example (messaging).**

Running an incident and communicating during it are inseparable. Every decision an Incident Commander makes depends on information getting in, and every order depends on being able to send it back out — so incident management and communications are not two separate jobs, they are one. When they are handled by tools that never talk to each other, the incident suffers: the message traffic, the resource assignments, and the Incident Command System (ICS) forms drift out of step at exactly the moment they most need to agree. FieldCommand IMS is built on the opposite idea — that communications and ICS incident management belong together, married in one tool, because during any incident, of any type, each one depends on the other.

This project began as an amateur-radio emergency-communications (EMCOMM) effort and grew into a complete incident management system, because that inseparability became impossible to ignore. It carries a second conviction just as strongly: amateur radio is not a hobbyist add-on to emergency management — it is often the communications of last resort that keeps working when nothing else does. An agency's primary communications are usually a public-service radio system and cellular service, and both can be overwhelmed or destroyed in moments. The recent hurricanes across Florida, Louisiana, and the Carolinas made this plain, and massive tornado outbreaks devastate public-service communications the same way: when the public-safety systems were overloaded or wiped out, it was amateur radio operators who moved the traffic that coordinated the response and saved lives. Incident Commanders and emergency managers sometimes forget how vital that capability is — so a deliberate goal of FieldCommand IMS is to keep amateur radio in front of them: ready, integrated, and available as the resilient backbone an incident can fall back on when the primary systems fail. Beyond serving any one incident, this system is intended to promote and advance the use of amateur radio in emergency management — putting it to work on everyday activations, growing the pool of trained and ready operators, and demonstrating on real incidents the capability that only amateur radio can provide.

And because the worst incidents take down the very infrastructure modern tools rely on — cellular networks, internet, and power — FieldCommand IMS is completely self-contained. It carries its own network, its own server, and its own tools, and it operates with no internet connection, no cellular service, and no outside infrastructure of any kind. The cell towers are down, the internet is gone, and you are running on a generator in a parking lot: that is exactly when you need incident management and communications the most, and exactly when FieldCommand IMS is designed to perform.

TIP In one sentence: FieldCommand IMS marries ICS incident management with the communications an incident depends on — public-service and amateur radio alike — in one self-contained system that keeps working when the internet and cell service are gone.

1.1 What FieldCommand IMS Is

CHANGED **Clarified: the Pi does NOT broadcast Wi-Fi; a separate ASUS router creates EMCOMM-NET (wording).**

FieldCommand IMS is a **complete ICS/NIMS all-hazards incident management system** with amateur radio and public-service Emergency Communications (EMCOMM) built into its core rather than bolted on. It manages the full lifecycle of any incident from initial response through demobilization using standard ICS forms and workflows. It runs on a Raspberry Pi 5 server. The Pi itself does **not** broadcast Wi-Fi — a separate wireless router (an ASUS travel router) creates the private network, named **EMCOMM-NET**, and the Pi connects to it. Any smartphone, tablet, or laptop that joins that network immediately has access to the full suite of 48 tools through a standard web browser — no app installation, no accounts, and no per-device configuration required.

The letters IMS stand for **Incident Management System**. The whole product is designed around one belief: communications and incident management are inseparable. Running an incident means running its Incident Action Plan (IAP), tracking its resources, and moving its message traffic — all at once. FieldCommand IMS marries the communications side and the planning side in a single tool so that a net check-in, a resource assignment, and an ICS form are all part of one connected picture rather than three separate systems that never talk to each other.

When internet connectivity is available, live features activate automatically: National Weather Service (NWS) weather alerts, APRS-IS, animated Next Generation Radar (NEXRAD) radar, and High Frequency (HF) propagation data. If connectivity is lost at any point, all core tools continue without interruption. The system degrades gracefully and recovers automatically.

1.2 How FieldCommand IMS Differs From Other ICS Platforms

Platforms such as WebEOC, E-Team, NIMSIAP, National Incident Management System (NIMS) Logic, and E-iSuite deliver powerful incident management capability — but every one of them requires a working internet connection and functioning server infrastructure. They are cloud-dependent by design. When a major disaster disables that infrastructure, they fail with it — at exactly the moment capability is most critical.

FieldCommand IMS is built on the opposite assumption: infrastructure will fail, and incident management capability must survive that failure. It carries its own infrastructure — server, network, storage, and tools — in a single deployable package. It requires no IT department, carries no licensing fees, and does not fail when the internet fails.

The feature that sets FieldCommand IMS apart from every platform in this category is its native integration of amateur radio and public safety communications directly into the incident management workflow. No other ICS platform includes built-in net control logging, Federal Communications Commission (FCC) callsign validation against the full national licensee database, Automatic Packet Reporting System (APRS) tactical mapping, Winlink radio email, JS8Call HF messaging, or Amateur Packet Radio Network (AMPRNet) gateway capability. These are not add-ons — they are core features fully integrated with the ICS platform so that radio traffic, net logs, and check-in data flow directly into ICS-309 communications logs, ICS-214 activity logs, and the Incident Action Plan (IAP).

PLATFORM	OFFLINE?	SELF-CONTAINED?	NATIVE EMCOMM?	COST
FieldCommand IMS	Fully offline	Own network + server	Full EMCOMM	Free / open source
WebEOC	Cloud only	Needs infrastructure	None	$10,000-50,000/yr
E-Team / Veoci	Cloud only	Needs infrastructure	None	Subscription
NIMSIAP	Internet required	Requires connectivity	None	Free
E-iSuite	Limited	Laptop, no network	None	License fee

1.3 System Architecture

A complete FieldCommand IMS deployment is a small stack of purpose-built parts. The table below lists each component, what it does, and the network address it uses by default. Only the first component — the FieldCommand Pi 5 server — is strictly required; everything else extends coverage, adds a radio gateway, or hardens the network. All of the tools in this manual are served by that one Pi.

COMPONENT	FUNCTION	DEFAULT IP
FieldCommand Pi 5 16GB	Primary server / all 48 web tools / 12+ background services / RAID-1 NVMe storage / EMCOMM-NET Wi-Fi AP	192.168.50.1
44Net Gateway Pi 5 (optional)	AMPRNet WireGuard tunnel / callsign-authenticated access / Part 97 access log / isolated from primary server	192.168.50.2
Wi-Fi router (primary)	DHCP / AiMesh controller / dual Wide Area Network (WAN) management / EMCOMM-NET Service Set Identifier (SSID) broadcast. Recommended: ASUS RT-BE58 Go.	192.168.50.254
AiMesh nodes (optional)	EMCOMM-NET coverage extension / seamless roaming. Recommended: additional ASUS RT-BE58 Go units.	DHCP assigned
UniFi Switch Lite 16 Power over Ethernet (PoE) (recommended)	Wired backbone / powers PoE devices / connects all Pi units, router, cellular antenna, and workstations	DHCP assigned
Operator workstations	Any device with a modern browser: smartphones, tablets, laptops, Raspberry Pi 500 desktops, Windows or macOS laptops	DHCP assigned

NOTE The default server address is 192.168.50.1. This is configurable during installation. All documentation uses this default — substitute your address wherever 192.168.50.1 appears if your deployment differs.

NOTE The Wi-Fi network is broadcast by the router, not by the Pi. The Pi is the server and the router is the radio that puts the EMCOMM-NET network on the air. Operators join EMCOMM-NET and then reach the Pi at 192.168.50.1. This split is why coverage can be extended with ordinary mesh nodes without touching the server.

1.4 Capability Summary

FieldCommand IMS groups its 48 tools into nine capability areas. Every area is covered in depth in its own chapter later in this manual; the table below is a one-look inventory of what the system does.

CAPABILITY AREA	TOOLS PROVIDED
Incident Management	Five-section ICS structure. Complete IAP form set: ICS-202 through ICS-221. Live T-card resource board. A growing library of pre-planned event templates. Incident archive, restore, and scenario/exercise mode.
ICS Forms & IAP	Full ICS form set with digital signature capture on all prepared-by and approved-by fields. One-click IAP Portable Document Format (PDF) compilation with title page and section dividers. Print center for on-site IAP packages.
Federal Emergency Management Agency (FEMA) Documentation	FEMA Public Assistance (PA) cost tracking: Force Account Labor with fringe, Equipment with 2025 FEMA rate schedule (45 built-in rates), Materials, and Contracts. Real-time cost dashboard. ICS-214 import. Project Worksheet text export.
Amateur Radio EMCOMM	Net control logger with FCC callsign auto-fill from offline 800,000+ licensee database. APRS tactical map. Winlink and Pat radio email. JS8Call HF digital. AMPRNet 44Net gateway. National Traffic System (NTS) radiogram generator.
Public Safety Comms	Trunked/P25 net logger with radio ID check-in. EMA member ID lookup. ICS-309 export. Observer mode for read-only net monitoring on any device.

CAPABILITY AREA	TOOLS PROVIDED
Situational Awareness	Animated NEXRAD radar (WAN required). Live NWS weather alerts. GPS-tracked resource map with status-coded SVG markers. Offline APRS tactical map. CalTopo / SARTopo GeoJSON import. HF propagation tool.
Personnel & Check-In	Member roster with certifications, radio IDs, and Quick Response (QR) code check-in codes. QR/barcode camera scan check-in using native BarcodeDetector API. ICS-211 check-in log. Digital ID card generator.
WAN & Connectivity	Dual WAN source support: any cellular modem, hotspot, or satellite dish. Preferred/fallback role configuration. Configurable detection per source. Automatic failover. Real-time WAN status with 30-second polling.
Reference & Offline Content	Kiwix offline Wikipedia and reference library. Radio frequency cheat sheets. Hospital and facilities directory. Repeater database. Channel library. NIMS resource typing library. ICS position checklists.

1.5 Every Incident Record Is Saved

FieldCommand IMS treats incident data as permanent. Everything an incident produces — net logs, ICS forms, T-cards, the IAP, cost records, resource assignments, roster snapshots, check-ins, and attachments — is written to the server and kept. Nothing is held only in a browser tab or only in memory, so a device that loses power or wanders off the network does not take the record with it. Any operator who reconnects sees the same live picture as everyone else.

The server stores this in a database on its mirrored (RAID-1) solid-state storage, so a single drive failure does not lose data. Completed incidents can be archived and later restored, and the whole data set is copied to an external backup drive. The practical result is that an incident worked on FieldCommand IMS becomes a durable, auditable record you can reopen, reprint, or hand to a served agency long after the event is over.

TIP This is the deliberate opposite of a throwaway tool. If you are building or changing an incident feature, the default is to persist it on the server, not to keep it in the browser only.

1.6 Offline First, Online When Available

The whole tool set is designed to run with zero internet. A small number of features depend on an outside connection and light up automatically when a Wide Area Network (WAN) — cellular or satellite — is present; if that connection drops, those features pause and everything else keeps running. The dashboard WAN status card shows the current state at a glance (covered in the Getting Started chapter). The table below shows the split.

FEATURE GROUP	RUNS OFFLINE?	NOTES
ICS platform, forms, IAP, T-cards	Always	No internet needed at any time
FEMA cost tracking and dashboards	Always	Rate schedule is built in
Net loggers (amateur and public safety)	Always	Local logging and export
FCC callsign lookup	Always	Reads the local offline licensee database
Roster, check-in, resource map	Always	Local data and local maps
NEXRAD weather radar	Internet only	Animated radar imagery needs a WAN

FEATURE GROUP	RUNS OFFLINE?	NOTES
NWS weather alerts	Internet only	Live alert feed needs a WAN
APRS-IS internet feed	Internet only	Off-air APRS still works over radio
HF propagation data	Internet only	Live space-weather data needs a WAN

NOTE A red WAN indicator never means FieldCommand IMS is down. It means only the internet-dependent extras are paused. Every ICS tool, form, net logger, and local map continues to work normally.

1.7 Amateur Radio and FCC License Gating

CHANGED **Reframed: amateur radio is central and license-gated (not "optional"); gated for FCC legality (wording).**

Amateur radio is central to what FieldCommand IMS is — not an add-on and not optional to its purpose. But amateur radio may only be operated by a properly licensed operator, so the amateur radio features are **license-gated**. The net control logger, APRS tactical map, Winlink and Pat radio email, JS8Call HF digital, the AMPRNet 44Net gateway, and the NTS radiogram generator are grouped under an Amateur Radio mode that turns on only when a station callsign has been entered during setup. This gating exists for two reasons: to keep the system legal with the Federal Communications Commission (FCC) by not placing transmit tools in front of unlicensed users, and to let a group that does not yet have amateur radio operators run everything else without the amateur screens in the way.

A group with no licensed operators leaves the callsign blank. The Amateur Radio mode then stays grayed out, and every other part of the system — the full ICS platform, forms, cost tracking, roster, check-in, resource map, and the public-safety net logger — works exactly the same. This is not meant to be the end state, though: a core goal of FieldCommand IMS is to promote and advance amateur radio in emergency management, so the app is built to grow with the group. The moment a licensed operator comes on board, a callsign can be entered — during setup or any time afterward — and the full amateur radio capability unlocks.

WARN Amateur radio transmissions are legal only when made by a properly licensed operator with privileges on the bands and modes in use. Turning on the amateur features does not grant that authority. Until your group has a licensed operator, leave the callsign blank; the gating keeps you compliant with the FCC while the rest of the system runs in full.

1.8 Who Uses FieldCommand IMS

FieldCommand IMS is a multi-user system. Every operator who joins the EMCOMM-NET network and opens a browser sees the same live incident. There are no per-device accounts to manage; roles below describe how people typically use the system, not logins the software enforces.

ROLE	HOW THEY USE THE SYSTEM
Incident Commander / Command Staff	Set objectives, approve the IAP, watch the whole picture from the dashboard and resource board.
Planning Section	Build and compile the IAP, manage ICS forms, run T-card resource status, and set up event templates.
Operations Section	Track assignments and resources on the ground, read the tactical and resource maps.
Logistics Section	Order and account for resources, manage the communications plan and reference material.

ROLE	HOW THEY USE THE SYSTEM
Finance / Administration	Record FEMA Public Assistance costs — labor, equipment, materials, contracts — and watch the cost dashboard.
Communications Unit / Net Control	Run the net loggers, validate callsigns and radio IDs, and move message traffic.
Any responder or observer	Check in by QR code, view read-only net and incident status from a phone or tablet.

1.9 How This Manual Is Organized

This manual follows the order you will use the system. Early chapters get you connected and configured; the middle chapters cover the day-to-day tools; the later chapters cover the ICS platform, FEMA documentation, radio features, and network hardware; and the final reference chapters answer questions and define terms.

- **Getting connected** — join EMCOMM-NET and complete Organization Setup (Chapters 2 and 3).
- **Everyday tools** — the dashboard, event templates, and net loggers for running an activation.
- **Incident management** — the ICS platform, the ICS form set, the IAP, the T-card board, and FEMA cost tracking.
- **Communications and situational awareness** — the amateur radio features, the two maps, weather, and personnel check-in.
- **Reference** — network hardware, troubleshooting and frequently asked questions, a glossary, and a menu reference.

> **TIP** Every chapter opens with a short plain-language summary, so a reader in a hurry can skim the top of a chapter and still succeed. If you are brand new, read Chapters 2 and 3 first, then jump to whichever tool you need.

1.10 Common Questions

QUESTION	ANSWER
Do I need to install an app?	No. Every tool is a web page. Join EMCOMM-NET and open a browser to 192.168.50.1.
Does it need the internet to work?	No. The full tool set runs offline. Only weather radar, NWS alerts, APRS-IS, and HF propagation need a WAN, and they resume automatically when one returns.
Do I need a login or account?	No. FieldCommand IMS is a no-login, open-LAN tool by design. Any device on EMCOMM-NET reaches the whole dashboard.
We have no licensed hams. Can we still use it?	Yes. Leave the callsign blank during setup. The amateur features stay off and everything else works normally.
What happens to our data after the incident?	It is saved on the server, mirrored across two drives, backed up to an external drive, and can be archived and restored later.
Which devices work?	Any device with a modern browser — a phone, tablet, laptop, or a Raspberry Pi desktop — on any operating system.
Does it cost anything?	The software is free and open source. Your only cost is the hardware in the architecture table above.

2 Getting Started — Connecting to FieldCommand

http://192.168.50.1

Every tool in FieldCommand IMS is a web page served from the Raspberry Pi. There is nothing to install on any device, no account to create, and no password to remember beyond the one for the Wi-Fi. You join the EMCOMM-NET wireless network, open a browser, type one address, and the full dashboard appears. It works the same way on an iPhone, an Android tablet, a Windows laptop, a Mac, or an operator workstation. This chapter walks you through that first connection, then shows you how to read the dashboard, switch between its three modes, and understand the status indicators you will see.

TIP In one sentence: connect to the EMCOMM-NET Wi-Fi, open http://192.168.50.1 in any browser, and every FieldCommand tool is there — no app, no login.

2.1 What You Need to Connect

Any modern device with a web browser and Wi-Fi can reach FieldCommand. You do not need the internet, a cellular signal, or any special software. The list below is everything that matters.

YOU NEED	DETAILS
A device with Wi-Fi	A smartphone, tablet, laptop, or an operator workstation. Any operating system works.
A modern browser	Chrome, Firefox, Safari, Edge, or the built-in browser on a phone or tablet. Nothing to download or update.
The EMCOMM-NET password	Printed on the equipment case label and in the Installation Guide. It is the only credential you enter anywhere.
Nothing else	No internet, no cellular data, no app store, no user account. The Pi carries its own network and its own copy of every tool.

2.2 First Connection

CHANGED **Clarified: power on the ASUS access router too, not just the Pi (wording).**

1. Power on **both** the FieldCommand Pi **and** the ASUS access router. The router is what broadcasts the Wi-Fi, so powering on the Pi by itself is not enough. Wait approximately 45 seconds for all services to start. The system is ready when the EMCOMM-NET Wi-Fi network appears in your device scan list.
2. On any smartphone, tablet, or laptop, open Wi-Fi settings and connect to **EMCOMM-NET**. The password is on the equipment case label and in the Installation Guide. No other credentials are required.
3. Open a browser and go to **http://192.168.50.1**. The FieldCommand dashboard loads immediately.
4. Bookmark the dashboard address (its Uniform Resource Locator, or URL). On a smartphone, tap **Share** then **Add to Home Screen** to create an icon that opens like a native app.
5. If the dashboard does not load, confirm you are connected to EMCOMM-NET and not to another Wi-Fi network. The Pi does not relay traffic to the internet, so if your device switches to a different network automatically, turn off auto-join on other networks during activations.

NOTE	The EMCOMM-NET Wi-Fi is broadcast by an external ASUS router, not by the Pi itself. If no EMCOMM-NET network appears at all, the router may still be starting or may be unplugged — see the Installation Guide network section.

2.3 Reading the Dashboard

Across the very top of every page is the **hero bar**. It never scrolls away and tells you, at a glance, who this station is and what time it is. Below it sits the mode switcher (covered next). Here is what each part of the hero bar shows.

HERO BAR ELEMENT	WHAT IT SHOWS
Callsign badge (top left)	The station callsign set during setup — for example a club callsign. Before setup is finished it reads **EMCOMM-NET**. If an organization name and logo were configured, they appear just below the badge.
INCIDENT MANAGEMENT SYSTEM	The product name and version (v1.0). This confirms you are on the FieldCommand dashboard and not another page.
Primary clock (large)	The main time display. In Amateur Radio mode this is **UTC** (Coordinated Universal Time), the worldwide amateur radio standard. In the other two modes it is your **local** time, which the Incident Command System (ICS) uses on forms.
Secondary clock (small)	The other time zone, shown smaller underneath — so both UTC and local time are always visible, whichever mode you are in.
Mode pill (top right)	A colored badge that names the mode you are currently in, so you always know which set of tools is on screen.

Under the hero bar, the working area shows the tool cards for the current mode on the left and a status column on the right. Each tool is a labeled card you tap or click to open. You never have to hunt through menus — everything for the job at hand is on one screen.

2.4 The Three Modes

FieldCommand groups its tools into three modes, chosen from the **mode switcher** bar just under the hero bar. A mode is only a filter on which tool cards show first. Switching modes never changes, hides, or deletes any data — every tool stays available, and any incident, log, or form you created is still there whichever mode you are in.

MODE	TOOLS SHOWN PROMINENTLY	BEST FOR
Amateur Radio	Net control logger, Federal Communications Commission (FCC) callsign lookup, Automatic Packet Reporting System (APRS) map, Winlink, JS8Call, Amateur Packet Radio Network (AMPRNet) gateway, High Frequency (HF) propagation, National Traffic System (NTS) radiogram	Amateur radio net operations, ARES/RACES activations
Public Safety	public-safety net logger, radio ID roster, resource map, Wide Area Network (WAN) status, weather radar, hospital directory, facilities	Public safety radio nets, public-service check-in operations
ICS	Full ICS platform, Incident Action Plan (IAP) forms, T-card resource board, personnel check-in, Federal Emergency Management Agency (FEMA) cost documentation, event templates, cost dashboard	Any active incident requiring ICS structure

2.5 Switching Modes and What Changes

The mode switcher has three buttons, each with an icon and a short subtitle listing what it covers. Changing modes takes one tap.

1. Tap or click one of the three buttons in the mode bar: **Amateur Radio**, **Public Safety**, or **ICS / Incident Command**.
2. The tool cards below switch to that mode. The colored mode pill in the hero bar changes to match, and the clock re-orders (UTC first in Amateur Radio mode, local time first in the other two).
3. Your choice is remembered on **this device** in the browser, so the next time you open the dashboard on the same device it returns to the mode you last used.

TIP	You can open the dashboard straight into a mode by adding it to the address: **http://192.168.50.1/?mode=ics** opens ICS mode, **?mode=amateur** opens Amateur Radio, and **?mode=starcom** opens the public-safety mode. This is handy for a big-screen display you always want in one mode.

Because the choice is saved per device, two operators can sit side by side on their own tablets, one in ICS mode and one in Amateur Radio mode, and neither affects the other. They are both looking at the same live data on the same server.

2.6 When Amateur Radio Mode Is Grayed Out

The Amateur Radio mode button is available only when a station callsign has been entered in the Setup wizard. A group with no licensed amateur radio operators leaves the callsign blank, and the button then shows a small padlock hint reading **Add a callsign in Setup to enable**. Tapping it does nothing, and the dashboard opens in ICS mode instead. Every incident-management and public-safety tool still works normally — only the amateur radio tools are held back.

WARN	A callsign, and the amateur radio features it unlocks, must only be used by a properly licensed operator with privileges on the bands and modes in use. You can add a callsign later in the Setup wizard at any time; the Amateur Radio mode lights up as soon as one is saved. Until setup is finished, all three modes are shown so you can explore the system before configuring it.

2.7 The Weather and Radar Row

At the top of the working area is a row that gives you a quick weather picture. It has two parts, and both depend on the internet — they light up when a WAN connection is present and wait quietly when there is none.

ITEM	WHAT IT DOES
NWS Weather Alerts	A live feed of National Weather Service (NWS) watches, warnings, and advisories for the area. A colored dot and a status line show whether it is current. The refresh symbol re-checks on demand. Offline, it reads **Waiting for internet connection**.
NWS Radar	A card that opens an animated NEXRAD (Next-Generation Radar) rain-and-storm loop for your area. Its small status badge shows whether the loop is reachable right now.

NOTE	These two are the clearest example of FieldCommand being offline-first. When the internet is gone they simply say so and stay out of your way — every other tool on the dashboard keeps working exactly as before.

2.8 WAN Status Indicators

Each mode includes a live **WAN Status** card in its status column (there is no top-of-page status bar). WAN stands for Wide Area Network — the path to the internet. The card re-checks about every 30 seconds and changes color on its own.

INDICATOR	MEANING	CORE TOOLS AFFECTED?
Green — Cellular	Primary internet source active	None — fully operational
Blue — Satellite	Satellite failover active	None — fully operational
Red — Offline	No internet — all core tools continue	None — fully operational
AMPRNet UP	44Net WireGuard tunnel active	N/A
AMPRNet DOWN	44Net tunnel down or gateway Pi unreachable	N/A

NOTE A red WAN indicator means the internet-dependent features are paused — NWS radar, APRS-IS, HF propagation. It does NOT mean FieldCommand IMS is unavailable. All ICS tools, forms, net loggers, roster, and local maps run normally offline.

2.9 Using It on a Phone or Tablet

FieldCommand is built to be used from whatever device an operator already carries. A few small habits make it feel like a real app and keep every operator on the same page.

Add it to your home screen. On a phone, open the dashboard, tap **Share** then **Add to Home Screen**. You get a tap-to-open icon and a full-screen view with no browser clutter.

Many people at once. Every operator who joins EMCOMM-NET can open the dashboard at the same time. They all see the same live incident data; a change one person saves appears for everyone.

Each device is independent. The mode you pick and any on-screen preferences are remembered on that device only, so a shared wall display and a handheld tablet can show different modes at the same time.

Stay on EMCOMM-NET. Turn off auto-join for home, cellular, and hotspot networks during an activation so your phone does not silently wander off the FieldCommand Wi-Fi.

2.10 Common Questions

QUESTION	ANSWER
Do I have to install an app?	No. FieldCommand is web pages served from the Pi. Any browser opens them. Adding it to your home screen only creates a shortcut.
Is there a login or password for the dashboard?	No login. The only password is for the EMCOMM-NET Wi-Fi. FieldCommand is deliberately a no-login tool on a private, isolated network.
Which address do I type?	Always **http://192.168.50.1**. Bookmark it or add it to your home screen so you never have to type it again.
Does it work with no internet?	Yes. That is the whole point. Only a few features (NWS radar and alerts, APRS-IS, HF propagation) need the internet; everything else runs fully offline.
Why is the Amateur Radio button grayed out?	No station callsign has been set. Open the Setup wizard and add a callsign to enable it (see 2.6). This is normal for a group with no licensed operators.
Which mode should I start in?	Whichever fits the job: Amateur Radio for ham nets, the public-safety mode for radio ID and unit tracking, ICS for running an incident. You can switch any time without losing data.
Two of us see different screens — is something wrong?	No. The mode is remembered per device, so each operator can be in a different mode. The underlying data is the same for everyone.

2.11 Keeping Time — One Clock for Everyone

CHANGED **New: all devices show and stamp the server clock (GPS offline / internet online); connect a GPS with sky view.**

The time you see on the dashboard — and the time stamped on every ICS form and every check-in — is the **FieldCommand server's clock**, not your own device's clock. This matters. At a busy check-in point several laptops, tablets, and phones may each have a slightly different clock, and you never want two devices to disagree about when someone signed in or when a form was prepared. FieldCommand avoids that by having every connected device read one clock: the server's. Even if a laptop's own clock is wrong, the times it writes onto forms and check-in are the server's.

The server keeps that clock accurate in two ways, and it uses whichever is available — automatically, with nothing for you to switch:

TIME SOURCE	WHEN IT IS USED
Internet time	When a Wide Area Network (WAN) — cellular or satellite — is connected, the server sets its clock from internet time servers, the same way any computer does.
GPS time	When there is no internet — the normal field case — the server takes time from a connected GPS receiver. GPS satellites carry atomic-clock time, so a GPS gives the server exact Coordinated Universal Time (UTC) with no internet at all.

So that the server has accurate time even when the internet and cell service are gone, **connect a USB GPS receiver to the FieldCommand server and place its antenna where it can see the sky** — near a window, or better, outdoors or on a rooftop. A GPS sitting indoors away from windows may never get a fix, and without a fix it cannot set the clock. Give it a few minutes on first power-up to lock on; once it has a fix, the server's clock is disciplined to GPS automatically. The Pironman server case also includes a small battery-backed clock (a coin-cell), so the server remembers the time across power-offs, and GPS or the internet then fine-tunes it at the next start-up.

TIP You can see that timekeeping is healthy from the dashboard: the small service dots include one labeled **Time Sync / Chrony**. If the clock ever looks wrong, make sure the GPS has a clear view of the sky (or that a WAN is connected), then give it a few minutes to correct itself.

3 Organization Setup & Station Configuration

http://192.168.50.1/setup.html

Before using FieldCommand IMS for the first time, complete the Organization Setup to configure your station identity, network defaults, and — if you use one — the public-safety net settings. These values flow into Incident Command System (ICS) form headers, net logs, cover pages, and printed documents, so it is worth setting them accurately once at the start. Open it from the dashboard, fill in the fields below, and click Save at the bottom.

NOTE The station callsign is OPTIONAL. A group with no licensed amateur-radio operators leaves it blank, and FieldCommand keeps the Amateur Radio features grayed out while everything else works normally. Add a callsign later to turn those features on.

3.1 Station & Organization Identity

These fields identify your group on forms, logs, and printed documents.

FIELD	WHAT TO ENTER
Club / Station Callsign	Your group or station FCC callsign (for example, K9ESV). Leave blank if you have no licensed operators — the amateur features then stay off by design.
Personal Callsign (Operator)	The individual operator or builder callsign (for example, KE4CON). Optional.
Organization Full Name	Required. The full name of your agency or group — shown in ICS form headers and on cover pages.
Organization Abbreviation	A short form for compact displays, footers, and net-log headers (e.g. MCESV).
Associated Agency Name	The served agency or authority having jurisdiction, if any (for example, a county emergency management agency).
Agency Abbreviation	The short form of the associated agency.
City / County / State	Your location, used on headers and printed documents.
Contact Email	A contact address shown where forms call for one. Optional.

3.2 Position, Network, and Regional Defaults

FIELD	WHAT IT CONFIGURES
Latitude / Longitude (decimal)	The station position (for example, 42.3247 / -88.3822). Used for HF propagation and APRS position reports.
Grid Square (Maidenhead)	The 4- or 6-character grid locator (for example, EN52wa). Used for propagation calculations and APRS.
Default Incident Name	Pre-fills the incident-name field when you create a new incident (for example, "County EOC Activation").

FIELD	WHAT IT CONFIGURES
Wi-Fi Network Name (SSID)	The network name FieldCommand records and displays (default EMCOMM-NET). Note: the ASUS router broadcasts this network — the Pi does not (see the Network Hardware chapter).
Wi-Fi Password	Recorded here for reference; set the matching value on the router itself.
Server Address	The address operators type in their browser (default http://192.168.50.1). Change only if your network requires it.
Time Zone	Sets the time zone for every timestamp in logs and forms (for example, America/Chicago). Set it to your deployment time zone.

CHANGED **New feature: "Use GPS to fill location" button on the Setup page.**

TIP A connected GPS can fill the location in for you. On the Setup screen, click **Use GPS to fill location** (the button just above the Latitude field). It reads the GPS plugged into the server first; if that has no fix, it uses the location of the device you are running Setup on. It fills Latitude, Longitude, and the Grid Square automatically. A USB GPS receiver plugs into a USB port on the FieldCommand server and appears to the system as /dev/gps0. If no GPS is available, just type the coordinates in by hand.

3.3 Public-Safety Net Settings

If your group runs a public-safety (non-amateur) net, set the system details here. They label the public-safety net logger and its check-in fields, and they do not require a callsign.

FIELD	WHAT TO ENTER
Primary System Name	Required for the public-safety logger. The radio system name (for example, a statewide system, a mutual-aid system, or a P25 zone).
System Type	The kind of system, chosen from the list.
Unit ID Field Label	Required. What each station is identified by on this system (for example, Radio ID, Unit #, Badge #, or Apparatus #). This label is used throughout the public-safety logger and check-in.
Dispatch Center Name	The dispatch center for this system, if applicable.
Secondary System Name	An optional second system (for example, a mutual-aid or interoperability system).

3.4 Offline vs. Online Feature Table

FEATURE	OFFLINE?	NOTES WHEN OFFLINE
ICS platform — all five sections	Full	No change
All ICS forms and IAP	Full	No change
T-card resource board	Full	No change
FEMA cost tracking	Full	No change
Event templates	Full	No change
IAP PDF compilation	Full	No change
Digital signatures on forms	Full	No change

FEATURE	OFFLINE?	NOTES WHEN OFFLINE
Net control loggers (both)	Full	No change
FCC callsign lookup	Full	Uses local SQLite database — no internet needed
Member roster and QR check-in	Full	No change
GPS resource map	Full	No change
Barcode / QR scan check-in	Full	No change
Offline tactical APRS map	Full	Map tiles local — RF APRS still works
Kiwix reference library	Full	No change
NWS weather alerts	WAN	Paused — last alert cached and displayed
NEXRAD animated radar	WAN	Tiles unavailable — offline banner shown
APRS-IS internet feed	WAN	RF APRS continues — internet feed pauses
HF propagation data	WAN	Last retrieved data shown until WAN returns
AMPRNet / 44Net tunnel	WAN	Tunnel drops — local network unaffected
Winlink Telnet sessions	WAN	RF Winlink (VARA/Pactor) continues

3.5 Troubleshooting

SYMPTOM	WHAT TO DO
The Amateur Radio features are grayed out	No station callsign is set. Open Setup, enter your Club / Station Callsign, and save. Amateur features stay off by design until a callsign is present.
My changes did not show up on forms or headers	Make sure you clicked Save at the bottom of the Setup page, then re-open the form — headers read the saved values.
The Wi-Fi name I set here is not being broadcast	This field only records the name for display. The ASUS router broadcasts the Wi-Fi — set the matching SSID and password on the router (see the Network Hardware chapter).
Timestamps are off by several hours	The Time Zone is wrong. Set it to your deployment zone (for example, America/Chicago) and save; new entries use the corrected zone.
I changed the Server Address and now devices cannot reach the app	Put it back to the default (http://192.168.50.1). Change it only if you have also reconfigured the router and the static IP to match.

4 Member Roster & Quick Response (QR) code Check-In Codes http://192.168.50.1/roster.html

The Member Roster is the central personnel database for your whole operation. It stores every member, mutual-aid visitor, and regular participant with their identifiers, license class, roles, certifications, equipment capabilities, an optional photo, and a personal check-in code. From this one page you build and search the roster, print laminate-ready photo identification (ID) cards, show a member's Quick Response (QR) check-in code, and run live check-in when an activation starts. Everything you enter here is saved on the server, so it survives a reboot and lands in the incident record and backups.

TIP In one sentence: enter your people once, and the roster feeds check-in, net logs, ID cards, and QR codes for every activation afterward.

4.1 The Two Views: Directory and Activation

The roster page has two tabs across the top, each showing a live count in parentheses:

- **Directory** - every member as a card. This is where you add, search, edit, import, export, and print. It is the everyday view.
- **Activation** - the live check-in board used when an incident is running: click a member to check them in, track who is Active, Standby, or Released, and add walk-ins (see 4.7 and 4.8).

Above the tabs sits the toolbar that acts on the whole roster:

CHANGED New: "Roster Report" PDF button (see the Roster Report row in this table).

BUTTON	WHAT IT DOES
Search box	Type any part of a name, callsign, Member ID, or Radio ID to filter the Directory instantly.
+ Add Member	Opens the blank member editor (see 4.4).
Import CSV	Loads members from a spreadsheet file (see 4.9). Hover the button to see the exact column list.
Export CSV	Downloads the whole roster as a spreadsheet file (see 4.9).
Roster Report	Opens a print-ready Portable Document Format (PDF) report of the whole roster, sorted by name - a clean table of name, callsign, radio ID, member ID, license, phone, email, roles, and certifications, and ends with a "Prepared / Reviewed By" sign-off block (Prepared By, Date/Time, Reviewed By / Section Chief, Date/Time) in the same style as the ICS forms, for accountability. Print it or save it as a PDF from your browser.
Print ID Cards	Opens a print-ready PDF of photo ID cards with a scannable QR for every member (see 4.5).
Print	Opens your browser print dialog for a paper copy of the on-screen roster.

4.2 Reading a Member Card

In the Directory, each member appears as a card. At a glance the card shows the member's photo (or the first letter of their first name if no photo is stored), their name, and colored badges for the pieces of identity they have:

ON THE CARD	WHAT IT MEANS
License class badge	The amateur license class, such as Technician or General, shown only if set.
Role badges	Any assigned roles, such as Net Control or Operator.
Callsign badge	The Federal Communications Commission (FCC) amateur callsign.
Radio ID badge	The trunked or Project 25 (P25) radio unit number.
Member ID badge	Your organization's internal member number.
Certification dots	Five small dots for ICS-100, ICS-200, ICS-700, EmComm I, and CPR/AED - green means held, gray means not held. Hover a dot to read which one.
QR button	Opens that member's personal check-in code (see 4.6).

NOTE Click anywhere on the card (except the QR button) to open the full editor for that member.

4.3 Member Fields

The Add and Edit windows use the same form, grouped into labeled sections. Only a name or a callsign is really needed to save a usable record; every other field is optional and can be filled in later.

4.3.1 Identifiers

FIELD	WHAT IT MEANS
Callsign	The FCC amateur radio callsign (for example W8XXX). Typed in capitals automatically. Leave blank for members who are not licensed.
Radio ID / Unit #	The public-service radio unit number if the member carries a P25, Digital Mobile Radio (DMR), or General Mobile Radio Service (GMRS) radio.
Member ID	Your internal member number - for example an ARES, RACES, or American Radio Relay League (ARRL) number. Used as the default check-in code.

4.3.2 Personal Information

FIELD	WHAT IT MEANS
First / Last Name	The member's name as it should appear on cards and logs.
License Class	Chosen from a list: Amateur Extra, General, Technician, Novice, Advanced, GMRS, or No amateur license.
Phone / Email	Contact details. Stored on the local server only.
Address	Mailing or home address, optional.
Grid Square	The Maidenhead grid locator for radio planning (for example EN90). Up to eight characters.
Emergency Contact	Who to call for this member in an emergency.
Notes	A free-text box for anything else worth recording.

4.3.3 Roles, Certifications, and Equipment

Three grids of checkboxes capture what a member can do. Tick every box that applies:

CHANGED **New: ICS command/general-staff roles added, plus a free-text "Other role" box.**

- **Roles** - the amateur and public-service roles Net Control (NCS), Operator, Liaison, and Emergency Coordinator (EC), plus the ICS command and general-staff positions Incident Commander (IC), Deputy Incident Commander, Safety Officer (SO), Public Information Officer (PIO), Liaison Officer (LOFR), Operations, Planning, Logistics, and Finance/Admin Section Chiefs, and the Communications Unit Leader (COML) and Technician (COMT). For any position not in the list, type it into the **Other role** box and click **Add** - it is saved with the member and shown as its own badge.
- **Certifications** - ICS-100, ICS-200, ICS-300, ICS-400, ICS-700, ICS-800, EmComm I, EmComm II, CPR/AED, First Aid, and CERT.
- **Equipment** - HF Radio, VHF/UHF, Digital, Packet, PACTOR Modem, VARA HF, VARA FM, APRS, Winlink, Go-Box, Generator, Battery/Solar, and Vehicle Mount.

4.4 Adding and Editing Members

1. Click **+ Add Member** in the toolbar.
2. Fill in at least a name or a callsign. Add the Radio ID, Member ID, and personal details as needed.
3. Tick the applicable Roles, Certifications, and Equipment boxes.
4. Click **Save Member**. The member appears in the Directory right away.
5. To change an existing member, click their card, edit any field, and click **Save Member** again.
6. To remove someone, open their card and click **Delete**, then confirm. This removes the roster entry but does not erase past net-log or check-in history.

4.5 Member Photos and Printable ID Cards

Each member record can hold one photo, used to print a laminate-ready ID card with a scannable QR check-in code. In the editor, under **Photo (for ID card)**, click **Take / upload photo** to use any phone or webcam picture - a head-and-shoulders shot on a plain background works best. The photo is resized automatically before it is stored, so you do not need to shrink it yourself. Click **Remove** to clear it.

There are two ways to print cards:

- **All cards at once** - click **Print ID Cards** in the toolbar. This opens a PDF of cards for every member. Walk-in and mutual-aid entries are included only if they have a photo.
- **One card** - open a member, then click the **ID Card** button next to Save to print just that person.

> **NOTE** A photo is optional. A member with no photo still gets a roster card and a QR code; they simply are not included in the printed photo-ID batch.

4.6 QR Check-In Codes

Every member has a personal QR code that lets them check in instantly at any activation using the Scan Check-In page (see Chapter 10). The QR button on each card opens a window with that member's code.

1. Click the **QR** button on any member card.
2. The window shows a scannable QR image, the member's name, and the code in plain text underneath. The QR image is generated by the FieldCommand server itself, so it works fully offline - no internet is required.
3. Click **Print** to open a clean print page with the QR, the name, and the check-in code, ready to laminate or tape to a badge.
4. Click **Save** to download the QR image so a member can keep it on their phone.

The code carried by the QR is the member's barcode identifier if one was set during import; otherwise it falls back to the callsign, and then to the Member ID. If the QR image ever fails to load, the plain-text code shown below it is a working manual-entry fallback - the operator can type it into the Scan Check-In page by hand.

4.7 Activation: Checking Members In

When an incident starts, switch to the **Activation** tab. It splits into two columns: **Roster - Click to Check In** on the left, and **Activated Personnel** on the right.

1. Use the **Search member to check in** box to find a person, or scroll the left list.
2. Click a member. A small prompt asks for their **Assignment** - type it (for example "Shelter Net") and confirm, or leave it blank.
3. The member moves to the Activated Personnel column with a status of **Active**.
4. Change anyone's status with the dropdown on their row: **Active**, **Standby**, or **Released**. Released personnel drop out of the active count.
5. Click **Export Log** at any time to download the activation record as a spreadsheet for the incident file.

4.8 Walk-In Check-In

A walk-in is someone who shows up to help but is not yet on the roster - a mutual-aid operator or a first-time volunteer. You can check them in without stopping to build a full record.

1. On the Activation tab, click **+ Walk-In Check-In**.
2. Enter what you know: **Callsign or Radio ID**, **Name**, **Organization**, and **Assignment**. A callsign or a name is enough.
3. Click **Check In Walk-In**. They appear in Activated Personnel with a yellow **WALK-IN** badge.
4. If you later want to keep them, click the **+ Roster** button on their row to save them as a permanent member you can then finish editing.

WARN Walk-ins are part of the live activation but are not saved to the permanent roster until you use + Roster. Do that before they leave if you want to keep the record.

4.9 Importing and Exporting via CSV

You can load an existing member list from a Comma-Separated Values (CSV) spreadsheet instead of typing everyone in. Click **Import CSV** and choose your file. The first row must be column headings.

Recognized columns: **member_id, callsign, radio_id, first_name, last_name, role, phone, email, grid**, plus optional **license_class** and **roles** (several roles separated by semicolons).

Import matches each row to a member by a natural key (Member ID first, then callsign, then Radio ID, then name), so re-importing an updated file updates the existing people instead of creating duplicates. Members already on the server who are not in the file are left untouched. To go the other way, click **Export CSV**: FieldCommand downloads the whole roster as a dated CSV file you can edit in a spreadsheet or hand to another server.

4.10 Troubleshooting

SYMPTOM	WHAT TO DO
The Directory says "Cannot reach API"	The core service on port 5050 is not answering. Check the dashboard health panel and make sure the fcc-lookup service is running, then reload the page.
A QR image will not appear	The plain-text code shown under it still works - type it into the Scan Check-In page by hand. If it keeps failing, confirm the fcc-lookup service is running.
A member is missing from the Print ID Cards batch	Cards print only for members who have a photo saved (and always for walk-ins only if they have one). Open the member, add a photo, and print again.

SYMPTOM	WHAT TO DO
Import CSV added duplicate people	The rows had no Member ID, callsign, or Radio ID to match on, so each became a new record. Add one of those columns and re-import to merge them.
A walk-in disappeared after the activation	Walk-ins are not permanent unless saved. Next time click + Roster on their row before they are released to keep the record.
The activation count looks wrong	Personnel marked Released are excluded from the active count by design. Set their status back to Active or Standby if they are still working.

5 Incident Management — Creating and Managing Incidents http://192.168.50.1/incident.html

All Incident Command System (ICS) work in FieldCommand IMS is organized around an **incident**. An incident holds the ICS forms, resource T-cards, personnel check-ins, cost tracking, and net log associations for one activation or exercise. You can have multiple incidents in the system simultaneously — only one is designated the **active incident** at any time.

5.1 Creating an Incident

1. Navigate to **Incident Management** from the dashboard.
2. Click **+ New Incident**.
3. Enter the incident name, incident number (if assigned by your agency), incident type, and initial operational period.
4. Select the incident commander from your roster, or type a name if the Incident Commander (IC) is not in the roster. For incidents operating under **Unified Command**, see Section 5.2 below.
5. Click **Create Incident**. The incident opens immediately and becomes the active incident. The incident name appears in the dashboard header and in all form headers.

The New Incident form has these fields:

FIELD	WHAT TO ENTER
Incident Name	Required. A clear name for the activation, such as "Fox River Flooding". Shown in the dashboard header and on every form.
Incident Number	Your agency's assigned number, if you have one (for example, 2026-0142). Optional.
Incident Type	Required. The category that best fits the activation — see the list in Section 5.3.
Jurisdiction	The responsible jurisdiction (for example, "McHenry County, IL"). Optional.
Incident Location / Address	Where the incident is centered. Optional.
Incident Commander	Pick a name from the roster or type one. For a multi-agency response, enter "Unified Command" (see Section 5.2).
Operational Period Duration	How long each operational period lasts (commonly 12 or 24 hours). Used as the default for later periods.
ICS Form Variant	Which edition of the ICS forms this incident uses. Leave the default unless your agency standardizes on a specific variant.
Initial Situation Summary	A short description of what is happening at the time of activation.

5.2 Command Structure — Single Incident Commander vs. Unified Command

ICS supports two command structures, and FieldCommand IMS accommodates both without any configuration change. The difference is a workflow convention, not a software setting.

STRUCTURE	WHEN USED	HOW TO CONFIGURE IN FIELDCOMMAND
Single IC	One agency or jurisdiction has clear authority. Most activations involving a single responding organization.	Enter the IC's name or callsign in the Incident Commander field when creating the incident. The ICS-203 org chart will show one IC at the top of the command structure.
Unified Command	Two or more agencies or jurisdictions share command authority — common in multi-agency responses, complex disasters, or incidents crossing jurisdictional boundaries. Each contributing agency provides an IC-equivalent to the UC group. All agencies still work from a single Incident Action Plan (IAP).	Enter "Unified Command" or the UC group designation in the Incident Commander field. List each UC member in the ICS-203 Organization Assignment List under the Unified Command section. All ICS forms, T-cards, and cost documentation continue to function identically — the UC structure is a labeling and documentation convention within the same incident record.

NOTE

Under Unified Command, each participating agency still develops its own objectives, but those objectives are reconciled into a single IAP. The Planning Section Chief role in the ICS-203 is typically filled by one agency on a rotating basis or by consensus. FieldCommand IMS supports multiple ICS-204 Assignment Lists — one per branch or division — which accommodates multi-agency operational structures naturally.

5.3 Incident Types

When creating an incident, pick the type that best matches the activation from a large categorized dropdown (about 32 options grouped into seven categories). The type labels the incident badge throughout the interface. All types use the same underlying ICS form set — the type is an organizational label, not a functional restriction.

CATEGORY	EXAMPLE TYPES
Natural Hazards	Winter Storm, Flooding, Tornado / Severe Weather, Earthquake, Wildfire, Heat Emergency, Drought
Technological	Hazmat / Chemical Spill, Transportation Accident, Structure Fire, Power Outage / Infrastructure, Dam Failure, Nuclear / Radiological
Human-Caused	Mass Casualty Incident, Active Threat, Civil Disturbance, Terrorism
Search & Rescue	Wilderness, Urban, Water, Missing Person — Dementia / Memory, Missing Person — Child
Public Health	Disease Outbreak / Pandemic, Mass Casualty — Medical, Public Health Emergency
Planned Events	Planned Event — Public Safety, Planned Event — EMCOMM Exercise, Drill / Training Exercise
Other	Mutual Aid Request, EOC Activation, Other / All-Hazards

Any incident can be flagged as a drill/exercise. Exercise / Scenario Reset (under incident settings) wipes exercise data while preserving the roster, channel library, and repeater database.

5.4 Operational Periods

ICS organizes work into **operational periods** — typically 12 or 24-hour blocks. FieldCommand IMS tracks the current operational period and uses it as the default for new T-cards, check-ins, and form entries. To advance to a new operational period, click **+ New Period** in the incident header. Prior period data is retained and viewable.

5.5 Archive, Restore, and Delete

After an incident closes, the full data package can be archived to a Universal Serial Bus (USB) drive for long-term storage and removed from the Pi's active storage.

ACTION	WHAT IT DOES	REVERSIBLE?
Archive to USB	Writes a complete JSON package (all forms, T-cards, check-ins, costs, net log associations) to /media/fieldcommand/backup/incidents/ on the labelled USB drive. Marks the incident as archived on the Pi.	Yes — restore any time
Restore from USB	Reads the JSON archive from the USB drive and re-inserts all data. Restores the incident to fully active status.	Yes
Hard delete from Pi	Permanently removes the incident and all associated data from the Pi SSD. Run only after confirming the archive is on USB.	Permanent

NOTE The USB drive must be labelled FIELDCOMMAND (all caps). Any USB drive with this label is recognized automatically. A LaCie Rugged or similar ruggedized 1TB USB-C drive is recommended for field use.

CHANGED **Renamed: "Beta Reset" is now "Exercise / Scenario Reset".**

5.6 Exercise / Scenario Mode

Exercises and training scenarios can be tagged as **Scenario** when the incident is created. Scenario incidents display a yellow badge throughout the interface so operators always know they are in training mode. A **Exercise / Scenario Reset** wipes all scenario data — incidents, forms, costs, check-ins, T-cards, and meetings — while preserving the roster, hospital directory, channel library, and repeater database. This returns the system to a clean state for the next exercise without losing any permanent configuration data.

5.7 Working with Multiple Incidents — Selecting, Switching, and Editing

The Incident Management screen lists every incident on the server. You can keep several at once, but only one is the **active** incident at a time — the one all new forms, T-cards, check-ins, and costs attach to.

Open / switch the active incident. Click any incident in the list to open it and make it active. Its name then shows in the dashboard header and on every form. If none exist yet, the screen reads "No active incidents".

Check which one is active. The active incident's name always appears in the dashboard header. Each card in the list also shows its current operational period (for example, "Period 2").

Edit an incident. Click **Edit** on the incident to change its name, number, type, commander, jurisdiction, location, or summary.

Advance the operational period. Click **Next Period** to open the period prompt, review the objectives carried forward, and start the next operational period (see Section 5.4).

5.8 Troubleshooting

SYMPTOM	WHAT TO DO
My forms or T-cards are attaching to the wrong incident	Check which incident is active — its name is in the dashboard header. Open the correct one from Incident Management to make it active, then continue.
I don't see my incident in the list	It may have been archived (Section 5.5); archived incidents are restored from the USB drive. Also confirm you are on the right server (192.168.50.1).
Archive to USB failed or the drive was not found	The backup drive must be labelled FIELDCOMMAND (all capitals) and plugged in. Re-insert it, wait a few seconds, and try the archive again.

SYMPTOM	WHAT TO DO
I can't delete an incident / I'm worried about deleting one	Hard delete is permanent. Archive to USB first (Section 5.5), confirm the archive is on the drive, and only then hard-delete from the Pi.
Objectives did not carry into the new operational period	Use Next Period to advance (not a brand-new incident). The carry-forward prompt is where you keep or edit the objectives for the new period.
My exercise/scenario data will not clear	Use Exercise / Scenario Reset in the incident settings. It wipes exercise data (incidents, forms, costs, check-ins, T-cards) but preserves the roster, channel library, and repeater database.

6 Pre-Planned Event Templates http://192.168.50.1/event_templates.html

A Pre-Planned Event Template is a complete, ready-made incident setup that you activate in seconds. Instead of building an incident from a blank page while the clock is running, you choose a template that already contains the objectives, the resource types you will need, the communications plan, and the Incident Command System (ICS) organization for that kind of event. Templates remove repetitive setup work, prevent the mistakes that happen when people build under pressure, and make every activation of the same event type look and run the same way.

TIP In one sentence: a template is a saved incident blueprint — pick it, name the incident, and everything is pre-loaded and ready to adjust.

6.1 What a Template Contains

Every template holds five kinds of pre-planned content. When you activate the template, each part is copied into the new incident so you can start work at once and adjust as the event develops.

PART	WHAT IT PRE-LOADS INTO THE INCIDENT
Objectives	The incident objectives — the ICS-202 "what we are trying to achieve" list. Plain-language goals for the operational period.
Resources	The kinds and quantities of resources you expect to need — teams, vehicles, equipment — so ordering and the T-card board start already populated.
Channels	The communications plan: named channels with receive and transmit frequencies, tone, mode, and the job each channel does.
Organization	The ICS command and general-staff roles, plus branch and division labels, so the org chart is drawn before anyone arrives.
Safety & flags	A safety message for the Safety Officer, plus optional flags that mark the template as a training scenario, a protected standard template, or a candidate for a future update.

CHANGED **New: 10 built-in templates now (4 new sample packs); the set grows via updates.**

6.2 Built-In Templates

FieldCommand IMS ships with ten built-in templates covering the most common incident types. Each can be used as-is or edited to match your local protocols.

TEMPLATE	PRE-CONFIGURED CONTENTS
Shelter Activation	Shelter management objectives, cot and supply resource types, registration and medical channels, Red Cross coordination section
Search & Rescue	Search and Rescue (SAR) objectives, field team and K9 resource types, search sector channels, base camp and medical branches
Severe Weather	Damage assessment objectives, utility and debris resource types, shelter and Emergency Operations Center (EOC) coordination channels, public information branch

TEMPLATE	PRE-CONFIGURED CONTENTS
Mass Gathering / Event	Crowd management objectives, medical and security resource types, venue and dispatch channels, medical and operations branches
HazMat / Spill	Decon and zoning objectives, HazMat team resource types, hot/warm/cold zone channels, safety officer emphasis
Planned Exercise / Drill	Training objectives, evaluator and observer resource types, exercise control channel, scenario-tagged ()
Flood Response	Swiftwater rescue and evacuation objectives, boat and high-water resource types, water-rescue and evacuation channels, rescue and mass-care sections
Wildland / Interface Fire	Structure-protection and LCES objectives, engine, crew, and air resource types, ground-operations and evacuation channels, fire-suppression branch
Winter Storm / Power Outage	Warming-center and welfare-check objectives, generator and high-clearance resource types, welfare and utility-liaison channels, mass-care section
Communications Support	Net and Winlink objectives, net-control and go-kit resource types, primary-net, tactical, and digital channels, communications-unit branch

NOTE The built-in set grows over time. A FieldCommand update can add new templates for situations other agencies encounter; they appear automatically after the update and never overwrite a template you have already customized.

6.3 The Two Views: Activate and Manage

The Event Templates screen has two views, chosen with the toggle near the top of the page:

- **Activate** — a gallery of templates for starting a real incident. This is the everyday view.
- **Manage** — the same templates shown as editable cards, for building, editing, reordering, exporting, importing, and deleting.

Across the top are the toolbar buttons that act on the whole library:

BUTTON	WHAT IT DOES
+ New Template	Opens the editor with a blank template.
Import JSON	Loads templates from a JSON file — from another FieldCommand server or a shared pack.
Export All	Downloads every template on this server as one JSON file — a backup or a share.
Export Update Candidates	Downloads only the templates you have flagged for future updates (see 6.7).

6.4 Activating a Template

1. Open **Event Templates** from the dashboard and make sure you are on the **Activate** view.
2. Browse the gallery and click the template you want. A configuration panel opens.
3. Enter the incident name, the date and time, and any other details you want to set before activating.
4. Click **Activate Template**. FieldCommand creates the incident and, from the template, auto-fills its core Incident Action Plan (IAP) forms — **ICS-202** (objectives and safety message), **ICS-203** (the organization assignments), and **ICS-205** (the communications plan) — and builds a **T-card** for each resource.
5. Use the success links that appear to open the ICS-202, ICS-203, or ICS-205 form, the T-card board, or the dashboard.

> **NOTE** Activation fills in every ICS form the template carries data for — the objectives and safety message (ICS-202), the organization (ICS-203), and the communications plan (ICS-205) — and builds the T-card board. From there you complete the rest of the Incident Action Plan (the assignment lists, medical plan, safety plan, and any other forms the incident needs) and assemble and print it in the IAP section of the app. Activation gives you the head start; the IAP section is where you finish building it.

6.5 Building or Editing a Template

In the Manage view, click a card to edit it, or click **+ New Template** for a blank one. The editor opens as a single form with the sections below. Every field is optional except the template name. Click **Save Template** when finished; all template data is stored on the server and is not overwritten by app updates.

6.5.1 Basic Information

FIELD	WHAT IT MEANS
Template Name	Required. The name shown on the card and in the gallery — for example, "Flood Response".
Incident Type Label	A short category shown under the name, such as "Flood — Water Rescue". Optional.
Icon (emoji)	A single emoji shown on the card for quick recognition, such as . Optional.
Sort Order	A number that sets the card position in the gallery — lower numbers appear first.
Summary / Description	One line describing the template, shown on the card.

CHANGED **New: Common Objective Library picklist in the template editor (hand entry still works).**

6.5.2 Objectives

The incident objectives, one per line. There are two ways to add them: pick from the **Common Objective Library** drop-down - a set of ready-made objectives grouped by category (Life Safety, Communications, Search and Rescue, and more) - which drops the objective in as a new line you can then edit, or click **+ Add Objective** and type your own by hand. Use the × to remove a line. Write them as plain-language goals; they become the ICS-202 objectives on the activated incident. The same Common Objective Library appears on the ICS-202 form itself, so your templates and your incidents draw from one shared list.

6.5.3 Safety Message

A short safety briefing for this event type. It carries into the incident for the Safety Officer and appears with the safety information for the operational period. Use it for the standing hazards and rules that apply every time you run this kind of event.

6.5.4 Resources

Each resource row has three fields:

FIELD	WHAT IT MEANS
Name	The resource label, such as "Swiftwater Rescue Team".
Type	The kind of resource, chosen from a list — see the reference table in 6.8.
Qty	How many of this resource the template expects — a starting number you can change on the incident.

6.5.5 Channels

Each channel row is a line in the communications plan:

FIELD	WHAT IT MEANS
Name	The channel purpose name, such as "Command" or "Water Rescue Tactical".
RX	Receive frequency — what the radio listens on, in megahertz (e.g. 146.520).
TX	Transmit frequency — what the radio sends on. On a simplex channel, RX and TX are the same.
Tone	The sub-audible tone (CTCSS/PL) if the channel needs one; leave blank for none.
Mode	The radio mode — see the reference table in 6.8.
Function	The job the channel does — Command, Tactical, Medical, and so on (6.8).

6.5.6 Organization

The ICS structure the incident starts with. Fill in the roles and the branch and division labels that fit this event type:

FIELD	WHAT IT MEANS
Operations Section Chief	The person or role leading tactical operations.
Safety Officer	The role responsible for personnel safety and the safety message.
Public Information Officer	The role handling public and media information.
Branch label	The name of the first operations branch, such as "Rescue & Evacuation".
Division A–D labels	Names for up to four divisions or groups under operations, such as "Swiftwater Rescue".

6.5.7 Flags

Three checkboxes sit at the bottom of the editor:

Training scenario () — incidents started from this template are marked as exercises, not real events.
Standard template () — pins the template with the built-ins and protects it from deletion (see 6.6).
Suggest for future updates () — flags the template as a candidate to ship with FieldCommand (see 6.7).

CHANGED **New: mark your own template "standard" (protected from deletion).**

6.6 Standard Templates and Deletion Protection

Templates come in three levels. The only difference is whether they can be deleted — all three are fully editable.

TYPE	CAN EDIT?	CAN DELETE?
Built-in (shipped with FieldCommand)	Yes	No — protected
Standard (you marked it)	Yes	No — protected
Ordinary custom	Yes	Yes — Delete button

Mark a template you rely on as **standard ()** so a busy operator cannot delete it by accident during an activation. To remove a standard template, open it and un-tick the standard box first; then the Delete button returns. Built-in templates cannot be deleted at all, but they can be disabled.

TIP

To hide a built-in template you never use, open it and turn off **Enabled**. It disappears from the Activate gallery but is kept in Manage, so you can turn it back on later.

CHANGED New: suggest templates for future updates + "Export Update Candidates"; maintainer drop-in packs.

6.7 Sharing Templates and Suggesting New Ones

Templates are plain JSON files, so they move easily between servers and agencies.

Export one — in the editor, click **Export JSON** to download that single template.

Export all — on the main screen, click **Export All** for the whole library as one file (a good backup).

Import — click **Import JSON** and choose a file. Imported templates are added; if an incoming template shares an identifier with one you already have, it is added under a new identifier instead of overwriting yours.

6.7.1 Suggesting a Template for a Future Update

When your group builds a template for a recurring local situation that other agencies would probably find useful too, you can suggest it for a future FieldCommand release without forcing it on anyone:

1. Open the template, tick **Suggest this template for future FieldCommand updates ()**, and save. This only flags it — nothing leaves your server yet.
2. On the main Event Templates screen, click **Export Update Candidates**. FieldCommand downloads one JSON file containing only your flagged templates.
3. Send that file to whoever maintains your FieldCommand build.

The maintainer reviews the suggestions and can add the worthwhile ones to a future update. Shipped templates then appear automatically on every server after that update — and because updates only add templates with new identifiers, they never overwrite a template you have customized. Not every local situation belongs in the shipped product, and that is fine: a template you never flag stays entirely on your own server.

NOTE

For maintainers (not an operator task): the mechanics of adding a suggested template - dropping the exported JSON into the python/seed_templates/ folder so it ships on the next update - are covered in the FieldCommand Programming Guide, in the Data Layer chapter under "Drop-In Template Packs" (and in python/seed_templates/README.md). If you maintain the build, that is where to look.

6.8 Reference — Channel and Resource Options

The drop-down choices in the channel and resource sections, in plain language.

6.8.1 Channel Function

OPTION	MEANING
Command	The command net — command staff and unit leaders coordinate here.
Tactical	A working channel for a specific job or team on the ground.
Medical	Medical coordination and patient traffic.
Logistics	Ordering, supply, transport, and support traffic.
Liaison	Coordination with outside agencies, the EOC, or utilities.
Other	Any purpose that does not fit the categories above, such as a data channel.

6.8.2 Radio Mode

MODE	MEANING
FM	Frequency Modulation — the standard voice mode on VHF/UHF (2 m, 70 cm).
NFM	Narrow FM — FM at reduced deviation, used where narrow channel spacing is required.
AM	Amplitude Modulation — for example, aircraft-band monitoring.
USB	Upper Sideband — a common voice mode on the higher HF bands.
LSB	Lower Sideband — a common voice mode on the lower HF bands.
DIG	A digital mode — packet, VARA, and similar data modes such as Winlink.

6.8.3 Resource Type

TYPE	MEANING
Personnel	Individual people or staffed positions.
Crew	An organized team that works as a single unit.
Engine	A fire engine or similar apparatus.
Vehicle	Any other vehicle — truck, bus, boat, or high-water vehicle.
Equipment	Gear and supplies — generators, cots, go-kits, tools.
Helicopter	A rotary-wing air resource.
Other	Anything that does not fit the categories above.

6.9 Troubleshooting

SYMPTOM	WHAT TO DO
A template I want is not in the gallery	It may be disabled. Switch to the Manage view — disabled templates still appear there. Open it and turn Enabled back on.
The Delete button is missing	The template is built-in or marked standard (). Un-tick the standard box to delete it, or disable it instead.
I imported a file but my edited template is unchanged	Import never overwrites an existing template. The incoming one was added under a new identifier — look for a second card with a similar name.
Export Update Candidates downloaded nothing	No template is flagged yet. Open a template, tick "Suggest for future updates ()", save, then export again.
Activating did not open the incident	That is expected — activation shows success links instead. Click a link to open the new incident, its IAP, or its T-cards.
My template changes disappeared after an update	They should not — template data is stored on the server and updates only add new templates. Confirm you are on the right server (192.168.50.1) and the Manage view.

7 Amateur Radio Net Control Logger http://192.168.50.1/netcontrol.html

The Amateur Radio Net Control Logger is the digital net log for licensed amateur radio operations - Amateur Radio Emergency Service (ARES), Radio Amateur Civil Emergency Service (RACES), Auxiliary Communications (AUXCOMM), SKYWARN, and any other amateur net. It replaces the paper net log with a live record: you open a net, log each station as it checks in, capture message traffic, and at the end export a finished ICS-309 Communications Log ready for the Incident Action Plan (IAP). It runs several nets at once, autofills each station's name from the offline Federal Communications Commission (FCC) database, and lets section chiefs watch a read-only copy from their own devices.

TIP

In one sentence: create a net, type each callsign as stations report in, and FieldCommand keeps the times, durations, and traffic - then hands you the ICS-309.

7.1 How the Screen Is Laid Out

The page has four working areas. Knowing where each one lives makes the rest of this chapter quick to follow.

AREA	WHAT IS THERE
Net tabs bar (top)	The **NET TABS** strip with a green **+ New Net** button, one badge per open net showing its name and check-in count, and on the right the **Observer Link**, **ICS-309**, and **Backup JSON** buttons.
Net header	The green-edged panel naming the selected net, with its **Opened**, **Closed**, and live **Duration** times, a **Drill Mode** checkbox, and the red **Close Net** button.
Entry form + tabs	The station check-in form, then three tabs - **Stations Logged**, **Traffic Log**, and **Roster Chips**.
Sidebar (right)	The **NET SUMMARY**, **ACTIVE NETS**, and **LAST 5 ENTRIES** panels that update as you log.

7.2 Opening a Net

1. Open **Net Control** from the dashboard and click the green **+ New Net** button in the NET TABS bar. The **Create New Net** window opens.
2. Type a **Net Name -** for example, "Thursday Evening Net". This is the only required field.
3. Choose a **Net Type** from the list (ARES Net, RACES Net, SKYWARN Net, EmComm Net, Traffic Net, Training Net, HF Net, VHF/UHF Net, or Digital Net).
4. Type the **Frequency / Mode** (for example, "146.520 FM"), or click the amber **Pick** button to choose a channel from the ICS-205 communications plan or the Channel Library (see 7.2.1).
5. Optionally pick a **Mode** and set an **open time**. Leave the open-time box blank to use the current moment; fill it in only to back-date a net that already started.
6. Type the **Net Control Callsign** of the operator running the net.
7. Click the green **Create Net** button. The net appears as a badge in the NET TABS bar, the header fills in, and the live Duration timer starts counting.

NOTE

You can run several nets at the same time - each gets its own badge. Click any badge to switch to that net. Every button on the top bar (Observer Link, ICS-309, Backup JSON) always acts on the net you currently have selected.

7.2.1 Picking a Channel Instead of Typing

The **Pick** button next to the Frequency / Mode box opens a searchable **Select Channel** list. It pulls channels from the incident ICS-205 (tagged **205**), the repeater database (tagged **RPT**), and the Channel Library (tagged **LIB**), and shows each channel name, receive frequency, mode, and sub-audible tone. Click a channel and its frequency and mode drop straight into the net. If no channels are set up, the app tells you to type the frequency by hand.

Where those channels come from: the **205** entries appear once the incident ICS-205 has channels on it, the **RPT** entries come from the repeater database, and the **LIB** entries are the ones you set up yourself on the Channel Library page — described next in 7.2.2.

7.2.2 Setting Up the Channel Library

CHANGED **New section + pointer: the Channel Library page (where the LIB channels are configured).**

The **LIB** channels in that picker come from the **Channel Library** — a page where you enter your agency's own radio channels once so they auto-fill the **ICS-205 Communications Plan** and appear in the net logger's channel picker on every device. Open it from the dashboard's **Channel Library** card (the radio icon). Because the library is saved on the server, every operator on every device sees the same channels.

TIP

A set of universal channels — National Simplex, APRS, NIFOG, and Mutual Aid — is always available automatically. You only need to add your agency-specific channels here: local repeaters, interop channels, and tactical frequencies.

There are three ways to fill the library:

WAY	WHAT TO DO
Add by hand	Click **+ Add Channel** and fill in the fields (listed below). Best for a handful of local channels.
Import a CSV	Click **Import CSV** to load channels from a spreadsheet. **Export CSV** saves the whole library to a file (channel_library.csv) for backup or to copy to another FieldCommand system.
Import from RadioReference	Click **Import from RadioReference**, enter your RadioReference username and password, look up your county by ZIP code, then fetch channels — optionally filtered by category (Fire Dispatch, EMS, Law, Interop, and so on). This requires a RadioReference Premium subscription; your credentials are used only for that one lookup and are not stored.

Each channel you add by hand has these fields:

FIELD	WHAT IT MEANS
Name	The channel's plain name, e.g. "McHenry County Command".
Alpha Tag	A short radio-display tag, up to 10 characters, e.g. "MHCO CMD".
Function	What the channel is for: Command, Tactical, Interop, Medical, Data, Amateur, Calling, Mutual Aid, Air, or Other.
RX Frequency (MHz)	The receive frequency — required, e.g. 155.3400.
TX Frequency (MHz)	The transmit frequency; leave blank if it is the same as receive (a simplex channel).

FIELD	WHAT IT MEANS
PL Tone / DCS	The sub-audible tone or digital code, e.g. "100.0 Hz" or "D023N".
Mode	FM, NFM, P25, DMR, D-STAR, C4FM, AM, SSB, or Analog.
Division / Group	Optional Incident Command System (ICS) assignment, e.g. "Div A", "Grp SAR", or "All".
Notes	Repeater location, coverage area, or special instructions.

Once saved, these channels appear tagged **LIB** in the Select Channel picker from 7.2.1 and auto-fill the ICS-205 Communications Plan — so a channel entered once is ready everywhere it is needed.

7.3 Logging Check-Ins

Once a net is selected, the check-in form appears above the tabs. Logging a station takes one field and one key.

1. Type the reporting station's callsign in the **Callsign** box. It forces upper case as you type.
2. When you finish the callsign, FieldCommand looks it up in the offline FCC database and fills the **Name** box automatically (see 7.4). Correct or add the name if needed.
3. Set the **status** drop-down if this is more than a plain check-in (the choices are Check-In, Traffic, Priority, Emergency, Net Control, Check-Out, Mobile, and Portable).
4. Optionally add a **Location/Grid**, a **precedence** (Routine, Welfare, Priority, or Emergency), and free-text **Remarks / Traffic**.
5. If this station is filling an incident job, choose an **ICS Position** from the grouped list and type the **Incident ID**; an **ICS-211** link then appears to open the Check-In List.
6. Press **Enter** or click the green **LOG ENTRY** button. The station appears in the Stations Logged list with a timestamp, and the sidebar counts update. Click **Clear** to empty the form without logging.

FIELD	WHAT IT MEANS
Callsign	The station identifier - the only field you truly must enter. Triggers the FCC lookup.
Name	The operator name. Filled from the FCC database; you can overwrite it.
status	How this station is reporting - check-in, traffic, emergency, mobile, and so on.
Location/Grid	Where the station is - a town, address, or Maidenhead grid square.
precedence	Message urgency - Routine, Welfare, Priority, or Emergency. It color-codes the row.
Remarks / Traffic	A short note about the station or a message it is passing.
ICS Position	The Incident Command System job the operator holds, chosen from a grouped list (Command, Operations, Planning, and so on). Optional.
Incident ID	The incident this check-in belongs to, so it can flow to the ICS-211 Check-In List. Optional.

7.4 The FCC Autofill Card

FieldCommand stores the entire United States amateur license database - more than 800,000 licensees - on the Pi itself, so lookups work with no internet. As soon as you type a valid callsign, a blue **autofill card** appears above the form showing what the FCC has on file.

CARD FIELD	WHAT IT SHOWS
Name	The licensed operator or club name on record.
Class	The license class - Technician, General, Amateur Extra, and so on.
Status	Whether the license is active, expired, or otherwise.
Location	The city and state on the license.
Expires	The license expiration date.

TIP The lookup only runs while the **Auto FCC lookup** checkbox at the bottom right of the form is ticked. Turn it off to log a special-event or tactical callsign the FCC would not recognize.

7.5 Managing Logged Stations

Every logged station is a row in the **Stations Logged** tab. The newest sits at the top. Each row shows the time, the callsign and name, a colored status badge, and a running **Duration**. Three buttons sit on the right of each row.

CONTROL / TAG	WHAT IT DOES
Check Out	Marks the station as leaving the net. The checkout time is stamped and the participation duration stops climbing. Once out, the button reads "Out" and is grayed.
+ Roster	Adds a station that is not yet on your roster to the permanent roster, assigning it a member ID. Shows only for stations not already on file.
x (remove)	Deletes the entry from this net - use it to clear a mistaken log.
WALK-IN tag	An amber flag on a station logged by name with no FCC-verified callsign.
ESV tag	A blue member-ID tag shown when the station matches your roster.

NOTE Participation duration is rounded **up to the nearest quarter-hour** - so a station on the net for 6 minutes counts as 15 minutes. This is deliberate: it matches how volunteer time is documented for reimbursement and cost recovery.

7.6 The Traffic Log

Formal message traffic - radiograms and welfare messages - belongs in the **Traffic Log** tab, separate from station check-ins. On the ICS-309 this traffic is the main content, so log it here rather than burying it in remarks.

1. Click the **Traffic Log** tab.
2. Fill the **From callsign**, the **To callsign/address**, a message **type** (Radiogram, Health & Welfare, Official, Priority, or Emergency), and a short **Message summary**.
3. Click **Log Traffic**. The message is stamped and listed newest-first, and the sidebar Traffic count goes up.

7.7 Roster Chips - One-Click Check-Ins

For a recurring net with the same regulars, the **Roster Chips** tab turns each member into a clickable button. Click the dashed **import roster** area (or drag a file onto it) and pick a **CSV or JSON** file of your members. Each becomes a chip showing callsign and name. Clicking a chip drops that station into the check-in form and runs the FCC lookup, so you only press Enter to log it. A CSV needs a header row with a **callsign** (or **call**) column and a **name** column, or separate **first_name** and **last_name** columns.

7.8 Drill Mode

When you are training rather than running a real event, tick the **Drill Mode** checkbox in the net header. A bold **DRILL / EXERCISE - NOT ACTUAL EMERGENCY** banner appears across the top, and any ICS-309 you export from that net is stamped with a large diagonal **DRILL EXERCISE** watermark. This keeps practice logs from ever being mistaken for real incident records. Un-tick the box to return the net to normal.

7.9 Closing a Net and Exporting the ICS-309

1. When the net is over, click the red **Close Net** button in the header and confirm. Every station still checked in is **automatically checked out** at the net-close time, so no duration is left open-ended.
2. The header now shows the final **Opened**, **Closed**, and total **Duration**.
3. Click the **ICS-309** button on the top bar. FieldCommand builds the finished Communications Log and both downloads it as a file and opens it in a print window.
4. Review or print it, then attach it to the Incident Action Plan.

The exported ICS-309 carries a summary block (net type, frequency and mode, open and close times, total duration, total check-ins, and total messages), the full **Message Traffic Log**, and the full **Station Check-In Log** with each operator's rounded-up participation duration and member ID. A closed net is not deleted - it moves into the **Closed nets** drop-down in the tabs bar, so you can reopen and re-export it any time.

NOTE The **Dead Man's Switch** (Chapter 12) watches net activity and sounds an alert if no check-in is logged within a set time window. It is meant for safety monitoring during Search and Rescue (SAR) and field operations, where radio silence can mean trouble.

7.10 Observer Link and Backup

Two more top-bar buttons help you share and protect a net. **Observer Link** copies a read-only web address for the selected net; hand it to a section chief or served agency and they watch the same log update on their own device - the observer page refreshes on its own and cannot change anything. **Backup JSON** downloads the entire net - every check-in, message, and roster chip - as a single data file you can archive or move to another server. Net data also lives on the server itself, so a browser refresh never loses a log.

7.11 Troubleshooting

SYMPTOM	WHAT TO DO
The callsign name does not autofill	Confirm the **Auto FCC lookup** checkbox is ticked, and type at least three characters. Club and brand-new callsigns may not be in the FCC database - just type the name by hand.
LOG ENTRY says "Select a net first"	No net is selected. Click a net badge in the NET TABS bar, or click **+ New Net** to create one, then log again.
The net badges say "Cannot reach API server"	The core API service is not answering. Check that you are on the 192.168.50.1 server and that the fcc-lookup service is running (see the Health dashboard).
Pick shows "No channels found"	No ICS-205 or Channel Library is set up for this incident. Type the frequency and mode by hand, or build the communications plan first.
A station's duration keeps climbing after it left	It was never checked out. Click **Check Out** on that row, or close the net - closing checks out everyone still on.

SYMPTOM	WHAT TO DO
The ICS-309 has a DRILL watermark on a real event	The net has Drill Mode on. Un-tick **Drill Mode** in the header, then export the ICS-309 again.

8 Public Safety Net Logger http://192.168.50.1/starcom.html

The Public Safety Net Logger is the tool for running a radio net where operators check in by radio ID or unit number instead of by amateur callsign. It is built for trunked radio, Project 25 (P25), Digital Mobile Radio (DMR), and conventional public safety systems, and for interoperability exercises and served-agency support where some participants may not hold an amateur license. It keeps a complete, time-stamped log of every unit and every message, calculates on-net time for cost recovery, and exports the whole net as a printable Incident Command System ICS-309 Communications Log. Everything you log is saved on the server the moment you enter it.

TIP

In one sentence: create a net, log each unit as it checks in, log the message traffic, then close the net and export the ICS-309.

8.1 How It Differs From the Amateur Net Logger

The Public Safety Net Logger and the Amateur Net Logger (Chapter 7) look and work almost the same. The difference is the primary identifier and whether a license check applies.

FEATURE	AMATEUR NET LOGGER	PUBLIC SAFETY NET LOGGER
Primary ID	Amateur radio callsign, auto-filled from the local database	Radio ID or unit number
License check	Federal Communications Commission (FCC) lookup confirms an active license	None required
Roster lookup	By callsign to name and member ID	By radio ID to name and member ID
ICS-309 station	Callsign in the station column	Radio ID in the station column
Typical use	Amateur emergency communications nets	Public safety agency nets, interoperability exercises

NOTE

A participant who holds both an amateur license and a public safety radio ID should check into the public safety net by radio ID only. If the same person also works a concurrent amateur net on the incident, they check into the Amateur Net Logger separately by callsign. This keeps both logs accurate.

8.2 The Screen at a Glance

The page has three working areas. Learn where each lives and the rest is easy.

AREA	WHAT IS THERE
Top bar	The **+ New Net** button, a row of badges for the active nets (click a badge to switch nets), and on the right the **Observer Link**, **ICS-309**, and **Backup** buttons.
Center column	The net header (name, type, open/close times, **Drill Mode** checkbox, and **Close Net** button), the entry form for logging units, and the two tabs **Units Logged** and **Traffic**.
Right sidebar	The **Net Summary** panel (net, type, unit count, traffic count, and times) and an **Active Nets** list.

8.3 Creating a Net

1. Click **+ New Net** at the top left. The **Create Starcom Net** box opens.
2. Fill in the fields (only the name is required), described in the table below.
3. Click **Create Net**. The net opens, its badge appears in the top bar, and the entry form becomes ready. Click **Cancel** to close the box without creating a net.

FIELD	WHAT IT MEANS
Net Name	Required. A short name for this net, such as "County Interop 1".
Net Type	The kind of net, chosen from a list: Dispatch Net, Tactical Net, Command Net, Mutual Aid Net, Search and Rescue Net, EOC Coordination Net, or Medical Net. (EOC is the Emergency Operations Center.)
Talkgroup	The radio technology in use: Digital Talkgroup, Conventional Digital, Conventional Analog, P25, or DMR.
Channel / Frequency	The channel name or frequency the net is running on, for example "TAC-4" or "155.340".
Dispatch Center	The dispatch center or agency the net answers to. It shows in the net header once set.

8.4 Logging a Unit Check-In

With a net open, use the entry form to log each unit. Only the Radio ID is required; fill in as much of the rest as the traffic gives you.

1. In **Radio ID / Unit #**, type the unit number or radio ID exactly as heard.
2. Add the **Unit Name / Callsign** if you have it.
3. Choose a **Status** and a **Precedence** (see 8.5).
4. Optionally add the **Talkgroup**, a **Channel / Frequency** (or click **Pick** to choose from the Channel Library, see 8.6), a **Location**, and **Remarks**.
5. Optionally set an **ICS Position for this incident** and an **Incident ID**; entering an Incident ID reveals an **ICS-211** link to the check-in list.
6. Click **LOG ENTRY**. The unit appears at the top of the **Units Logged** tab and is saved at once. Click **Clear** to empty the form without logging.

FIELD	WHAT IT MEANS
Dispatch Center	Sets or updates the dispatch center shown in the net header.
Radio ID / Unit #	Required. The primary identifier for the unit.
Unit Name / Callsign	A friendlier name, or the operator callsign if licensed.
Status	What the unit is doing right now (see 8.5).
Talkgroup	The talkgroup or channel type the unit is on.
Channel / Frequency	The specific channel; the **Pick** button pulls from the incident ICS-205 and Channel Library.
Precedence	How urgent the traffic is (see 8.5). Sets the color stripe on the log entry.
Location	Where the unit is; shown with a pin marker on the entry.
Remarks	Any free-text note about this unit or message.

FIELD	WHAT IT MEANS
ICS Position	Optional. The Incident Command System role this unit fills, chosen from grouped Command, Operations, Planning, Logistics, and Finance lists.
Incident ID	Optional. Ties the entry to a specific incident and shows the ICS-211 Check-In List link.

8.5 Status, Precedence, and Talkgroup Options

The drop-down choices on the entry form, in plain language.

8.5.1 Status

OPTION	MEANING
Check-In	The unit is joining the net.
Traffic / Priority Traffic	The unit is passing a message, routine or urgent.
Emergency	Life-safety traffic that takes priority over everything else.
Dispatch	A dispatch assignment to a unit.
Check-Out	The unit is leaving the net.
En Route / On Scene / Available / Out of Service	Standard unit-status reports used to track where a unit is in its assignment.

8.5.2 Precedence

OPTION	MEANING
Routine	Normal traffic. No color emphasis.
Welfare	Health-and-welfare traffic. Green stripe.
Priority	Important, time-sensitive traffic. Amber stripe.
Emergency	Life-safety traffic. Red stripe, most prominent.

8.5.3 Talkgroup

The talkgroup list on the entry form offers Digital Talkgroup, Conventional Digital, Conventional Analog, P25 Direct, DMR Direct, and GMRS (General Mobile Radio Service). Pick the one that matches how the unit is transmitting.

8.6 Picking a Channel From the Library

Rather than typing a frequency, click the **Pick** button next to the **Channel / Frequency** field. FieldCommand reads the active incident communications plan and opens a **Select Channel** list.

1. Click **Pick**. A search box and a list of channels appear.
2. Each row is tagged by source: **205** (from the incident ICS-205 plan), **RPT** (from the repeater database), or **LIB** (the Channel Library).
3. Type in the search box to filter, then click a channel. Its name and frequency fill the field. Click **Cancel** to close without choosing.

NOTE	If you see "No channels found," the incident has no ICS-205 plan yet. Type the channel by hand, or build the communications plan first (Chapter 6 and the ICS forms).

8.7 Logging Message Traffic

The **Traffic** tab keeps the formal message log — the heart of the ICS-309. Click the **Traffic** tab, then fill the row at the top:

FIELD	WHAT IT MEANS
From Unit / Radio ID	Who originated the message.
To Unit / Address	Who the message is for.
Type	The handling: Dispatch, Tactical, Resource Request, Status, Priority, or Emergency.
Message summary	A short plain-language summary of the message.

Click **Log** to record it. The message appears at the top of the traffic list with its time stamp, and the traffic counter updates in the header and sidebar.

8.8 Checking Units Out and On-Net Time

Each logged unit shows its check-in time, its status badge, any member ID pulled from the roster, and a running **Duration**. When a unit leaves, click its **Check Out** button; the button changes to show the unit is out and the duration freezes. Any unit still on the net when you close it is checked out automatically.

NOTE	Durations are rounded UP to the nearest quarter hour (a unit on for one minute counts as 0:15). This matches how reimbursable personnel time is usually claimed. The member ID shown on an entry comes from the roster when the radio ID or name matches a member.

8.9 Drill Mode

Before an exercise, tick the **Drill Mode** checkbox in the net header. A bold banner reading DRILL / EXERCISE - NOT ACTUAL EMERGENCY appears across the top, and the exported ICS-309 is stamped DRILL / EXERCISE. This prevents an exercise log from ever being mistaken for a real event. Un-tick it to return to normal.

8.10 Closing, Exporting, and Sharing

Three buttons at the top right, plus **Close Net** in the header, finish the net.

CONTROL	WHAT IT DOES
Close Net	Ends the net, stamps the close time, and automatically checks out every remaining unit. You are asked to confirm first.
ICS-309	Builds the printable ICS-309 Communications Log — the message traffic log and the unit check-in log, with net times and totals — and opens it to print or save as a file.
Observer Link	Copies a read-only, auto-refreshing web link to this net that you can share with any device on the network (the Observer view, Chapter 9).
Backup	Downloads the entire net as a JSON data file for your own records.

TIP

Closing a net does not delete it. All net data stays on the server and is part of the incident record, so you can reopen the ICS-309 or the backup at any time.

8.11 Troubleshooting

SYMPTOM	WHAT TO DO
The entry form is missing	No net is selected. Click a net badge in the top bar, or click + New Net to create one. The form appears once a net is open.
"Select a net first" when I click Log Entry	Same cause. Pick or create a net first, then log the unit.
"Enter a Radio ID" warning	The Radio ID / Unit # box is empty. It is the one required field on the entry form; type the unit identifier and log again.
The Pick button says no channels found	The incident has no ICS-205 communications plan for period 1. Type the channel by hand, or build the ICS-205 first.
A unit shows no member ID	The radio ID or name did not match anyone on the roster. That is normal for outside agencies; the log is still complete. Add the person to the roster if they belong to your group.
My exercise log looks like a real event	Turn on Drill Mode before the exercise. It stamps the banner and the ICS-309 so the log cannot be mistaken for an actual emergency.

9 Observer Mode — Read-Only Net View http://192.168.50.1/observer.html

Observer Mode is a read-only, self-refreshing window into any active net. It shows the same check-in list and traffic log the Net Control operator sees, but it has no buttons that change anything - nobody watching an observer screen can add a check-in, close the net, or edit a single field. That is exactly why it exists. A section chief, a served-agency liaison, an Emergency Operations Center (EOC) duty officer, or a served-agency representative can keep the net picture up on their own device without ever touching, or accidentally disturbing, the live Net Control workstation. Any device already on the EMCOMM-NET Wi-Fi opens the page in a browser - no app to install, no login, any operating system.

TIP

In one sentence: hand someone the observer link and they can watch the net live, read-only, refreshing itself every fifteen seconds.

9.1 Opening the Observer Page

There are two ways in. Most observers arrive by a shared link that already points at one specific net; anyone who opens the page with no net chosen gets a picker of every net on the server.

1. The everyday way: the Net Control operator clicks **Observer Link** on the active net, which copies a web address (Uniform Resource Locator, or URL) ending in **?net=** and the net number. They share it - over the net by voice, by Winlink or JS8Call message, or by reading it out. You open that link and the page goes straight to that net.
2. The pick-it-yourself way: open **http://192.168.50.1/observer.html** with no net on the end of the address. The page shows **Select a Net to Observe** and lists every net (see 9.3). Click the one you want.
3. Bookmark the link once you are on the net you want - it reopens the same net every time.

9.2 The Header Bar

A colored strip runs across the very top of every observer screen. It never scrolls away, so the read-only reminder and the clock are always in view.

WHAT YOU SEE	WHAT IT MEANS
An eye icon and the words **OBSERVER MODE - READ ONLY**	The permanent reminder that this screen cannot change anything. It is on every observer page.
A second line under it	The name of the net you are watching (with "(Starcom)" added when it is a public-service net). It reads "Loading..." for a moment while the page fetches the net.
A large green clock on the right	The current time in Coordinated Universal Time (UTC), the 24-hour standard used on every ICS form. It ticks once a second.
Auto-refresh in N s	A countdown to the next automatic reload. It starts at 15 and counts down.
A thin bar directly below the strip	A progress bar that drains from full to empty over the fifteen seconds, a visual twin of the countdown number.

9.3 The Net Picker

When you open the page without choosing a net, the picker lists every net on the server so you can select one. Each row is one net.

ON EACH ROW	WHAT IT TELLS YOU
Net name (bold)	The name the operator gave the net, such as "Thursday Evening Net".
Type and entry count	The net type and how many stations have been logged - for example, "ARES Net - 12 entries".
ACTIVE or **CLOSED** badge	Green ACTIVE means the net is still open; gray CLOSED means it has ended (you can still read a closed net).
STARCOM badge	Blue badge marking a public-service net (shown only when it applies).
DRILL badge	Amber badge marking an exercise, not a real event (shown only when it applies).

Click a row to open that net. The address bar updates to the direct link, which you can then bookmark or share.

9.4 The Net Header

Once a net is open, a panel at the top of the view identifies it and shows its state.

ELEMENT	WHAT IT SHOWS
Net name (large)	The name of the net being observed.
Meta line	The net type and its identifying number - for example, "ARES Net - ID: 4" - with "STARCOM" added on a public-service net.
Status badge	Green **ACTIVE** while the net is open, gray **CLOSED** after it ends.
Updated HH:MM UTC	The time of the most recent refresh, so you can confirm the view is current.

WARN If the net is a drill, a bold banner reading "DRILL / EXERCISE - NOT ACTUAL EMERGENCY" appears across the top of the view. It is there so no one mistakes exercise traffic for a real event.

9.5 The Stations Logged Column

The left half of the view is the check-in list, headed **STATIONS LOGGED** with a live count in parentheses. The newest check-in sits at the top. Each station is one card:

ON EACH CARD	WHAT IT MEANS
Callsign (bold)	The station callsign as Net Control logged it.
Name	The operator name, when it was captured (often auto-filled from the offline FCC database).
Status badge	The station status the operator recorded - it is color-coded by message precedence (see 9.6).
Time in UTC	The check-in time as hours:minutes:seconds, 24-hour UTC.
Location	The station location, when one was entered (shown after the time).
Remarks	Any note the operator added for that station, shown under the line.

If no one has checked in yet, the column reads **No stations logged**.

9.6 Precedence Colors

Each station card carries a colored left edge that reflects the precedence (urgency) of its traffic, so an observer can spot an emergency at a glance without reading every line.

PRECEDENCE	LEFT-EDGE COLOR	MEANING
EMERGENCY	Red	Immediate danger to life or property - the highest urgency.
PRIORITY	Amber	Important traffic that needs prompt handling but is not life-threatening.
WELFARE	Green	Health-and-welfare traffic - status of people, reassurance messages.
ROUTINE	Plain gray	Normal traffic with no special urgency (the default).

9.7 The Traffic Log Column

The right half of the view is headed **TRAFFIC LOG** with its own live count. It lists formal message traffic passed on the net, newest first. Each entry is one card:

ON EACH CARD	WHAT IT MEANS
From and to callsigns	The sending station and the receiving station, shown as "FROM to TO".
Time in UTC	When the traffic was logged, hours:minutes:seconds UTC.
Type	The kind of traffic the operator recorded, shown after the time.
Note	A short description of the message, when one was entered.

When nothing has been passed, the column reads **No traffic**.

9.8 Staying Current - Auto-Refresh

The page reloads its data by itself every fifteen seconds. You do not touch anything - the countdown in the header reaches zero, the page pulls the latest check-ins and traffic, the "Updated" time changes, and the countdown starts over. That is why an observer screen left up on a wall monitor stays honest without an operator babysitting it. The clock in the header ticks every second on its own, separate from the fifteen-second data refresh.

NOTE Observer Mode does not repeat the net frequency or mode, and it has no elapsed-net timer - it shows the net name, type, and identifying number instead. For the full net controls and timers, use the Net Control page (Chapter 7).

9.9 What Observers Can and Cannot Do

Observer Mode is read-only by design. There is no way, from this page, to change the net - the buttons simply are not there.

OBSERVERS CAN	OBSERVERS CANNOT
See every check-in live, newest first	Add, edit, or remove a check-in
See the net status and last-updated time	Open or close the net
Read the full traffic log	Add or change a traffic entry
Watch it refresh itself every 15 seconds	Change any net setting

OBSERVERS CAN	OBSERVERS CANNOT
Bookmark and reopen the observer link	Reach the Net Control workstation view
Switch nets from the picker	Delete or export net data

9.10 Sharing the Observer Link

The link is generated on the Net Control side, so a busy observer never has to build a web address by hand.

1. On the Net Control page (Chapter 7), select the net you want people to watch.
2. Click **Observer Link**. The web address for that net is copied to the clipboard.
3. Paste and share it - in a group chat, a Winlink or JS8Call message, an email on the LAN, or read it aloud over the net.
4. Each observer opens the link on any device already joined to the EMCOMM-NET Wi-Fi. It lands directly on that net, read-only, and begins refreshing.

9.11 Troubleshooting

SYMPTOM	WHAT TO DO
The page says "Cannot reach API server at 192.168.50.1:5050"	The core service is not answering. Confirm the device is on EMCOMM-NET Wi-Fi, then check that the fcc-lookup service is running on the Pi (Chapter on System Health). Reload once it is back.
A pop-up says "Net not found"	The link points at a net number that no longer exists on this server. Open http://192.168.50.1/observer.html with no net on the end and pick the net from the list.
The picker shows "No nets found"	No net has been created yet. Open one on the Net Control page first, then reopen the observer link.
New check-ins are not appearing	The page refreshes every fifteen seconds - wait for the countdown, or reload the page. If it still lags, confirm Net Control is actually logging to that same net number.
The screen is stuck on an error after a brief hiccup	A momentary network drop can leave the view showing an error. Reload the page - it recovers and resumes refreshing.
I see edit buttons and want to change something	You will not - Observer Mode has none by design. To make changes, use the Net Control workstation (Chapter 7).

10 Barcode & Quick Response (QR) code Scan Check-In

http://192.168.50.1/scan_checkin.html

The Scan Check-In page turns any phone, tablet, or laptop into a fast personnel check-in station. Point the camera at a member's Quick Response (QR) code or barcode, or type an identifier by hand, and the person is looked up in the roster, their details fill in, and one tap records the check-in against the current incident. It uses the web browser's own built-in Barcode Detector -- no app to install, no library to download, no internet -- so it works fully offline on EMCOMM-NET. It reads QR codes plus Code 128, Code 39, European Article Number (EAN) 13 and 8, Data Matrix, Portable Document Format 417 (PDF417), and Aztec barcodes.

TIP In one sentence: scan the code or type the identifier, glance at the auto-filled details, and tap Check In -- the person is on the incident roster in seconds.

10.1 Four Ways to Read a Person In

The page offers four input methods and picks the ones your device can actually do. A staffed check-in station usually uses the live camera or a USB scanner; a walk-up on a plain phone uses Take Photo or manual entry.

METHOD	HOW IT WORKS	BEST FOR
Live camera	The rear camera runs as a viewfinder and detects a code automatically at four frames per second. The same code is ignored for three seconds so one card is not read twice. Needs a secure Hypertext Transfer Protocol Secure (HTTPS) link.	A staffed, high-volume check-in station
Take Photo of QR	Snaps one still photo with the device camera and reads the code out of that picture. Works over a plain (non-HTTPS) link where the live viewfinder is blocked.	A phone or tablet on a plain http connection
Manual ID entry	Type a member identifier, callsign, or radio identifier and press Enter or click Look Up. Same roster lookup as a scan.	Walk-ins with no code; when no camera is available
USB or Bluetooth scanner	A hardware scanner acts like a keyboard: it "types" the code and presses Enter into the manual box, which stays focused so the lookup fires hands-free.	A fixed desk station reading printed badges

10.2 The Screen at a Glance

From top to bottom the page is laid out in the order you use it:

AREA	WHAT IT IS
Camera viewfinder	The black scan window with a white corner frame and a moving blue scan line. A status line at the bottom tells you what the scanner is doing.
Camera controls	The **Start Camera** and **Stop Camera** buttons and a camera-picker drop-down (front or rear) for the live scan.
Take Photo of QR	A button that opens the device camera to snap one still, with a short note explaining it reads the person's QR card.

AREA	WHAT IT IS
Manual ID entry	A single text box ("KE4CON or ESV-001 or 412") with a **Look Up** button.
Check-in form	The details that appear once a code resolves -- name, identifier, agency, position, and resource type -- with the green **Check In** button.
Recent Check-Ins This Session	A running list of the last check-ins made on this device since the page opened.
Incident badge	The strip at the very bottom naming the incident and operational period these check-ins are being filed under.

10.3 Choosing the Incident First

Every check-in is filed against one incident and one operational period. The badge at the bottom of the page shows which. When it reads **Incident: (name) - Period (number)** you are ready. If it reads **No incident selected**, the check-ins have nowhere to land -- open the page from the incident's ICS-211 Share Uniform Resource Locator (URL) link, or select an incident on the dashboard first, so the page opens already pointed at the right incident.

10.4 Checking Someone In with the Live Camera

1. Open **Scan Check-In** from the dashboard. Confirm the incident badge at the bottom names the right incident.
2. Click **Start Camera**. If the device has more than one camera, choose the rear-facing one from the camera drop-down first.
3. Hold the member's QR code or barcode inside the white corner frame. The status line reads "Scanning..." and then "Scanned: (code)" when it catches one. On phones that support it, the device gives a short buzz.
4. If the person is on the roster, their details fill in with green borders and a green banner names them and marks them a roster member or a visitor. Glance over the fields and correct anything that is wrong.
5. If the person is not on the roster, a red "not in roster" banner appears and their scanned code drops into the identifier field. Type the name and agency by hand.
6. Click **Check In**. A full-screen "CHECK-IN COMPLETE" confirmation shows the name and the check-in time, and the phone buzzes.
7. Click **Scan Next Person** to clear the form for the next member. Use **Stop Camera** when the line is done.

10.5 Take Photo and Manual Entry (No Live Camera)

The live viewfinder needs a secure HTTPS link. On a plain http connection, or on a browser that blocks it, use one of these instead -- both reach the same lookup.

Take Photo of QR -- click the button, snap one clear photo that fills the frame with the code, and the page reads it from the still image. If it reads "No QR code found", hold steady, get closer, and retake.

Manual ID entry -- type the person's callsign, member identifier, or radio identifier in the box and press Enter or click **Look Up**. A USB or Bluetooth scanner feeds this same box automatically.

10.6 How the Roster Lookup Works

Whatever the code says -- scanned, photographed, or typed -- it is trimmed, forced to capital letters, and matched against the roster. The lookup tries these fields in order until it finds a match:

ORDER	FIELD MATCHED	EXAMPLE VALUE
1st	barcode_id (the badge or card code)	ESV-001
2nd	member_id (roster membership number)	ESV-001
3rd	callsign (Federal Communications Commission license)	KE4CON
4th	radio_id (tactical or unit number)	412

NOTE A found member auto-fills their name, identifier, and agency, and pre-selects a suggested Incident Command System (ICS) position when the roster carries one. Nothing is filed until you click Check In, so you always get a chance to correct the details first.

10.7 The Check-In Form

Once a code resolves, the form appears. Only the name is required; fill in as much of the rest as you know.

FIELD	WHAT IT MEANS
Full Name *	Required. The person's first and last name. Auto-filled for a roster member.
Callsign / ID	The person's amateur callsign or unit identifier, such as "KE4CON" or "Unit 412".
Agency	The organization the person represents -- their home group or a mutual-aid agency.
ICS Position	The role they are filling on this incident, chosen from the drop-down in 10.7.1. Left as "Select position..." if unknown.
Resource Type	What kind of resource they are, chosen from the drop-down in 10.7.2.

10.7.1 ICS Position Choices

POSITION	WHO IT IS
Incident Commander (IC)	The person in overall charge of the incident.
Safety Officer (SOFR)	Responsible for personnel safety.
Operations / Planning / Logistics / Finance-Admin Section Chief	The four section chiefs leading tactics, the plan, support, and cost.
Branch Director / Division-Group Supervisor (DIVS)	Mid-level supervisors under Operations.
Net Control / Amateur Radio Operator	The communications-unit roles.
Emergency Medical / Volunteer / Other	Medical personnel, general volunteers, and anything not listed above.

10.7.2 Resource Type Choices

TYPE	MEANING
Personnel	An individual person or staffed position.
Amateur Radio	A licensed amateur radio operator resource.
Engine	A fire engine or similar apparatus.
Crew	An organized team that works as one unit.

TYPE	MEANING
Vehicle	Any other vehicle -- truck, bus, or boat.
Equipment	Gear and supplies rather than people.
Other	Anything that does not fit the categories above.

10.8 Confirmation and Session History

When you click **Check In**, the record is sent to the incident and a green full-screen confirmation shows the person's name and the time they were logged. The check-in is now on the incident's ICS-211 check-in list and cannot be lost -- incident data is saved on the server and backed up. Each check-in also drops onto the **Recent Check-Ins This Session** panel, which keeps the last ten made on this device so a single operator can see who they have just processed. That panel is a convenience list for this session only; the durable record lives on the server. Click **Scan Next Person** to clear the form and carry on.

10.9 Which Devices Can Scan

The scanner relies on two browser features: the Barcode Detector (to read a code) and a secure context (to run the live camera). What a device can do depends on which it has.

DEVICE / BROWSER	WHAT WORKS
Chrome or Edge over HTTPS (Android or desktop)	Everything -- live camera, Take Photo, manual entry, and USB or Bluetooth scanners.
Chrome or Edge over plain http	Live camera is disabled; Take Photo, manual entry, and hardware scanners all work.
Apple iOS Safari, or Firefox	No code reading at all -- the scan buttons are disabled. Manual entry and a USB or Bluetooth scanner still work.

NOTE When the browser cannot read codes, the page says so on the status line and steers you to manual entry, so a check-in station is never fully blocked -- someone can always type the identifier.

10.10 Troubleshooting

SYMPTOM	WHAT TO DO
Start Camera is grayed out	You are on a plain http link or an unsupported browser. Use **Take Photo of QR** or manual entry, or open the page over the secure HTTPS link.
The status line says the browser cannot read QR codes	It is likely an iPhone Safari or Firefox. Type the member identifier, callsign, or radio identifier in the manual box, or use a USB or Bluetooth scanner.
"No QR code found in that photo"	Hold the phone steady, fill the frame with the code, and retake. If it still fails, type the identifier in the manual box instead.
A member scans as "not in roster"	The code did not match any barcode, member, callsign, or radio identifier. Fill in the name and agency by hand and check them in; add them to the roster later.
The bottom badge says "No incident selected"	The page is not pointed at an incident. Open it from the incident's ICS-211 Share URL, or select an incident on the dashboard, then reopen Scan Check-In.

SYMPTOM	WHAT TO DO
"Check-in failed" or a server error appears	The device briefly lost the EMCOMM-NET link. Confirm the Wi-Fi connection and try Check In again; the form keeps your entries.

11 Federal Communications Commission (FCC) Callsign Lookup http://192.168.50.1/callsign.html

The Callsign Lookup tool puts the entire national Federal Communications Commission (FCC) amateur radio license database in your hands with no internet connection. Over 800,000 active licensees are stored locally in a SQLite database on the Pi, so you can confirm a station name, verify a license class and status, or find a licensee by name or location in a fraction of a second - even when every outside network is down. The same database quietly powers the Net Control Logger, the Scan Check-In tool, and the Incident Command System (ICS) message forms, filling in names automatically as you work. This chapter covers the standalone lookup page and how the shared database is used everywhere else.

TIP In one sentence: type a callsign and press Enter to see the licensee, or use Advanced Search to find people by name, city, state, class, or grid square - all offline.

11.1 Reading the Screen

You reach the tool from the top navigation bar (**Callsign Lookup**) or from the dashboard. The page has four parts, stacked top to bottom:

AREA	WHAT IS THERE
Search hero (top)	The **CALLSIGN LOOKUP** banner, the line "FCC Amateur Radio License Database - Offline SQLite - Updated weekly", a large callsign entry box, and the **Look Up** button.
Result card	Appears under the search box after a lookup. Shows the licensee in large type with a color-coded license-class tag, an Active or Expired status dot, and a grid of details.
Advanced Search	A panel headed **Advanced Search** for finding licensees by name, city, state, class, or grid square when you do not know the callsign. Results appear as a table below it.
Recent Lookups	A row of buttons, one per callsign you looked up recently, so you can re-check a station with one click.

11.2 Looking Up a Single Callsign

1. Click the callsign box (it shows the example **W8XYZ** until you type). Type a callsign - it is forced to capital letters automatically, so case does not matter.
2. Press **Enter** or click **Look Up**. Results are not live-as-you-type; nothing happens until you submit.
3. The result card scrolls into view. A match shows the callsign in large amber letters with the licensee details described in 11.3.
4. If there is no match, the card turns red and reads "Callsign ... not found in FCC database" with a suggestion to search by name instead or to confirm the database is up to date.

NOTE You can also arrive with a callsign already loaded: another page can link here with ?call=W8XYZ on the end of the address, and the lookup runs on its own the moment the page opens.

11.3 Understanding the Result Card

A found record shows the callsign, a colored **license-class** tag, a status dot (a green dot with **Active**, or a red dot with the status word when the license is not active), and a grid of detail fields. Each field is labeled; a dash means the database has no value for it.

FIELD	WHAT IT MEANS
Name	The licensee - a person's name, or an entity/club name.
City, State	The mailing city and two-letter state from the license.
Zip Code	The mailing ZIP code on file.
Country	The country of the license; almost always US.
Grant Date	The date the current license term was granted by the FCC.
Expiration Date	When the license expires. Shown in red if the date has already passed.
FRN	The FCC Registration Number - the identifier that ties all of a person's FCC licenses together.
License ID	The FCC internal license record number, used to build the online lookup link.
Grid Square	The Maidenhead grid locator (for example EN80), shown only when the record has one.

11.3.1 License-Class Colors

The class tag is color-coded so you can read the privilege level at a glance:

TAG	CLASS	COLOR
T	Technician	Green
G	General	Amber
E	Amateur Extra	Red
N	Novice	Blue
P	Advanced	Purple

11.4 The Result Buttons

Three buttons sit at the bottom of a found result card:

BUTTON	WHAT IT DOES
FCC ULS	Opens the official FCC Universal Licensing System page for this license in a new browser tab. This is the one action that needs the internet - it reaches out to the FCC website, so it works only when a Wide Area Network (WAN) is present.
+ Add to Roster	Jumps to the Roster page with this callsign already filled in, so you can create a roster entry from the FCC record without retyping it.
Copy	Copies the callsign to the clipboard so you can paste it into a log, a form, or a message.

11.5 Advanced Search - Finding a Licensee by Name or Place

When you do not have the callsign - you know the name of a person who checked in, or you want everyone with a license in a town - use the **Advanced Search** panel. Fill in any one or more fields and the tool finds every matching record (up to 100 at a time).

FIELD	WHAT IT MEANS
Last Name / Entity Name	A person's last name, or a club or entity name.
First Name	A person's first name.
State	A drop-down of all states; leave on **All States** to search everywhere.
License Class	A drop-down: **All Classes**, Technician, General, Amateur Extra, Novice, or Advanced.
City	The mailing city.
Grid Square	A Maidenhead grid locator (for example EN80); forced to capital letters as you type.

1. Type into one or more of the fields above. The more fields you fill, the narrower the result.
2. Click **Search**. If you leave every field blank, the tool reminds you to enter at least one criterion.
3. Matches appear in a results table below the panel, with a count such as "12 results found" shown at the right of the button row.
4. Click **Clear** to empty all the search fields and wipe the results table so you can start a fresh search.

11.6 The Search Results Table

Advanced Search lists its matches in a table. Each row is one licensee:

COLUMN	WHAT IT SHOWS
Callsign	The callsign, shown in amber. Click it to run a full single-callsign lookup and see the complete result card for that station.
Name	The licensee name, or the first and last name joined together.
Class	A short class label - Tech, General, Extra, Novice, or Advanced.
City	The mailing city.
State	The two-letter state.
Status	Green **Active** for a current license, red **Expired** otherwise.
Expires	The license expiration date.

TIP The list shows at most 100 records at once. If your search returns more than that, add another field - a state, a city, or a class - to narrow it down to the person or group you actually want.

11.7 Recent Lookups

Every single-callsign lookup that finds a match is remembered on this device and shown as a button in the **Recent Lookups** row at the bottom of the page (the most recent up to twenty). Click any of those buttons to look the station up again instantly. The list is stored in this browser only, so a different tablet or laptop keeps its own recent list, and clearing the browser's stored data empties it.

11.8 Automatic Lookup in Other Tools

You rarely need to open this page by hand, because the same offline FCC database is queried automatically wherever a callsign is entered:

WHERE	HOW IT WORKS
Net Control Logger	When you enter a callsign in the check-in field, the name and license class fill in automatically. A red border warns you when the callsign is not found or the license has expired.
Scan Check-In	After a Quick Response (QR) code scan or a manual entry, if the value is a callsign the FCC record fills the name field (falling back to the roster if the callsign is not in the database).
ICS-213 General Message	The "From Callsign" field fills the operator name from the FCC database, so the message header is complete without extra typing.

11.9 Keeping the Database Current

The FCC releases license changes constantly, so the local copy is meant to be refreshed about once a week. Updating it does not require the Pi to be online during an incident - you download the FCC data once, when you do have internet, and import it into the Pi.

1. On a computer with internet, download the FCC Universal Licensing System (ULS) amateur database export from wireless.fcc.gov/uls.
2. Copy the downloaded file to the Pi and run the database import script as described in the Installation Guide.
3. The tool will then serve the newer records immediately; no restart of the whole system is needed.

NOTE The banner line under the title reads "Updated weekly" as a reminder of the intended schedule. If a station you know is licensed does not appear, the local copy is most likely just older than that station's license - refresh it when you next have internet.

11.10 Troubleshooting

SYMPTOM	WHAT TO DO
A callsign I know is valid shows "not found"	The local database may be older than that license. Refresh the FCC data (11.9), or use Advanced Search by name to confirm the record exists under a different spelling.
"Could not reach local FCC database" / API error	The core Application Programming Interface (API) server on port 5050 is not responding. Confirm the fcc-lookup service is running and reload the page.
Nothing happens when I type	Lookup is not live-as-you-type. Press Enter or click Look Up to run the search.
The FCC ULS button does nothing / shows an error	That button opens the official FCC website and needs the internet. Without a WAN it cannot load; use the offline details on the result card instead.
Advanced Search says "Enter at least one search criterion"	Every field was left blank. Fill in at least one field - a name, city, state, class, or grid - then click Search.
My Recent Lookups list is empty on another device	Recent Lookups is stored per browser. A different tablet or laptop keeps its own list, and clearing browser data empties it - this is normal.

12 Dead Man's Switch http://192.168.50.1/deadmans.html

The Dead Man's Switch is a safety watchdog for your radio nets. It watches how long it has been since a net logged any activity, and if that quiet stretch grows past a time limit you set, it sounds an alarm and flashes a red banner across the screen. It is built for field operations - especially search and rescue - where radio silence beyond a set interval may be the first sign that a field team is in trouble. You do not have to remember to watch the clock; the page watches it for you and interrupts you the moment a net goes quiet too long.

TIP In one sentence: it is a countdown timer per net that raises an alarm when a net stops checking in - so a silent team never goes unnoticed.

12.1 How It Works and the Four States

Every open net has its own countdown running toward a threshold - a number of minutes of allowed silence. Each time that net records activity (a check-in logged in Net Control), the countdown resets to full. If activity stops, the countdown runs down, and the net moves through four states shown by the big ring at the top of the page and by the colored badge on each net card.

STATE	WHAT IT MEANS
DISARMED	Green. The net is not being watched right now. The ring reads **ALL CLEAR**. Nothing will alarm until you arm it.
ARMED	Amber. The net is live and being watched. The ring reads **MONITORING**. The countdown is running and activity keeps resetting it.
WARNING	Red. Silence has passed the warning mark (by default 75 percent of the threshold). This is your early heads-up that time is running out.
TRIGGERED	Red and pulsing. The full threshold elapsed with no activity. The alarm sounds and the top banner flashes **DEAD MAN'S SWITCH TRIGGERED**.

The top ring always shows the **worst** state across all your nets, so a single triggered net turns the whole page red even if others are calm.

12.2 Opening the Page

Open **Dead Man's Switch** from the top navigation bar, or go straight to http://192.168.50.1/deadmans.html. Best practice is to leave it open on a dedicated monitor or tablet at the Net Control position or the Emergency Operations Center (EOC) duty station, where the person watching the nets will see and hear it. Make sure that device has its speaker on and turned up - the alarm is audible.

WARN The alarm sound is produced by the browser on the device showing this page. If the screen is off, the tab is closed, or the volume is muted, no one hears it. Keep the page open and awake at a staffed position.

12.3 Setting the Default Configuration

The **DEFAULT CONFIGURATION** panel near the top sets the values used for new nets you arm. Change them to suit the operation, then click **Save Defaults**. These settings are remembered on this device.

SETTING	WHAT IT MEANS
Default Threshold (minutes)	How many minutes of silence are allowed before a net triggers. Default 30; allowed 1 to 480. Common field values are 10, 15, or 30 minutes.
Warning at (% of threshold)	How far into the countdown the amber-to-red WARNING appears, as a percentage of the threshold. Default 75; allowed 50 to 95. At a 30-minute threshold, 75 percent warns at 22.5 minutes of silence.
Poll Interval (seconds)	How often the page asks the server for fresh net activity. Default 15; allowed 5 to 60. Lower is more responsive but asks the server more often.
Alert Sound	The sound played when a net triggers: **Beep** (a softer tone), **Alarm** (a higher, more urgent tone), or **None (visual only)** - the banner still flashes but nothing sounds.

NOTE The countdown timers on the page tick every second on their own; the Poll Interval only controls how often the page refreshes the real activity data from the server, not how smoothly the numbers count down.

12.4 Where Monitored Nets Come From

The Dead Man's Switch does not create nets. It watches the nets you open in **Net Control** (Chapter 11). When you start a net there, it appears under **MONITORED NETS** on this page as a card. If no nets are open, the page shows "No active nets being monitored" and tells you to open a net in Net Control and arm the switch. Each net you have open gets its own card and its own independent countdown.

12.5 Reading a Net Card

Each net card shows that net's live status at a glance. The colored bar down the left edge and the badge in the corner match the four states in 12.1.

CARD ELEMENT	WHAT IT SHOWS
Net name and badge	The net's name, with a state badge (DISARMED, ARMED, WARNING, or TRIGGERED).
Big timer	The time remaining until this net triggers, counting down as minutes:seconds. It is colored to match the state.
"remaining of N min threshold"	A reminder of the total threshold this net is counting down from.
Progress bar	A bar that fills as silence grows - empty just after activity, full at trigger.
Last activity	The clock time (UTC) of the most recent logged activity on this net.
Elapsed	How many minutes have passed since that last activity.
Warning at	The minute mark and percentage at which this net moves to WARNING.
Triggered	Shown only after a trigger: the UTC time the alarm fired.

12.6 Arming, Resetting, and Disarming a Net

The buttons at the bottom of each card change with the net's state. You will only ever see the buttons that make sense for what the net is doing right now.

BUTTON	WHEN IT APPEARS / WHAT IT DOES
Arm	Shown when the net is DISARMED. Starts watching the net using the current default threshold. The net turns amber (ARMED) and the countdown begins.
Reset	Shown when the net is armed, warning, or triggered. Restarts the countdown from full - use it to clear a warning or a trigger once you have confirmed the team is fine, or after any manual check-in.
Disarm	Shown when the net is armed, warning, or triggered. Stops watching the net entirely and returns it to DISARMED. You are asked to confirm first.

The everyday routine for a net is short:

1. Open the net in Net Control so its card appears here.
2. On the card, click **Arm**. The card turns amber and the countdown starts.
3. As the net runs, each check-in logged in Net Control resets the countdown automatically - you do nothing.
4. If a net reaches WARNING or TRIGGERED, investigate. If the team is confirmed safe, click **Reset** to restart the countdown.
5. When the net closes, click **Disarm** and confirm.

12.7 When a Net Triggers

A trigger is meant to be impossible to miss. When any net hits its threshold: the red **DEAD MAN'S SWITCH TRIGGERED - NET INACTIVITY ALARM** banner appears and flashes at the top of the page, the alarm sound plays (unless Alert Sound is set to None), the top ring turns red and reads **TRIGGERED**, and the net's own card pulses red. Treat it as a real event until proven otherwise:

1. Look at which net card is red - the "Triggered" line shows the time it fired.
2. Attempt contact with the field team or net on that net immediately.
3. If you reach them and all is well, click **Reset** on that card to silence the alarm and restart the countdown.
4. If you cannot reach them, follow your group's overdue-team procedure. The alarm stays up until you Reset or Disarm the net.

> **TIP** Reset silences the alarm and gives the net a fresh full countdown; Disarm silences it and stops watching that net altogether. Reset when the net is continuing; disarm only when the net is truly closing.

12.8 The Activity Log

Below the net cards, the **ACTIVITY LOG** timeline records what happened on this page - each arm, reset, disarm, and configuration change, with a UTC time and the net it applied to. It is a quick running record of operator actions during the watch. Click **Clear** above the log to empty it. This log lives in the browser on this device for the current session; it is a convenience view, not the permanent incident record - the net check-ins themselves are saved by Net Control.

12.9 Practical Tips for the Watch

Match the threshold to the risk. A short SAR sweep in bad terrain may warrant a 10-minute threshold; a stable shelter net can run at 30 or more.

Keep the page on a staffed, awake screen. A blanked or muted device is a silent alarm.

Reset promptly after any manual check-in so a routine gap does not grow into a false trigger.

Disarm nets you have closed so the ring reflects only nets that are genuinely being watched.

12.10 Troubleshooting

SYMPTOM	WHAT TO DO
The page says "No active nets being monitored"	No net is open. Open a net in Net Control (Chapter 11); it will appear here as a card, then click Arm.
A net alarmed but the team is fine	A routine gap outran the threshold. Click Reset on that card to restart the countdown, and consider a longer threshold for this operation.
No sound when a net triggers	Check that Alert Sound is not set to None, the device volume is up, and the tab is open. Some browsers need a click on the page first before they will play sound.
The countdown does not reset on check-ins	Confirm check-ins are being logged in Net Control for that net, and that the page shows "Updated" (not "Server offline") on the MONITORED NETS line.
"Server offline" appears on the refresh line	The page cannot reach the server. Confirm you are on the EMCOMM-NET Wi-Fi and can open the dashboard at 192.168.50.1; the page retries on its own once reachable.
My default settings were not remembered	Click Save Defaults after changing them. Settings are stored per device, so a different tablet or a cleared browser starts from the defaults again.

13 Tactical Automatic Packet Reporting System (APRS) Map

http://192.168.50.1/tactical.html

The Tactical Map is your live picture of who and what is on the air. It is a full-screen Leaflet map that plots every Automatic Packet Reporting System (APRS) station it can hear, drawn from as many as three feeds at once: Direwolf decoding signals straight off the radio, the YAAC (Yet Another APRS Client) program, and the internet APRS-IS feed when a Wide Area Network (WAN) is available. It is the primary situational-awareness display for tracking field teams, vehicles, and mobile resources, and it doubles as a place to send and receive short APRS text messages and to drop your own labeled markers on the map.

TIP In one sentence: the Tactical Map merges every APRS source into a single live map of stations, color-coded by how recently each one was heard.

WARN Offline caveat: the map is fully offline only when the Leaflet map library and the base map tiles are served locally from the Pi. In the current build, Leaflet and the default base tiles load from the internet (a Content Delivery Network), so without a WAN the base map may not draw. Station markers and your overlays still plot on whatever base is available. Pre-download tiles with **sudo bash download_tiles.sh** for a true no-internet map.

13.1 Reading the Screen at a Glance

The screen has three parts: a thin top bar, the map itself, and a sidebar on the right. Along the top bar, left to right, you will see the home icon (returns to the dashboard), the **TACTICAL MAP** title, a live **Stations** count, a **Refresh** button that re-polls every source, a **KML** button that exports what is on the map, and a UTC clock. Down the left edge of the map is a column of layer toggle buttons. The right sidebar holds five tabs.

SIDEBAR TAB	WHAT IT SHOWS
Stations	The searchable, filterable list of every station on the map, newest first.
Msgs	Received APRS text messages and a form to send one.
Markers	Your own hand-placed map markers (EOC, shelter, command post, and so on).
Sources	The connection status and station counts for Direwolf, YAAC, and APRS-IS.
Settings (gear icon)	Your station location, source addresses, and all display options.

13.2 The Three Data Sources

The map can listen to three feeds at once and merges them into one deduplicated list, so a station heard by more than one source appears only once. Open the **Sources** tab to see each feed, its address, and how many stations it is contributing right now. Each source has its own color, used for the marker border and the list badges.

SOURCE	COLOR	WHERE IT COMES FROM	NEEDS WAN?
Direwolf (RF)	Green	Live APRS off the radio. Direwolf decodes it; APRS Command serves it to the map. Always available with no internet.	No

SOURCE	COLOR	WHERE IT COMES FROM	NEEDS WAN?
YAAC	Blue	The YAAC program running on the network, offering its own station feed over a local port.	No
APRS-IS	Purple	The worldwide internet APRS feed by way of the aprs.fi service.	Yes

At the bottom of the Sources tab a **Merged Dataset** box tallies the total unique stations and how many came from each source alone or from more than one. Each source panel has a **Poll** button to fetch it on demand; the YAAC panel also has **Open UI** and a **Setup** guide. To use APRS-IS, paste an aprs.fi Application Programming Interface (API) key into the key box; leave it blank to keep that feed off.

13.3 Map Layer Toggles

The column of buttons down the left side of the map turns individual layers on and off. A lit button means the layer is showing. This lets you declutter the map, for example hiding the busy internet feed to see only what you are hearing on the radio.

BUTTON	WHAT IT TOGGLES
Direwolf	The green RF stations decoded off the radio.
YAAC	The blue stations from the YAAC feed.
APRS-IS	The purple stations from the internet feed.
Overlays	Your own hand-placed markers from the Markers tab.
My Station	The marker for your own location.
Range Ring	A dashed distance circle around your station (radius set in Settings).
Track	The movement trail of a station you chose to follow.
Repeaters	The repeater database overlay, loaded from the FieldCommand server.
CalTopo / SARTopo	An imported CalTopo / SARTopo GeoJSON overlay (see 13.9).

13.4 Station Markers -- Color, Symbol, and Age

Each station is drawn with its standard APRS symbol (a car, a house, a weather flag, and so on) or, if you prefer, a plain colored dot. Two things carry meaning: the fill color tells you how recently the station was heard, and the marker border tells you which source it came from.

FILL COLOR	MEANING
Green	Fresh -- heard within the age threshold (default 15 minutes).
Amber	Aging -- heard between the threshold and four times it (about 15 to 60 minutes).
Gray	Old -- not heard for more than an hour.
Blue	Your own station marker.
Magenta	A manual overlay marker you placed.

The border color repeats the source scheme: green for Direwolf, blue for YAAC, purple for APRS-IS, and white when the same station was seen by more than one source. Click any marker to open a popup listing its symbol, type, comment, speed, course, altitude, path, frequency (when reported), and when it was last heard. The popup also has a **Track** button and a **Message** button for that station.

13.5 The Stations List and Filters

The **Stations** tab lists every station on the map, sorted with the most recently heard at the top. Type in the search box to match a callsign or comment. Two drop-downs narrow the list further:

FILTER	CHOICES
Source	All sources, Direwolf, YAAC, or APRS-IS.
Type	All types, Mobile, Fixed, Weather, Digi (digipeater), or iGate.

Clicking a station in the list centers the map on it and opens its popup. A small colored badge on each row shows which source or sources heard it.

13.6 Placing Manual Markers

APRS shows you what is on the air, but many important places -- the Emergency Operations Center (EOC), a shelter, a command post -- are not beaconing. Use the **Markers** tab to drop your own labeled markers on the map.

1. Open the **Markers** tab in the sidebar.
2. Type a **Label** such as "North Shelter" or "Command Post".
3. Choose a **Type** from the list: EOC, Shelter, Staging Area, Command Post, Medical, Repeater, Fire / Hazmat, Law Enforcement, or Custom -- each has its own icon.
4. Set the location: either type the **Latitude** and **Longitude**, or click **Pick on Map** and then click the spot on the map (the cursor becomes a crosshair and fills in the coordinates for you).
5. Add optional **Notes**, then click **+ Add Marker**. The marker appears at once and is listed under **Placed Markers**.

Placed markers are remembered between sessions. Use **Clear All Markers** at the bottom of the tab to remove them all (it asks you to confirm first).

13.7 APRS Messages

The **Msgs** tab receives short APRS text messages addressed to your station and lets you send one. Received messages show the sender, the recipient, the text, and the time; unread ones are marked with a green edge. To send a message:

1. Open the **Msgs** tab and find the **Send APRS Message** form at the bottom.
2. Enter the destination callsign in **To callsign** (for example, a field team's radio callsign).
3. Type your message in the text box. APRS limits a message to **67 characters**, so keep it short.
4. Choose how to send it in the **Via** drop-down: **Via Direwolf** (over the radio) or **Via YAAC**.
5. Click **Send**. The status line below the form confirms the result.

> **TIP** You can also start a message from a station's map popup: click its **Message** button and the To field is filled in for you.

13.8 Settings -- My Station, Sources, and Display

The gear tab holds every setting. It is grouped into three sections plus a legend. Your entries are saved in the browser, so this map remembers them the next time you open it on the same device.

13.8.1 My Station

Enter your **Callsign**, **Latitude**, and **Longitude**. If the Pi has a Global Positioning System (GPS) fix, the map fills these in from the live position automatically. **Center on Station** jumps the map back to you.

13.8.2 APRS Sources

FIELD	WHAT TO ENTER
RF source host	The address of the computer running APRS Command that serves the Direwolf stations. Use "localhost" for a Pi-side bridge, or the laptop's Internet Protocol (IP) address, for example 192.168.50.42.
RF source port	The port that feed listens on. Default 8080.
YAAC port	The port the YAAC feed listens on. Default 8082.

13.8.3 Display

OPTION	WHAT IT DOES
Auto-refresh (seconds)	How often the map re-polls: 15s, 30s, 60s, 2 min, or Manual only. Use a short interval when resources are actively moving.
Map Tiles	Shows whether offline tilesets are loaded. Switch the base map with the layer control in the top-right corner of the map.
Station age threshold (min)	The number of minutes that counts as "fresh" (green). Default 15.
Range ring radius (km)	The size of the dashed distance circle around your station. Default 50 km.
Show APRS symbols	Choose emoji-style symbols or plain colored dots for less clutter.
Station labels	Show or hide the callsign text above each marker.

CHANGED **Renamed: "SARTopo" is now "CalTopo / SARTopo" throughout.**

13.9 Tracking, Range Ring, Repeaters, and CalTopo / SARTopo Overlays

Four extra layers add context beyond the live stations:

- **Track** -- click **Track** in a station's popup (or the Track button on the left) to follow one station and draw its movement trail as it beacons.
- **Range Ring** -- turns on a dashed circle at the radius set in Settings, so you can judge distance from your station at a glance.
- **Repeaters** -- overlays the repeater database served by the FieldCommand server, handy for choosing a machine to work through.
- **CalTopo / SARTopo** -- shows search sectors, assignments, and zones you imported from CalTopo / SARTopo. Export your map as GeoJSON, import it on the CalTopo / SARTopo Import page, then turn on the CalTopo / SARTopo layer here. The overlay stays put across refreshes until you clear it.

13.10 Exporting to KML

Click **KML** in the top bar to download every station currently on the map as a Keyhole Markup Language (KML) file. Open it in Google Earth or share it with another agency to hand off the tactical picture. Each placemark carries the station callsign, its comment, and which source or sources heard it.

13.11 Troubleshooting

SYMPTOM	WHAT TO DO
The map is blank -- no base map draws	You have no internet and offline tiles are not installed. Run "sudo bash download_tiles.sh" on the Pi, then reload. Station markers still plot even without a base map.
No stations appear at all	Check the Sources tab. If Direwolf shows a count of "--", confirm the RF source host and port in Settings point at the computer running APRS Command, then click Poll. With no radio traffic there is simply nothing to hear.
The purple APRS-IS feed shows nothing	APRS-IS needs the internet and an aprs.fi API key. Confirm a WAN is up and paste a valid key into the aprs.fi key box on the Sources tab.
All my stations are gray	Gray means "not heard in over an hour." Either traffic has stopped, or your device clock is wrong -- age is measured against the clock, so a bad clock ages every station. Check the UTC clock in the top bar.
A sent message never confirms	The message must go out over a live source. Make sure the Via source (Direwolf or YAAC) is connected on the Sources tab, keep the text to 67 characters, and watch the status line under the Send button for the result.
My placed markers disappeared	Markers are saved in this browser only. Opening the map on a different device or after clearing browser data starts fresh. Re-add them, and avoid clearing site data for 192.168.50.1.

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14 GPS-Tracked Resource Map http://192.168.50.1/resource_map.html

The Resource Map is a live picture of where your resources actually are. It reads the same T-cards you build on the Operations board (Chapter 16) and draws each one that has a Global Positioning System (GPS) position as a color-coded pin on a map, colored to match its current status. Resources that do not have a position yet are listed in the sidebar so the Operations Section can see at a glance what still needs to be placed. You set a position three ways -- from a phone standing at the resource, by clicking the map, or by typing coordinates -- and the pin appears immediately for every operator on EMCOMM-NET.

TIP In one sentence: pick an incident, and the map shows every resource that has a GPS fix as a status-colored pin, with the un-placed ones waiting in the sidebar.

14.1 Where the Positions Come From

The Resource Map does not keep its own list of resources. It draws the **T-cards** for the incident you choose -- the same resource cards created and tracked on the Operations T-card board. Each pin carries the resource name, its type, its status, its current assignment, and how many personnel it holds, all pulled straight from that card. Setting a position on this page writes the coordinates back onto the T-card, so the resource carries its last known location everywhere in the app.

NOTE A resource has to exist as a T-card before it can appear here. If a resource is missing from the map and the sidebar, add it on the Operations board first, then come back and place it.

14.2 The Top Bar

The controls that act on the whole map run across the top of the page:

CONTROL	WHAT IT DOES
Incident selector	The drop-down that reads "Select incident..." until you choose one. Pick which incident's resources to show. Your choice is remembered, so the same incident loads automatically the next time you open the page.
My Location	Drops an amber marker at your own device GPS position and zooms the map to it. Handy for checking the map against a spot you can see.
Refresh	Reloads every resource position from the server right now. Use it after someone else has moved or placed a resource.
Repeaters	Toggles an overlay of amateur repeaters from the Repeater Database on and off (see 14.7). The button border turns gold while the overlay is on.
Auto-refresh 30s	A checkbox. When ticked, the map reloads positions by itself every 30 seconds -- turn it on when resources are moving and updating live.
T-Cards / Dashboard	The two links at the far right jump to the Operations T-card board and back to the main dashboard.

14.3 Marker Colors and the Legend

Each positioned resource is a teardrop pin holding the first letter of its resource type, painted the color of its status. The legend in the lower-left corner of the map is your key. The status colors are:

COLOR	STATUS IT MEANS
Green	Available -- ready to be assigned.
Navy	Assigned -- working an assignment.
Blue	Staging -- checked in and waiting at a staging area.
Red	Out of Service -- not available (also used for En Route on the sidebar dot).
Amber	En Route -- moving to an assignment (shown amber on the map pin).

The legend also reminds you that an "X" means a resource has no GPS position yet, and that clicking any pin lets you edit it. Resources with no position are not drawn on the map -- they wait in the sidebar until you place them.

14.4 The Resources Sidebar

The panel down the right side lists every resource on the T-card board, placed or not. Its header shows a running count -- for example "3/7 have GPS" -- so you can see how much of the incident is positioned. Each row shows the resource name with a colored status dot, then a line of details, and, if it is placed, its coordinates:

ROW ELEMENT	WHAT IT SHOWS
Status dot	A small colored dot before the name, matching the status colors above.
Name	The resource name from its T-card.
Details line	The resource type, the status, the assignment (in italics) if one is set, and the personnel count if it is above zero, separated by dots.
Coordinates line	For a placed resource: its latitude and longitude, any location label, and the time the position was last updated.
Left edge bar	A green stripe on the left marks a resource that has a GPS position; a dim gray stripe and the words "No GPS -- click to set position" mark one that does not.

Click any row to open the Set Position window for that resource. The **Set All...** button in the sidebar header is a reminder prompt -- it tells you to click each resource in turn, or to use the map-click method, to place them one by one.

14.5 Setting a Resource Position

Click a resource in the sidebar, or click an existing pin on the map and choose **Update Position** in its popup. Either opens the Set Position window, titled with the resource name. Three methods are offered, and you can mix them -- for example, use the map click to get close, then fine-tune the numbers by hand:

METHOD	HOW TO USE IT	BEST FOR
Use Device GPS	Click the **Use Device GPS** button. The browser asks permission, then fills in the latitude and longitude from this device and shows the accuracy in meters.	An operator standing at the resource with a phone or tablet.

METHOD	HOW TO USE IT	BEST FOR
Click Map to Place	Click **Click Map to Place**. The window closes and the pointer becomes a crosshair; click the spot on the map. The window reopens with the coordinates filled in.	Placing a resource by a location you can see on the map.
Type coordinates	Type the decimal **Latitude** and **Longitude** straight into the two fields (for example 42.30890 and -88.43560).	A position read from a paper map, a GPS receiver, or a radio report.

To place a resource:

1. Open the Set Position window by clicking the resource in the sidebar (or a pin, then **Update Position**).
2. Fill in the coordinates using any of the three methods above.
3. Optionally type a **Location Label** -- a plain description such as "Division Alpha staging area" that shows under the coordinates and in the pin popup.
4. Click **Save Position**. The pin appears on the map at once and the sidebar row gains its green edge and coordinates.

NOTE The app checks your numbers before saving: latitude must be between -90 and 90 and longitude between -180 and 180, and both must be real numbers. If they are not, it says "Enter valid coordinates" or "Coordinates out of range" and nothing is saved.

14.6 Updating or Clearing a Position

To move a resource, open it again and set a new position the same way -- the new coordinates replace the old ones and the update time refreshes. To remove a position entirely, open the resource and click **Clear GPS**. The pin disappears from the map and the resource drops back to the un-placed list in the sidebar. The T-card itself is not deleted; only its location is cleared. The **Clear GPS** button appears only for a resource that already has a position; **Cancel** closes the window without changing anything.

14.7 The Repeater Overlay

Click **Repeaters** to lay the amateur repeaters from the Repeater Database over the map -- useful for planning which machine a moving resource can reach. Each repeater is a small dot, colored by its mode, with a popup that shows the output frequency, callsign, offset, tone, and town. Repeaters flagged for emergency communications are drawn in gold and their popups list the ARES, RACES, or SKYWARN tags. The dot colors are:

DOT COLOR	REPEATER TYPE
Gold	Flagged for ARES, RACES, or SKYWARN emergency communications.
Green	Standard analog FM.
Blue	D-STAR digital.
Orange	C4FM / System Fusion digital.
Purple	Digital Mobile Radio (DMR).
Red	Project 25 (P25) digital.

Click **Repeaters** again to hide the overlay. The overlay is empty until you have imported repeater data on the Repeater Database page -- if none is loaded, the app tells you to import a RepeaterBook file first.

14.8 My Location and Auto-Refresh

The **My Location** button uses this device's own GPS to zoom the map to where you are and drop a bright amber circle labeled "Your location." It does not change any resource -- it is only for orienting yourself. Tick **Auto-refresh 30s** when an incident is moving so the map keeps itself current without your clicking **Refresh**; clear the checkbox to stop it. Because every operator reads the same server, a position one person sets shows up for everyone on the next refresh.

14.9 Troubleshooting

SYMPTOM	WHAT TO DO
The sidebar says "Select an incident above" and nothing loads.	Choose an incident in the top-bar drop-down. The map only shows resources once an incident is selected.
A resource I expect is not on the map or in the sidebar.	It has no T-card. Add it on the Operations T-card board first, then return -- the map only lists resources that exist as T-cards.
The Use Device GPS button does nothing or reports an error.	The browser needs permission and a real location fix. Allow location access when asked, make sure the device has GPS, and try outdoors; otherwise type the coordinates by hand.
I clicked Save Position but it refused.	The coordinates were blank, not numbers, or out of range. Latitude must be -90 to 90 and longitude -180 to 180. Re-enter valid decimal values and save again.
The Repeaters overlay is empty or will not load.	No repeater data is imported. Open the Repeater Database page and import a RepeaterBook file, then toggle the overlay again.
Someone placed a resource but I do not see it.	Click Refresh, or tick Auto-refresh 30s. Each device reads its own copy until it reloads from the server.

15 Incident Command System (ICS) Platform Overview — Five-Section Structure http://192.168.50.1/incident.html

The Incident Management screen is the front door to every incident FieldCommand IMS runs. From this one page you create an incident, pick its type and command details, and reach every Incident Command System (ICS) form for that incident. ICS is the standard command structure defined by the National Incident Management System (NIMS); it organizes any incident into five sections — Command, Operations, Planning, Logistics, and Finance / Administration. Everything on this page is grouped by those five sections, and every form you open writes back to the same incident record, so a change made in one section shows up everywhere else at once.

TIP In one sentence: open Incident Management, create or pick an incident, then click the ICS forms you need — all five sections work from one shared incident record.

15.1 The Incident Management Screen

Open **Incident Management** from the dashboard, or browse straight to **http://192.168.50.1/incident.html** on any EMCOMM-NET device. Across the top is a dark header with an **ICS** badge, the words **INCIDENT MANAGEMENT**, a **+ New Incident** button on the right, and a **Dashboard** link back to the home page. Below the header the screen is split into two columns:

AREA	WHAT IT SHOWS
Left column (list)	Two lists of incidents on this server: **Active Incidents** at the top and **Closed Incidents** below. Each incident is a clickable card.
Right column (workspace)	The workspace for the incident you have selected. Until you pick one it reads "NO INCIDENT SELECTED" with a **+ Create New Incident** button.

15.2 The Incident List

Every incident card in the left column shows a colored status dot (green for active, gray for closed) and a summary line so you can find the right incident at a glance. The card carries these details:

ITEM	WHAT IT MEANS
Incident name	The name you gave the incident, or "Unnamed Incident" if none.
Type badge	The incident type, shown as a small rounded badge.
Number	Your agency tracking number, shown with a "#" when one was entered.
Period	The current operational period number (Period 1, Period 2, and so on).
Started	The date the incident was created.
IC	The Incident Commander name, when one was set.

Click any card to load that incident into the workspace on the right.

15.3 Creating a New Incident

Click **+ New Incident** (top right) or the **+ Create New Incident** button in the empty workspace. The **CREATE NEW INCIDENT** window opens.

1. Type the **Incident Name** (required) — for example "McHenry County Winter Storm Response 2026". If your setup saved a default name, it is filled in for you.
2. Enter an **Incident Number** if your agency assigns one (optional), such as "2026-0142".
3. Choose an **Incident Type** (required) from the drop-down — see the type list in 15.4.
4. Fill in **Jurisdiction** (for example "McHenry County, IL") and **Incident Location / Address** (a street address or Global Positioning System (GPS) coordinates). Jurisdiction may be pre-filled from your setup.
5. Set the **Incident Commander** — start typing to pick a name from the roster, or type any name or callsign for someone not on the roster.
6. Pick the **Operational Period Duration** (12 hours is the standard default; 8, 24, or 6 hours are also offered) and the **ICS Form Variant** (see 15.5).
7. Add an **Initial Situation Summary** if you have one, then click **CREATE INCIDENT**. The incident appears in the Active list and opens in the workspace. Click **Cancel** to close without creating.

NOTE Only the Incident Name and Incident Type are required. Everything else can be left blank now and filled in later with the Edit button.

15.4 Incident Types

The **Incident Type** drop-down groups the all-hazards types into seven categories. Pick the closest match; it sets the badge on the card and helps organize your records.

CATEGORY	EXAMPLE TYPES
Natural Hazards	Winter Storm, Flooding, Tornado / Severe Weather, Earthquake, Wildfire, Heat Emergency, Drought.
Technological	Hazmat / Chemical Spill, Transportation Accident, Structure Fire, Power Outage / Infrastructure, Dam Failure, Nuclear / Radiological.
Human-Caused	Mass Casualty Incident, Active Threat, Civil Disturbance, Terrorism.
Search & Rescue	Wilderness, Urban, and Water search; Missing Person (Dementia / Memory) and Missing Person (Child).
Public Health	Disease Outbreak / Pandemic, Mass Casualty (Medical), Public Health Emergency.
Planned Events	Planned Event (Public Safety), Planned Event (EMCOMM Exercise), Drill / Training Exercise.
Other	Mutual Aid Request, EOC Activation, Other / All-Hazards.

15.5 ICS Form Variant — FEMA, USCG, or NWCG

Different agencies use slightly different editions of the ICS forms. The **ICS Form Variant** picker offers three pills — **FEMA** (Federal Emergency Management Agency, the all-hazards default), **USCG** (United States Coast Guard), and **NWCG** (National Wildfire Coordinating Group). Your setup default is selected automatically. The chosen variant is shown on the workspace above the forms, and you can change it later with the **Change** link there.

15.6 The Incident Workspace

Selecting an incident fills the right column. At the top a header repeats the incident name, its type badge, the location, jurisdiction, Incident Commander, and the start time in Coordinated Universal Time (UTC), plus the summary if one was entered. For an active incident three buttons sit at the right of this header:

BUTTON	WHAT IT DOES
Next Period	Advances the incident to the next operational period (see 15.7).
Edit	Opens the full incident editor to change any detail.
Close Incident	Ends the incident and moves it to the Closed list (see 15.9).

A closed incident shows the word CLOSED here instead of the buttons.

15.7 Operational Periods

An operational period is one planning block of an incident — often a 12-hour shift. Below the header the **OP** bar shows the current period number ("OP 1"), when it started, a **GENERAL INFO** button that opens the ICS-201 initial briefing for this period, and an **Export IAP** button that opens the Incident Action Plan (IAP) for printing.

When it is time to roll over to the next shift, click **Next Period** in the header. The **ADVANCE OPERATIONAL PERIOD** window opens with an **Objectives carried forward to next period** box. Type the objectives that continue into the new period and click **Advance Period**. The period counter increases and the new period becomes current; every ICS form now works against that new period.

15.8 The ICS Forms Navigator — Five Sections

The heart of the workspace is a grid of form buttons grouped under colored section dividers, one group per ICS section. Each button shows the form number (such as "ICS-202"), its plain name, and which agency variants it supports. Click a button to open that form; forms save automatically. The table below lists the forms exactly as they are grouped on the page.

SECTION	FORMS ON THE PAGE
Command	ICS-201 Incident Briefing, ICS-202 Incident Objectives, ICS-207 Organization Chart, ICS-208 Safety Message/Plan.
Operations	ICS-204 Assignment List, ICS-211 Check-In List, ICS-219 Resource Status (T-Cards), ICS-210 Resource Status Change.
Planning	ICS-203 Organization Assignment, ICS-209 Incident Status Summary, ICS-215 Operational Planning Worksheet, ICS-215A IAP Safety Analysis.
Logistics	ICS-205 Radio Communications Plan, ICS-205A Communications List, ICS-206 Medical Plan, ICS-213RR Resource Request.
Finance / Admin	ICS-214 Activity Log, ICS-220 Air Operations Summary, ICS-221 Demobilization Check-Out.
Communications Unit	ICS-213 General Message and ICS-309 Communications Log. ICS-309 and the ICS-211 check-in are handled through the Net Logger.

NOTE

The section a form is grouped under on this screen is where you reach it, not always where it is developed. For example, the Communications Unit and Medical Unit sit in the Logistics service branch and develop ICS-205, 205A, 206, and 309; the Planning Section then assembles those into the IAP. See Chapter 17 for the full development-responsibility table.

15.9 Meeting Scheduler, Activity Log, Editing and Closing

Under the Planning section a **Meeting Scheduler** link plans meetings, agendas, required attendees, and minutes for the incident. At the bottom of the workspace the **ACTIVITY LOG** feed lists recent actions with a time, the section, and what happened, so anyone can see the incident history at a glance.

To change incident details, click **Edit** in the header — it opens the full incident editor. To end an incident, click **Close Incident**; you are asked to confirm because closing cannot be undone. A closed incident stays on the server in the Closed list as a permanent record — nothing an incident produces is thrown away.

WARN Incident data is permanent. Every incident, form, T-card, and activity entry is saved on the server and backed up. Closing an incident archives it; it does not delete it.

15.10 Troubleshooting

SYMPTOM	WHAT TO DO
The incident list shows "ICS platform offline"	The ICS platform service (port 5055) is not responding. Check that the ics-platform service is running (see the Health Monitor), then reload the page.
CREATE INCIDENT does nothing / a warning appears	Incident Name and Incident Type are both required. Fill in the name and pick a type from the drop-down, then click CREATE INCIDENT again.
I do not see my new incident in the list	It may already be active but the list did not refresh — reload the page. If it was closed, look in the Closed Incidents list lower in the left column.
The Next Period / Edit / Close buttons are missing	That incident is closed; a closed incident shows the word CLOSED instead. Closed incidents are read-only records.
The wrong form edition (FEMA vs USCG vs NWCG) is showing	Click the Change link above the form buttons and enter FEMA, USCG, or NWCG to set the variant for this incident.
The Incident Commander picker is empty	The roster has no members yet, or the core API (port 5050) is unreachable. You can still type any name or callsign by hand; import the roster later.

16 Incident Command System (ICS) Operations Section — T-Card Resource Board

http://192.168.50.1/ics/operations.html

The Operations page is where the Operations Section keeps track of every resource working the incident — every engine, crew, individual, helicopter, and piece of equipment — and shows at a glance what each one is doing right now. It is a digital version of the paper ICS T-card rack that hangs on the wall in a traditional command post: one card per resource, sorted into columns by status. But because it lives on the server, every operator on EMCOMM-NET sees the same board at the same time, a card can be dragged from column to column, and the board stays in step with your ICS-204 Assignment Lists. Everything you enter here is saved on the server, not just in the browser, so it survives, backs up, and shows on every device.

TIP In one sentence: this is your live resource status board — add a card for each resource, drag it to the right column as its status changes, and open a card to record who is on it.

16.1 The Three Views

Across the top of the page are three tabs. They all show the same resources — just in different shapes for different jobs. Click a tab to switch views.

TAB	WHAT IT SHOWS
T-Card Board	The everyday view. One card per resource, sorted into four status columns. Drag a card between columns to change its status.
Resource List	The same resources as a full-width table, with more columns visible at once. Good for scanning, printing, or finding one resource quickly.
Assignments	Only the resources currently set to Assigned, shown as the working ICS-204 assignment picture.

16.2 The T-Card Board

The board has four columns, left to right: **Available**, **Staged**, **Assigned**, and **Out of Service**. A small count next to each column title tells you how many resources are in it. Each card shows the resource type icon and type, the resource ID, the name, its current assignment, and a personnel count when people are listed on it. A color-coded legend near the panel title reminds you which color means which status.

COLUMN	WHAT IT MEANS
Available	On scene and ready to be assigned. This is the default for a new resource.
Staged	Standing by at a staging area, held in reserve until needed.
Assigned	Working a task now, tied to a division, group, or sector.
Out of Service	Not usable right now — broken down, resting, or demobilized.

To move a resource, click and hold its card, drag it to the target column, and let go. The card lands in the new column, its color changes to match, and the change is saved to the server for everyone. When you drag a card into **Assigned**, a small box pops up asking for the Division or Group (for example, "Division Alpha" or "Group

SAR"); type it and click OK, or leave it blank. Dragging a card out of Assigned clears its division automatically.

NOTE

The **Sync from ICS-204s** button at the top of the board pulls resources listed on your ICS-204 Assignment Lists onto the board — creating a card for any that are missing and marking them Assigned to their division. The board also runs this sync on its own a moment after the page loads. See 16.9 for how the two-way sync works.

16.3 Adding a Resource

1. Click **+ Add Resource** at the top right of the page. The Add Resource box opens.
2. Fill in **Resource ID** — a short label such as "E-101". This is required.
3. Choose a **Type** from the drop-down (see the table in 16.4). This is required.
4. Enter a **Name / Description** — for example, "Engine 101 - F-250 Brush Truck". This is required.
5. Fill in **Capability** if it helps — a free-text note such as "Type III, 5 persons".
6. Set the starting **Status** (Available, Staged, Assigned, or Out of Service). Most new resources start as Available.
7. Add an **Assignment / Location** and a **Contact / Callsign** if you know them (for example, "Division A - Sector 3" and a callsign or channel).
8. Click **Add Resource**. The card appears on the board and the record is saved to the server. Click **Cancel** instead to close without saving.

FIELD	WHAT IT MEANS
Resource ID *	Required. Short unit designator, such as E-101 or Team Alpha.
Type *	Required. The kind of resource, chosen from the drop-down (16.4).
Name / Description *	Required. The plain-language name or description of the unit.
Capability	Optional note on what the resource can do, such as type and crew size.
Status	Which column the card starts in.
Assignment / Location	Where the resource is or what task it has.
Contact / Callsign	How to reach it on the radio — a callsign or channel.

16.4 Resource Types

The **Type** drop-down offers six categories. The type sets the icon shown on the card and groups like resources together; it does not restrict what you can do with the resource.

TYPE	WHAT IT COVERS
Engine / Vehicle	A fire engine, brush truck, or any other apparatus or vehicle.
Crew / Team	An organized team that works together as one unit.
Individual	A single person or staffed position.
Helicopter	A rotary-wing air resource.
Equipment	Gear and supplies — generators, pumps, go-kits, tools.
Other	Anything that does not fit the categories above.

16.5 Opening a Resource and Changing Its Status

Click any card on the board (or any row in the Resource List) to open the resource detail panel. The title shows the resource ID and name, and the panel opens on the **INFO** tab. The Info tab shows the fields below and lets you edit the assignment. Along the bottom of the panel are one-click status buttons and a Delete button.

ITEM	WHAT IT MEANS
Type	The resource type and its icon.
Status	The current status chip.
Leader	The supervisor or crew boss for this resource.
Contact	The radio callsign or channel for the resource.
Home Agency	The agency or mutual-aid source the resource belongs to.
Personnel (count)	How many people are listed on the resource (set on the Personnel tab).
Assignment / Division / Sector	An editable box for the current task. Type a new value and click Update Assignment.

16.5.1 Two ways to change status

1. Fastest way: on the board, drag the card into the column you want.
2. From the detail panel: open the resource, then click one of the bottom buttons — **Available**, **Staged**, **Assigned**, or **Out of Service**. The panel closes and the card moves.
3. To change the task without changing status, type into the **Assignment / Division / Sector** box and click **Update Assignment**.
4. To remove a resource entirely, click **Delete** and confirm. This cannot be undone.

16.6 The Personnel Tab

On a paper ICS T-card, the back lists the individual people on that resource — not just a headcount, but names, roles, and contacts. FieldCommand does the same with a **PERSONNEL** tab on the detail panel. The number in parentheses next to the tab name is the current headcount, and adding or removing a person updates the personnel count shown on the card automatically. The name field is linked to the Member Roster, so as you type, matching people appear as blue suggestion chips you can click to auto-fill.

FIELD	WHAT IT MEANS
Name *	Required. The person's full name. Type at least two letters to see roster suggestions; click a suggestion to fill the name, callsign, and agency.
ICS Position on Resource	The role this person fills on this resource, chosen from the drop-down (16.7). This is their job on the resource, not their spot on the overall org chart.
Agency	Home agency or mutual-aid source. Auto-filled from the roster when matched.
Contact / Callsign	Radio callsign, channel, or phone for this person. Auto-filled from the roster when a licensed operator is matched.

1. Open a resource and click the **PERSONNEL** tab.
2. Under **+ ADD PERSONNEL**, begin typing a name in the **Name** field. Click a blue suggestion chip to auto-fill from the roster, or just type a name by hand.
3. Pick the **ICS Position on Resource** from the drop-down.
4. Fill in **Agency** and **Contact / Callsign** if they did not auto-fill.

5. Click **Add to Resource**. The person appears in the list and the count updates.
6. To remove someone, click the remove (x) button on their row and confirm.

> **NOTE** The Personnel tab answers "who is on this resource?" It is separate from the ICS-211 check-in list, which answers "who has checked into this incident?" A mutual-aid crew might all be listed on their engine's Personnel tab while only the crew boss checks in on the ICS-211.

16.7 ICS Position on Resource — Options

The **ICS Position on Resource** drop-down offers the roles below, plus **Other** for anything not listed.

POSITION	PLAIN MEANING
Crew Boss	Leads a crew that works as one unit.
Single Resource Boss	Supervises a single resource such as one engine.
Division Supervisor	Runs a geographic division.
Task Force Leader	Leads a task force of mixed resource types.
Strike Team Leader	Leads a strike team of the same resource type.
Paramedic / EMT	Provides emergency medical care.
Firefighter	Fireground personnel.
Emergency Radio Operator	A licensed operator handling radio traffic.
Net Control	Runs a radio net.
Operator / Driver / Operator / Equipment Operator	Runs a vehicle or a piece of equipment.
Logistics Support	Provides supply, transport, or support to the resource.

16.8 The Resource List and Assignments Views

The **Resource List** tab shows every resource in one wide table. Click any row to open the same detail panel, or use the **Edit** button in the Actions column.

COLUMN	WHAT IT SHOWS
Resource ID	The short designator.
Type	The resource type and icon.
Name / Description	The plain-language name.
Capability	The capability note, if any.
Status	A color-coded status chip.
Assignment	The current task or division.
Contact	The callsign or channel.
Actions	An Edit button that opens the detail panel.

The **Assignments** tab, titled **Current Assignments (ICS-204)**, lists only the resources set to Assigned, in Resource, Type, Assignment, and Contact columns — your at-a-glance ICS-204 picture. The **+ Add Assignment** button opens the same Add Resource box so you can add a resource without leaving this view.

16.9 Keeping the Board and the ICS-204 in Sync

The T-Card Board and the Planning Section's ICS-204 Assignment Lists describe the same resources, so FieldCommand keeps them in step both directions. When you drag a card, the app looks for that resource on any ICS-204 for the incident and updates it — for example, writing the new division when a card goes to Assigned, or noting "OUT OF SERVICE" when a card goes Out of Service. Going the other way, **Sync from ICS-204s** reads the assignment lists and brings those resources onto the board, creating a card for any that are not there yet. Because ICS-211 self check-ins also land in the same resource store, a person who checks in can show up here automatically.

TIP

The board keeps a local copy in the browser so it paints instantly, then refreshes from the server, which is the authoritative record. If you are briefly offline, the board still shows the last known picture and syncs again when the connection returns.

16.10 Troubleshooting

SYMPTOM	WHAT TO DO
A card will not drag	Click and hold directly on the card body, then drag. A single quick click opens the detail panel instead — that is normal.
I clicked Add Resource but nothing was saved	Resource ID, Type, and Name / Description are required. Fill all three, then click Add Resource again.
Sync from ICS-204s did nothing	The sync needs a selected incident and at least one ICS-204 with resources listed. Confirm an incident is selected and that Planning has built a 204 first.
A resource shows on the board that I did not add	It was pulled in by the ICS-204 sync or an ICS-211 self check-in. Open it to see its details, or delete it if it does not belong.
The personnel count on the card looks wrong	Open the resource, click the Personnel tab to reload the real list, then close it — the card count refreshes from the server.
My changes are not showing on another operator's screen	Each device refreshes from the server on load and after a change. Have them reload the Operations page; if it still lags, check that the ICS platform service (port 5055) is running from the Health page.

17 Incident Command System (ICS) Planning Section & Incident Action Plan (IAP) Assembly http://192.168.50.1/iap.html

The Incident Action Plan (IAP) is the written work plan for one operational period - the block of time (often 12 hours) that one shift of responders works to a single set of goals. It is the Planning Section's job to assemble it. The IAP Assembly page pulls together the Incident Command System (ICS) forms your team has filled out, lets you choose which ones belong in the package, shows you at a glance how complete the plan is, and turns the whole thing into a printed plan or a portable file you can carry off-site. You do not build a form here - you build the finished plan out of forms.

TIP

In one sentence: this page is the assembly line - tick the forms you want, watch the completeness meter, then Print or Save the plan for the next shift.

17.1 The Assembly Screen at a Glance

Open the page and the dark blue header across the top reads INCIDENT ACTION PLAN - ASSEMBLY, with the current incident name shown beside it (or "No incident selected" if you have not chosen one). The rest of the screen is split into two columns.

AREA	WHAT IS THERE
Header buttons (top right)	Four controls: Print IAP, Save as PDF, Save as HTML, and an Incident link that returns you to the incident page. These act on the whole plan.
Left column	The IAP Form Checklist for the chosen operational period - one card per ICS form, with a checkbox, the form number and title, its ICS section, and its status.
Right column	A live IAP Cover Page preview, the IAP OPTIONS box (operational period and form variant), the COMPLETENESS meter, and a QUICK LINKS list of the first forms.

17.2 The IAP Form Checklist

The left column lists every ICS form FieldCommand knows how to place in an IAP, in assembly order. Each row shows the form number (for example ICS-205), its plain title, and the ICS section that owns it - color coded so you can scan by section. The six forms marked Required are pre-checked because a complete IAP is not valid without them. The table below is the full set the page carries.

FORM	TITLE	SECTION	REQUIRED?
ICS-201	Incident Briefing	Command	Yes
ICS-202	Incident Objectives	Command	Yes
ICS-203	Organization Assignment List	Planning	Yes
ICS-204	Assignment List	Operations	Yes
ICS-205	Radio Communications Plan	Logistics	Yes
ICS-205A	Communications List	Logistics	No
ICS-206	Medical Plan	Logistics	Yes

FORM	TITLE	SECTION	REQUIRED?
ICS-207	Organization Chart	Command	No
ICS-208	Safety Message / Plan	Command	No
ICS-209	Incident Status Summary	Planning	No
ICS-213RR	Resource Request	Logistics	No
ICS-214	Activity Log	All	No
ICS-215	Operational Planning Worksheet	Planning	No
ICS-215A	IAP Safety Analysis	Command	No
ICS-210	Resource Status Change	Planning	No
ICS-211	Incident Check-In List	Command	No
ICS-213	General Message	All	No
ICS-216	Radio Requirements Worksheet	Logistics	No
ICS-217A	Comms Resources Available	Logistics	No
ICS-218	Support Vehicle / Equip Inventory	Logistics	No
ICS-219	Resource Status Cards (T-Board)	Planning	No
ICS-220	Air Operations Summary	Operations	No
ICS-223	Wildlife Protection Measures	Planning	No
ICS-224	Crew Performance Rating	Finance	No
ICS-225	Personnel Performance Rating	Finance	No
ICS-226	Work Capacity Test Record	Finance	No
ICS-233	Open Action Tracker	Planning	No
ICS-234	Work Analysis Matrix	Planning	No

17.3 Reading a Form Card

Every card carries a colored status pill on the right that tells you whether the form is ready. The status is read live from the server for the operational period you have selected, so it changes the moment someone saves that form.

STATUS PILL	WHAT IT MEANS
Saved (green)	The form has been filled in and saved for this incident and period. It is ready to include in the IAP.
Required - not yet saved (red)	This is one of the six required forms and it has no saved content yet. The plan is not complete until it is filled in.
Not started (amber)	An optional form that has not been saved. Include it only if the incident needs it.

At the far right of each card is an Open link (or Edit, if the form is already saved). Clicking it leaves this page and opens that ICS form in the form editor, carrying the same incident and period along so you land in the right place. Fill the form in there, including any Prepared By or Approved By signature, then return to this page - the card updates to Saved.

17.4 Choosing Which Forms to Include

The checkbox on the left of each card, and the card itself, control whether the form goes into the plan. Clicking anywhere on a card toggles it.

1. Make sure the correct operational period is chosen in the IAP OPTIONS box (see 17.6) - the checklist and every status pill follow that choice.
2. Leave the six Required forms checked. They are pre-selected for you.
3. Click any optional card to add it to the plan. A checked card turns green to show it is included; click again to remove it.
4. A typical amateur or public-service IAP includes ICS-202, 203, 204, 205, 205A, 206, 207, and 208 - objectives, organization, assignments, comms plan, comms list, medical plan, org chart, and safety message.

WARN You can include a form that is not yet saved, but it will be blank in the finished plan. The completeness meter is your guard against printing an empty required form.

17.5 The Cover Page Preview

The IAP Cover Page box at the top of the right column shows exactly what the front page of your plan will look like, built from your organization settings and the active incident. It updates by itself - there is nothing to fill in here.

COVER LINE	WHERE IT COMES FROM
Logo	Your organization logo, if one was uploaded in Settings.
Organization name	The short organization name or callsign from your configuration.
Incident name	The name of the active incident.
Operational Period	The period number chosen in the IAP OPTIONS box.
Incident Commander	The Incident Commander recorded on the incident (a dash if none is set).

17.6 IAP Options - Period and Form Variant

The IAP OPTIONS box holds two dropdowns that shape the whole plan.

OPTION	WHAT IT DOES
Operational Period	Choose Period 1 through 5. Changing it reloads the checklist and every status pill for that period, and updates the cover page. Each period is its own set of saved forms.
ICS Form Variant	FEMA, USCG (United States Coast Guard), or NWCG (National Wildfire Coordinating Group). This tags the plan for the form family your agency uses; the label prints on the cover and contents pages. For most all-hazards use, leave it on FEMA.

17.7 The Completeness Meter and Quick Links

The COMPLETENESS box shows a colored bar and a line of text such as "4 of 6 required forms saved (67% complete)". The bar is red below 60 percent, amber up to 99 percent, and green only when all six required forms are saved. Use it as your go / no-go check before you print: if it is not green, a required form is still blank.

Below it, the QUICK LINKS box lists the first several forms with a green check when saved or a dash when not, each a one-click shortcut into that form's editor - a fast way to jump to the forms most plans start with.

17.8 Exporting the Finished IAP

The three header buttons turn your selected forms into the finished plan. Choose by where the plan needs to go and whether a printer is on hand.

BUTTON	WHEN TO USE IT	WHAT HAPPENS
Print IAP	A printer is on EMCOMM-NET or attached to your device.	Opens a print-ready cover page, a contents table, and links to every included form in a new browser tab, then opens the print dialog. Allow pop-ups if nothing appears.
Save as PDF (recommended)	No printer on site, or the plan must go off-site, to the Emergency Operations Center (EOC), or into the archive.	Sends your form list to the FieldCommand server, which builds a proper 8.5 by 11 inch Portable Document Format (PDF) and downloads it as `IAP_[Incident]_Period[N]_[Date].pdf`. Opens on any device.
Save as HTML (fallback)	The server is unreachable, or you want a lightweight file with no server call.	Downloads a self-contained web-page file straight from the browser. Open it and use File - Print. Page breaks can vary by browser, so prefer PDF when layout matters.

1. Confirm the operational period, the form variant, and your checked forms.
2. Watch the completeness meter turn green before you commit the plan.
3. To print now, click **Print IAP**, then pick the site printer in the dialog.
4. To carry the plan off-site, click **Save as PDF**. The button shows Generating while the Pi builds it (a few seconds), then the PDF downloads on its own.
5. Only if the PDF button fails, click **Save as HTML** and print that file from any browser with File - Print.

> **NOTE** The PDF is built on the Pi and downloads to whatever device you are on - the printer does not have to be attached to that device. The file is self-contained and can be printed anywhere later. For the full compiler with embedded signatures and section dividers, use the IAP Compile page (iap_compile.html).

17.9 Doctrine Note - Forms That Cross Section Lines

The checklist colors ICS-205, ICS-205A, and ICS-206 as Logistics forms, which surprises some operators because they appear in the IAP that Planning assembles. That is correct ICS doctrine: the forms are developed in Logistics but distributed through Planning. Knowing this tells you who to ask when one needs fixing.

FORM	WHO DEVELOPS IT
ICS-205 Radio Comms Plan	Developed by the Communications Unit Leader (COML) under Logistics, then handed to Planning for the IAP. To correct it, go to the COML - not the Planning Section Chief.
ICS-205A Comms List	Also developed by the COML under Logistics. A supplemental contact directory, usually attached to the IAP.
ICS-206 Medical Plan	Developed by the Medical Unit Leader (MEDL) under Logistics. The Safety Officer must review and concur before it is finalized.

17.10 Troubleshooting

SYMPTOM	WHAT TO DO
Header reads "No incident selected" and every card is blank.	You opened the page without an active incident. Go back through the Incident link, select or start an incident, then open IAP Assembly again so the incident id is passed.
A form you saved still shows "Not started" or "Required - not yet saved".	Check the Operational Period dropdown - status is per period. Set it to the period you saved the form in. If still wrong, reopen the form, confirm it saved, and reload.
Completeness meter will not turn green.	One of the six required forms (ICS-201, 202, 203, 204, 205, 206) is still blank. Find the red pill in the checklist, click Open, fill and save that form, then return.
Print IAP does nothing / no new tab appears.	The browser blocked the pop-up. Allow pop-ups for this site and click Print IAP again, or use Save as PDF, which needs no pop-up.
Save as PDF shows an error or stays on "Generating".	The Pi server did not answer. Confirm you are on EMCOMM-NET and the Pi is running, then retry. As a stopgap, use Save as HTML and print that file from a browser.
The PDF opens but a form page is empty.	That form was included while unsaved. Uncheck it, or open it, add its content and save, then export the IAP again.

18 Federal Emergency Management Agency (FEMA) Public Assistance (PA) Cost Documentation http://192.168.50.1/fema_costs.html

When a disaster is federally declared, the Federal Emergency Management Agency (FEMA) can reimburse eligible response costs through its Public Assistance (PA) program. But FEMA only pays for costs you can prove -- who worked, what equipment ran, what you bought, at what rate, on what dates, tied to this disaster. The FEMA PA Cost Documentation screen is where you capture all of that while the incident is happening, so the paperwork is already built when the reimbursement claim comes due. It records the three cost buckets FEMA recognizes -- Force Account Labor (your own people), Force Account Equipment (your own vehicles and gear), and Materials and Contracts -- adds up the eligible total, and produces a Project Worksheet summary you hand to your state emergency management agency.

TIP In one sentence: log your labor, equipment, and purchases here as the incident runs, and the app tallies the FEMA-eligible total and exports a Project Worksheet summary for you.

18.1 Opening the Screen and Picking an Incident

Open **FEMA PA Cost Documentation** from the dashboard. Everything on this page is tied to one incident, so the first thing to do is choose it. At the top right of the page header is the **Select incident...** drop-down. Pick your incident from the list; the page then loads that incident's labor, equipment, and material entries and fills the totals bar. Your choice is remembered, so the next time you open the page the same incident is already selected.

WARN If the drop-down still says "Select incident..." and the tables read "No entries," you have not chosen an incident yet -- nothing will save until you do. Create the incident first (Chapter 8) if it is not in the list.

18.2 Reading the Totals Bar

Directly under the header is the totals bar -- four running numbers that update the instant you add, edit, or delete an entry. This is your at-a-glance picture of how much the incident has cost so far in FEMA-eligible terms.

TOTAL	WHAT IT COUNTS
Force Acct Labor	The sum of every labor entry, hours times rate, with fringe benefits added in.
Force Acct Equipment	The sum of every equipment entry, hours times the FEMA hourly rate.
Materials / Contracts	The sum of every material, supply, contract, and rental entry.
Total Eligible Costs	All three buckets added together -- the headline number, shown in green, set off by a divider on the right.

A small reminder printed beside the totals says to review every entry with your Finance Section Chief before submission. If the loaded FEMA equipment rates are more than a year old, an amber reminder banner appears above the tabs with a **Review Rates** link that jumps to the FEMA Equipment Rates page (Chapter 19).

18.3 The Four Tabs

The work is split across four tabs, chosen with the row of tab labels below the totals bar. Click a tab to switch views; the totals bar stays visible the whole time.

TAB	WHAT YOU DO THERE
Force Account Labor	Log hours for your own employees and volunteers.
Force Account Equipment	Log hours for your own vehicles and equipment at FEMA rates.
Materials / Contracts	Log purchases, supplies, rentals, and contracted work.
Project Worksheet	Enter the applicant and disaster details and review the eligible cost summary.

18.4 Force Account Labor

"Force account" means your own personnel -- not a contractor. FEMA generally reimburses **overtime** for your regular employees and, depending on your state's policy, may reimburse volunteer time. On this tab, click **+ Add Labor Entry** to open the labor form. Each field maps to what FEMA wants on the record.

FIELD	WHAT IT MEANS
Employee Name	Required. The person's name, entered "Last, First."
Position / Title	Their job title or ICS position, such as "Emergency Manager."
Department	The agency or unit they belong to, such as "McHenry County OEM."
Date Worked	The calendar day these hours were worked.
Regular Hours	Straight-time hours worked on the incident.
Overtime Hours	Overtime hours -- the primary FEMA-reimbursable category for paid staff.
Regular Rate ($/hr)	The straight-time hourly wage from payroll.
OT Rate ($/hr)	The overtime hourly wage.
Fringe Benefits %	Fringe (taxes, insurance, retirement) as a percentage of wages. Defaults to 30. FEMA requires fringe be documented.
Preview Total	Calculated live as you type -- (reg hrs x reg rate + OT hrs x OT rate) x (1 + fringe%).
Notes	Free text -- the assignment, EOC watch, or any justification.

1. Click **+ Add Labor Entry**.
2. Fill in the name (required), position, department, and date.
3. Enter regular and overtime hours and their rates. Watch the **Preview Total** update.
4. Adjust **Fringe Benefits %** if your rate differs from the 30 default.
5. Click **SAVE**. The entry appears in the labor table and the totals bar jumps.

The saved rows show as a table with columns Employee, Position / Title, Dept, Date, Reg Hrs, OT Hrs, Reg Rate, OT Rate, Fringe %, Labor Cost, Total w/Fringe, and Notes. Click the pencil (edit) button at the end of any row to reopen that entry; from the edit form you can change any field and click **SAVE**, or click **Delete** to remove it.

18.4.1 Importing from ICS-214

Rather than retype names, click **Import from ICS-214**. FieldCommand reads the completed ICS-214 Activity Logs for the selected incident and creates a labor entry for each unit leader, tagged with the operational period it came from. The imported rows come in with placeholder hours and a zero rate -- you must open each one and fill in the real hours and pay rates. A message tells you how many leaders were imported.

18.5 Force Account Equipment

This tab logs your own equipment -- trucks, generators, pumps, chainsaws -- at the hourly rates published in the FEMA Schedule of Equipment Rates. Click **+ Add Equipment Entry** to open the form.

FIELD	WHAT IT MEANS
Equipment Type	Required. What it is, such as "Pickup Truck (< 1 Ton)."
Agency Unit ID	Your own identifier for the unit, such as "E-412" or "Unit 7."
FEMA Schedule Code	The code from the FEMA rate schedule (for example 2-7-1).
Date Used	The day the equipment ran on the incident.
Hours Used	How many hours it operated.
FEMA Rate ($/hr)	The eligible hourly rate. Type it, or click **Lookup** to pick from the loaded schedule.
Operator Name	Who ran it, if the operator is tracked separately from labor.
Preview Total	Calculated live -- hours x rate.
Notes	Mileage, fuel, or purpose.

The **Lookup** button opens the **FEMA Equipment Rate Lookup** picker. Type in the search box to filter by equipment name, code, or category; each row shows the equipment, its rate, and its unit. Click **Use** on a row to drop that rate into the form (and the description and code, if those fields are empty). A **Manage rates** link in the picker opens the FEMA Equipment Rates page for editing the schedule itself. The saved equipment table lists Equipment, Unit ID, FEMA Code, Date, Hours, Rate/Hr, Operator, Total, and Notes, each row with a pencil (edit) button.

18.6 Materials and Contracts

This tab captures anything you bought or contracted for the incident. Click **+ Add Material / Contract** to open the form.

FIELD	WHAT IT MEANS
Description	Required. What it is, such as "Barricades, traffic cones, sandbags."
Category	One of Materials, Supplies, Contracts, Rental, or Other.
Vendor / Contractor	Who you bought it from or who did the work.
Purchase / Invoice Date	The date on the receipt or invoice.
Quantity	How many units. Defaults to 1.
Unit	The unit of measure -- each, ton, cubic yard, hour.
Unit Cost ($)	The price per unit.

FIELD	WHAT IT MEANS
Total Cost	Calculated live -- quantity x unit cost.
PO / Contract #	The purchase order or contract number, such as "PO-2026-001."
Notes	Receipt number or justification.

NOTE Attach or keep the purchase order, invoice, and receipt for every material entry. FEMA requires source documentation for each purchase; the note field and PO number here are your pointer back to the paper.

The saved table shows Description, Category (as a colored tag), Vendor, Date, Qty, Unit, Unit Cost, Total, PO #, and Notes, each row with a pencil (edit) button for changes or deletion.

18.7 The Project Worksheet Tab

The Project Worksheet (PW) is FEMA Form FF-104-FY-21-112 -- the form your state emergency management agency uses to package a reimbursement. This tab does not replace the official form; it gathers the header details and rolls up the costs so preparing the real PW is straightforward. Fill in the top fields, and the cost summary below fills itself from the entries you made on the other tabs.

FIELD	WHAT IT MEANS
Applicant Name	The organization claiming the costs.
Disaster / Declaration Number	The FEMA disaster number, in the form FEMA-DR-XXXX-IL.
Incident Name	Auto-filled from the selected incident; editable.
Work Category	One of FEMA's categories A through G (Debris Removal, Emergency Protective Measures, Roads and Bridges, Water Control Facilities, Buildings and Equipment, Utilities, or Parks/Recreational/Other).
Work Description / Scope	A plain-language description of the work performed, the facilities affected, and why it was necessary.
Work Start Date / Work End Date	The span of the work being claimed.

Below the header is the **Eligible Cost Summary** -- Force Account Labor, Force Account Equipment, Materials and Contracts, and the green **Total Eligible Project Cost**. These mirror the totals bar and cannot be typed over; change a cost by editing the underlying entry on its tab. An amber notice at the bottom repeats the rules: this summary does not replace the official form, all costs must go through your state agency with supporting documentation, and costs must be tied to the disaster and not covered by insurance.

18.8 Exporting the Project Worksheet Summary

When you are ready to hand off the numbers, click the **Export PW** button in the page header. FieldCommand builds a plain-text file and downloads it. The file leads with the applicant, disaster number, incident, category, and work period, then the scope of work, then a cost summary, then line-by-line detail for every labor, equipment, and material entry, and closes with a generated timestamp and a reminder to submit through your state agency. Open it in any text editor or spreadsheet, or paste it into the FEMA Grants Portal.

WARN The export reflects whatever is on screen at that moment. If the header fields are blank, the file shows placeholders like "[Enter DR number]" -- fill the Project Worksheet tab first so the export is complete.

18.9 Troubleshooting

SYMPTOM	WHAT TO DO
Tables all say "No entries" and nothing saves	No incident is selected. Choose one from the "Select incident..." drop-down at the top right first; every entry is tied to an incident.
"Import from ICS-214" says "Select an incident first"	Pick the incident in the header drop-down, then click Import again. Import only works on the selected incident's ICS-214 logs.
Imported labor rows show 8 hours and a $0 rate	That is expected -- the import brings in unit leaders as placeholders. Open each row with the pencil button and enter the real hours and pay rates.
An amber banner warns the FEMA rates are outdated	The loaded equipment schedule is over a year old. Click "Review Rates" (or "Manage rates" in the Lookup picker) to update the schedule on the FEMA Equipment Rates page (Chapter 19).
Preview Total or the cost summary stays $0.00	A rate or hours field is blank or zero. Labor needs hours and a rate; equipment needs hours and a FEMA rate; materials need quantity and unit cost.
The exported file shows "[Enter DR number]" or other placeholders	Those header fields were empty at export time. Fill in the Project Worksheet tab (applicant, disaster number, category, scope) and export again.

19 Federal Emergency Management Agency (FEMA) Equipment Rate Schedule http://192.168.50.1/fema_rates.html

The FEMA Equipment Rate Schedule page holds the list of dollar figures FieldCommand uses to put a price on your own equipment when you document the cost of an incident. When the Federal Emergency Management Agency (FEMA) reimburses a disaster, it does not pay whatever you ask for a piece of equipment you already own -- it pays a set published rate for each type of machine, per hour or per day it was used. Those published figures are the **Schedule of Equipment Rates**, and this page is your local copy of them. FieldCommand ships with the standard set of about 44 equipment categories already loaded from the 2025 schedule, so cost documentation works the day you turn the app on. This chapter shows you how to read the rates, search them, correct one, add a rate FEMA does not list, and roll the whole schedule forward when FEMA publishes a new year.

TIP In one sentence: this page is the price list for applicant-owned equipment -- the per-hour or per-day figures the FEMA Force Account Equipment costs are calculated from. It covers equipment only; personnel labor is priced separately.

WARN These rates cover **equipment only**. Labor -- the pay and fringe for the people operating the equipment -- is tracked as a separate cost and is not included in any figure on this page. The page header says so directly: "Labor NOT included."

19.1 Opening the Page and Reading the Header

Open **FEMA Equipment Rates** from the Finance section of the dashboard. The blue **FEMA** badge and the title **FEMA Equipment Rate Schedule** sit across the top, with the reminder line "Eligible reimbursement rates for applicant-owned equipment - Labor NOT included" beneath them. On the right of that header bar are three controls you will use throughout this chapter.

CONTROL	WHAT IT DOES
Update Rate Year (calendar button)	Opens the Update FEMA Rate Year window. Use it once a year, after FEMA publishes a new schedule, to stamp the new year onto every rate (see 19.8).
+ Add Custom Rate	Opens a blank Add Custom Rate window so you can add a piece of equipment the standard schedule does not list (see 19.6).
FEMA Costs (back link)	Returns you to the FEMA Cost Documentation page, where the rates on this page are actually put to work costing an incident.

19.2 The Rate Year and Source Bar

Directly under the header is a thin information bar that tells you which schedule you are looking at and where the official numbers come from.

ITEM	WHAT IT MEANS
Rate Year	The year of the FEMA schedule these figures represent -- for example, 2025. This is the single most important thing to check before an incident, because you must use the rate that was in effect at the time of your disaster declaration.

ITEM	WHAT IT MEANS
Source	A link to the official fema.gov "Schedule of Equipment Rates" page. It opens in a new browser tab and needs an internet connection; it is the master list you check your local copy against.
Labor reminder	A standing note that labor costs are tracked separately and these rates cover equipment only. Always use the rate in effect when your disaster was declared.

19.3 The Stale-Rate Reminder Banner

When the loaded rates are getting old, a yellow reminder banner appears above the information bar. It is the app nudging you to check fema.gov for a newer schedule and update your local copy. If the banner is not showing, your rates are considered current and nothing is wrong. When you see it, do not panic -- the numbers still work; they simply may no longer match the latest FEMA publication. Follow the yearly-update steps in 19.8 to clear it.

19.4 Reading the Rate Table

The main table lists every rate, grouped under a colored category header row (for example, "Generators" or "Vehicles"). Each equipment row has these columns.

COLUMN	WHAT IT MEANS
Code	The FEMA code for that equipment line, such as "6-1-4". Blank shows as a dash. This is the identifier FEMA uses in its own schedule.
Description	The plain description of the equipment, such as "Generator, 10-25 kW". This is what you match your real equipment against.
Unit	How the rate is charged -- hour, day, mile, or each. It tells you what one unit of the rate buys.
Rate	The dollar figure per unit, shown in green (for example, $27.50). This is the number multiplied by usage to price the equipment.
Year	The schedule year this specific rate belongs to. Normally it matches the Rate Year in the bar above; a mismatch flags a rate you updated by hand out of step.
Notes	Any extra note attached to the rate -- a size band, a condition, or a reminder. Often blank.
Edit (pencil button)	The small pencil button at the end of each row. Click it to open that rate in the edit window (see 19.5).

19.5 Searching and Filtering

With dozens of rates loaded, two controls above the table help you find one fast.

1. Type into the **Search equipment...** box to filter the table live. It matches the description, the FEMA code, and the category, so typing "generator", "6-1", or "vehicles" all narrow the list as you type.
2. Use the **All categories** drop-down to show only one category at a time. Choose a category to hide everything else; choose "All categories" again to bring the full list back.
3. The small count next to the drop-down (for example, "12 rates") tells you how many rows match what you have typed and chosen, so you know the filter is working.

TIP Search and category filter work together. If a rate you expect is missing, clear the search box first and set the drop-down back to "All categories" -- one of the two is probably still narrowing the list.

19.6 Editing an Existing Rate

When FEMA changes a figure, or you find one that does not match the official schedule, correct it in place.

1. Find the rate using search or the category filter, then click the **pencil** button at the end of its row. The **Edit Rate** window opens with the current values already filled in.
2. Change the figure in the **Rate ($/unit)** box, and adjust any other field that needs it.
3. Click the green **SAVE** button. The window closes and the table refreshes with the new figure immediately.

The edit window is the same form used to add a rate. Its fields are:

FIELD	WHAT IT MEANS
Description	Required. The plain name of the equipment, such as "Generator, 10-25 kW".
Category	The group the rate is filed under, such as "Generators" or "Vehicles". It sets which colored header row the rate appears beneath.
FEMA Code	The official FEMA code, such as "6-1-4". Optional, but worth filling in so your copy matches the published schedule.
Unit	How the rate is charged -- pick hour, day, mile, or each from the list.
Rate ($/unit)	The dollar figure for one unit. Type numbers only; the app formats it as currency.
Rate Year	The schedule year this figure belongs to. It defaults to the current Rate Year.
Notes	Optional free text -- a size band, a condition, or a reminder about the rate.

The **Unit** drop-down offers four choices:

UNIT	MEANING
hour	The rate is charged for each hour the equipment is used.
day	The rate is charged for each day the equipment is used.
mile	The rate is charged for each mile driven -- used for some vehicles.
each	The rate is a flat charge per item, not tied to time or distance.

19.7 Adding a Custom Rate

The standard schedule does not list every machine a local group owns. When you have a piece of equipment FEMA does not cover with a standard line, add your own rate.

1. Click **+ Add Custom Rate** in the top-right of the header. The **Add Custom Rate** window opens with empty fields.
2. Fill in at least the **Description** -- it is the only required field. Add the **Category**, **FEMA Code**, **Unit**, **Rate**, **Rate Year**, and **Notes** as needed.
3. Click **SAVE**. The new rate is stored locally and appears in the table under its category right away, ready to use in cost documentation.

NOTE If you click SAVE with the Description empty, the app stops you with a "Description required" message. Fill it in and save again.

19.7.1 Deleting a Rate

To remove a rate you no longer want, open it with the pencil button and click the red **Delete** button in the edit window. The app asks "Delete this rate?" first; click OK to confirm. The rate is removed from the active list.

Delete a rate only when you are sure no past incident cost record still depends on it.

19.8 Updating the Rate Year After FEMA Publishes

FEMA publishes a fresh Schedule of Equipment Rates from time to time. When it does, you roll your local copy forward. The year-update tool stamps the new year onto every rate at once -- but it is important to understand what it does and does not change.

1. Click **Update Rate Year** (the calendar button) in the header. The **Update FEMA Rate Year** window opens.
2. Type the new year in the **New Rate Year** box.
3. Click **Update Year Tag**. The app stamps that year onto all existing rates and confirms how many it updated.
4. Then open fema.gov (use the Source link) and correct the individual dollar figures that actually changed, editing each one as in 19.6.

WARN

The Update Year tool changes only the **year label** on your rates -- it does NOT change any dollar amount. FieldCommand cannot download the new figures for you. You still have to compare against fema.gov and edit the individual rates that changed. Update the year first, then the amounts.

19.9 How These Rates Feed Cost Documentation

This page is a reference list; you do not tie rates to a specific incident here. The work of costing an incident happens on the **FEMA Cost Documentation** page (the FEMA Costs back link), where each piece of Force Account Equipment you log is priced against the matching rate from this schedule and multiplied by the hours or days it ran. Those equipment totals then roll up into the incident financial picture on the Cost Tracking Dashboard (Chapter 20). Keeping this schedule accurate and current is what makes those downstream totals defensible when you submit for reimbursement.

19.10 Troubleshooting

SYMPTOM	WHAT TO DO
The table says "Loading rates..." and never fills in	The ICS platform service (port 5055) is not answering. Check that the ics-platform service is running from the Health page, then reload the page.
A yellow reminder banner is showing at the top	Your rates are getting old. Check fema.gov for a newer Schedule of Equipment Rates, then use Update Rate Year and edit the changed figures (see 19.8).
A rate I know exists is not in the table	A search term or the category drop-down is still filtering. Clear the Search box and set the drop-down back to "All categories".
I clicked SAVE and got "Description required"	The Description field was empty. Type a description -- it is the one field every rate must have -- and save again.
I updated the year but the dollar amounts look the same	That is expected. Update Rate Year only changes the year label. Edit each changed figure by hand from the official fema.gov schedule.
The Source link will not open	That link needs internet. On an offline field network it will not load; check it from a device with a WAN connection before the incident.

20 Cost Tracking Dashboard http://192.168.50.1/cost_dashboard.html

The Cost Tracking Dashboard is the incident financial picture on one screen. It reads every Federal Emergency Management Agency (FEMA) Force Account cost entry you have logged and every daily rate you have set on a T-card, adds them all up, and shows the running total, where the money is going, how fast it is being spent, and what the incident is likely to cost if it keeps going at the current pace. It is a **read-only** view -- you do not type costs here; you type them on the FEMA PA Cost Documentation page (fema_costs.html) and on the T-cards, and this dashboard reflects them. The Finance/Administration Section Chief lives on this page during a multi-day incident, and the Incident Commander glances at the big Total Costs number from across the room.

> **TIP**
> In one sentence: enter costs on the FEMA cost page and rates on the T-cards, then watch this dashboard total them, chart them, and project them forward -- automatically, refreshed every two minutes.

20.1 Opening the Dashboard and Choosing an Incident

The page opens showing "Select an incident to view cost dashboard." Nothing is calculated until you pick which incident you want to look at.

1. Open **Cost Dashboard** from the main dashboard.
2. At the top right, open the **Select incident...** drop-down and choose the incident. The dashboard fills in immediately.
3. The incident you pick is remembered, so the next time you open the page it loads that same incident for you.
4. To force an immediate update without waiting for the two-minute cycle, click the **Refresh** button next to the drop-down.

> **NOTE**
> The page reloads its own numbers automatically every two minutes, so a screen left up on a wall monitor stays current on its own. You do not have to keep clicking Refresh.

Two shortcuts sit in the top bar for jumping to where costs are actually entered: the **Edit Costs** button and the **Dashboard** back link. Edit Costs opens the FEMA PA Cost Documentation page; the back link returns you to the main FieldCommand dashboard.

20.2 The Six Summary Tiles

Across the top of the dashboard are six large tiles. Each is sized to be readable from a distance so command staff can see the headline figures without leaning in.

TILE	WHAT IT SHOWS
Total Costs	The grand total of every documented cost on the incident. The small line underneath counts how many resources and how many personnel are on the T-card board (for example, "8 resources - 24 personnel").
Force Acct Labor	The total of all Force Account Labor entries -- your own personnel hours, including any fringe. The small line counts how many labor entries make up the figure.
Force Acct Equipment	The total of all Force Account Equipment entries -- your own owned equipment run at FEMA equipment rates. The small line counts the equipment entries.

TILE	WHAT IT SHOWS
Materials / Contracts	The total of all materials and contracted-cost entries -- purchased supplies and outside contracts. The small line counts the material entries.
Elapsed Time	How long the incident has been open, shown in minutes, hours, or days. This is the clock the burn rate and projections are measured against. Shows a dash if the incident has no open time recorded.
Burn Rate	How fast money is being spent -- the dollars-per-hour figure. The small line restates it as a per-day and per-12-hour-period figure. Shows a dash with "Enter costs above to calculate" until there are both costs and elapsed time.

20.3 The Cost Breakdown Bars

Below the tiles, the **Cost Breakdown** card draws one horizontal bar for each of the three cost categories, so you can see at a glance which category is driving the spend. Each bar is labeled with the exact dollar amount and the percentage of the total it represents.

BAR	WHAT IT REPRESENTS
Force Account Labor	Your personnel labor costs -- usually the largest bar on a staffed incident.
Force Account Equipment	Your owned equipment costs at FEMA equipment rates.
Materials / Contracts	Purchased materials and contracted services.

NOTE The percentages are shares of the current total, so they always add up to roughly 100 percent. If one bar fills most of the width, that is where your money is going -- a fast sanity check that the cost mix looks right for this kind of incident.

20.4 Resources on Incident

The **Resources on Incident** card, on the left of the middle row, groups the T-card resource board by resource type and shows what each type is contributing. The small line at the top reminds you: "T-card resources. Set daily cost to include in burn rate estimate." The **Edit rates** link jumps to the FEMA cost page where daily rates are set.

COLUMN	WHAT IT MEANS
Resource / Type	The resource type name (for example, Crew, Engine, Personnel), with a small line under it counting how many resources of that type are on the board.
Count	How many resources of this type are currently on the T-card board.
Personnel	How many people are assigned across the resources of this type.
Est Daily $	The estimated cost per day for this type, from the daily rates set on the T-cards. Shows a dash when no rate has been set.

When at least one resource is present, a **Totals** line appears at the bottom of the card with the total personnel count and the total estimated cost per day. If the board is empty, the card reads "No resources on T-card board."

TIP A resource shows a dash in the Est Daily $ column until you give it a daily rate on its T-card. Resources with no rate still count toward personnel accountability, but they contribute nothing to the burn rate -- so set rates on the resources you want reflected in the cost projection.

20.5 Cost Projection

The **Cost Projection** card, on the right of the middle row, extrapolates the current burn rate forward to four fixed horizons so you can plan and request budget before you run out. It reads "Based on entered costs and elapsed time. Projections assume current burn rate continues."

HORIZON	WHAT IT ESTIMATES
Next 12hr (1 period)	Projected total after one more 12-hour operational period.
Next 24hr (1 day)	Projected total after one more day.
Next 72hr (3 days)	Projected total after three more days.
Next 7 days	Projected total after a full week.

Each row shows the projected running total, with the added amount for that horizon in parentheses. Until there are both costs and an incident start time, the card reads "Enter costs and incident start time for projections."

WARN These are straight-line estimates -- they assume you keep spending at the same rate you have so far. A demobilizing incident will come in under the projection; a ramping one will run over it. Treat the numbers as planning figures and verify them with the Finance Section Chief before committing to a budget request.

20.6 Budget Tracker

The **Budget Tracker** sits at the bottom of the Cost Projection card. Enter an authorized dollar figure and it shows how much of that budget you have used, in a colored bar, plus how much is left.

1. In the **Budget Limit** box, type the authorized dollar amount -- for example, 50000 for a $50,000 authorization.
2. Click **Set** (or the bar updates as you type).
3. Read the bar: the left label shows the amount spent of the budget, and the right label shows the percentage spent.
4. Read the line under the bar: it shows either "$X remaining" in green, or "$X over budget" in red when spending has passed the limit.

The bar changes color as a warning: **green** below 70 percent, **amber** at 70 percent or more, and **red** at 90 percent or more. Once you cross the limit, the remaining line flips to red and reads "over budget."

20.7 Cost by Operational Period

The **Cost by Op Period** card at the bottom breaks the FEMA Force Account costs out by the operational period each entry was tagged to. This is the layout you want for after-action documentation and for building a FEMA Project Worksheet (PW). It reads "From FEMA force account entries tagged by period."

COLUMN	WHAT IT MEANS
Period	The operational period label -- "Period 1", "Period 2", and so on. Entries with no period tag are grouped under "Unassigned".
Labor	Force Account Labor cost logged in that period, including fringe.
Equipment	Force Account Equipment cost logged in that period.
Materials	Materials and contract cost logged in that period.

COLUMN	WHAT IT MEANS
Total	The row total for the period -- labor plus equipment plus materials.

If nothing is tagged yet, the card reads "No period data -- tag FEMA cost entries with an op period." To populate it, set the operational period on each entry when you record it on the FEMA cost page.

20.8 Where the Numbers Come From

Everything on this dashboard is a reflection of data entered elsewhere. Knowing the source of each figure tells you exactly where to go to correct it.

DASHBOARD FIGURE	ITS SOURCE
Total Costs, category totals, breakdown bars, per-period table	The FEMA PA Cost Documentation page (fema_costs.html) -- the Labor, Equipment, and Materials entries you record there.
Resources on Incident, personnel counts, Est Daily $	The T-card board -- the resources on it, the people assigned to them, and the daily rates set on each card.
Elapsed Time, Burn Rate, projections	Calculated from the incident open time and the totals above.
Budget Tracker	The budget figure you type into this page, compared against the Total Costs.

NOTE The footer states the scope plainly: costs shown are from FEMA Force Account entries only, and all projections are estimates to verify with the Finance Section Chief. If a real cost is missing from the dashboard, it has not been entered on the FEMA cost page yet.

20.9 Troubleshooting

SYMPTOM	WHAT TO DO
The dashboard is blank and says "Select an incident"	No incident is chosen. Open the "Select incident..." drop-down at the top right and pick one.
Total Costs shows $0 but I know we have spent money	Nothing has been recorded yet. Click Edit Costs to open the FEMA PA Cost Documentation page and enter the labor, equipment, and material costs.
Burn Rate and projections show a dash	They need both entered costs and an incident open time. Confirm the incident has a start time recorded and that at least one cost has been entered.
A resource shows a dash under Est Daily $	That T-card has no daily rate set. Open the resource on the T-card board (or click Edit rates) and enter a daily cost.
The Cost by Op Period card says "No period data"	The FEMA cost entries are not tagged to a period. Open each entry on the FEMA cost page and set its operational period.
The numbers look stale	The page refreshes every two minutes on its own. To update now, click the Refresh button; if it still looks wrong, confirm you are on the correct incident.

21 Personnel Accountability http://192.168.50.1/accountability.html

Personnel accountability is a fundamental National Incident Management System (NIMS) and Incident Command System (ICS) safety requirement. At any moment during an incident, the Incident Commander (IC) and the Safety Officer must be able to answer two questions with certainty: **Who is on this incident?** and **Where is each person right now?** Failure to maintain accountability has cost responders their lives at wildland fires, structural collapses, and other incidents where conditions changed fast and people could not be located. The Personnel Accountability page (accountability.html) is the Safety Officer's single screen for keeping both answers current.

> **TIP**
> In one sentence: this is the Safety Officer roll-call board -- it lists everyone on the incident, lets you run a timed head count (a PAR), and flags anyone who is not fully accounted for.

21.1 The Two Levels of Personnel Accountability

FieldCommand tracks people two ways at once, because either one alone leaves a gap. The page pulls both together so the Safety Officer sees a single, cross-checked picture.

LEVEL -- TOOL	WHAT IT TRACKS	MAINTAINED BY
Incident level ICS-211 Check-In List	Who has formally checked into this incident, when they arrived, and whether they have checked out. Every person on the incident must have an ICS-211 entry regardless of agency, role, or resource assignment.	Check-in recorders at entry points; the Resources Unit Leader (RESL) keeps the master list.
Resource level T-Card Personnel	Who is assigned to each specific resource (crew, engine, unit), and therefore where each person is supposed to be right now.	The Operations Section and crew supervisors, who build each T-card personnel list.

> **WARN**
> Both levels are required -- neither alone is enough. The ICS-211 tells you everyone who is on the incident. The T-card list tells you which resource each person is on and where they should be. The Cross-Reference tab compares the two and catches gaps in both directions.

21.2 The Screen at a Glance

The page opens with a red header reading **PERSONNEL ACCOUNTABILITY** and the sub-line "Safety Officer tool - All personnel - All resources". A control bar runs across the top; below it, four tabs switch between the working lists.

The control bar holds these buttons and menus:

CONTROL	WHAT IT DOES
Select incident...	Choose which incident this board is for. Your choice is remembered the next time you open the page.
Period 1	Pick the operational period. Accountability is tracked per period.
Refresh	Reloads the counts and lists from the server right now.

CONTROL	WHAT IT DOES
Conduct PAR	Starts a Personnel Accountability Report -- a timed roll call (see 21.5).
Reset PAR	Clears every PAR confirmation and starts a fresh PAR cycle.
Print	Opens a clean print view of the current list (buttons and tabs are hidden).
Auto 60s	A checkbox that reloads the board automatically every 60 seconds.
Dashboard	Returns to the main dashboard.

The four tabs are:

ICS-211 CHECK-IN LIST -- everyone who has checked into the incident.

T-CARD PERSONNEL -- personnel grouped by the resource they are assigned to.

UNACCOUNTED -- only the people not yet confirmed in the current PAR.

CROSS-REFERENCE -- gaps between the two lists (see 21.9).

21.3 The Summary Tiles and PAR Badge

A strip of six count tiles sits under the header, giving the whole-incident picture at a glance. To their right, the **PAR badge** shows when the last PAR was run and whether everyone was accounted for.

TILE	WHAT IT COUNTS
Total Personnel	Everyone known to this incident and period, checked in or out.
Checked In	People currently checked in (still on the incident).
Checked Out	People who have been checked out and are no longer counted as present.
PAR Confirmed	People confirmed accounted for in the current PAR cycle.
Unaccounted	Checked-in people not yet confirmed in the current PAR. This is the number to watch -- it should reach zero.
On T-Cards	People listed on resource T-cards.

NOTE When a PAR is running and anyone is still unaccounted, a red **UNACCOUNTED PERSONNEL** banner appears near the top with the count. It clears itself the moment the last person is confirmed.

21.4 The ICS-211 Check-In List Tab

This tab lists everyone who has completed ICS-211 check-in for the selected incident and period. (Check-in itself is done on the separate Incident Check-In and camera-scan pages, covered in the check-in chapter.) The list sorts unconfirmed checked-in people to the top during a PAR so the ones needing attention are always in view.

Each person row shows:

ITEM ON THE ROW	WHAT IT MEANS
Name	The person's name.
Callsign / ID badge	A blue tag with their callsign or member ID, when one is on file.
Agency - Position - Resource	The person's agency, ICS position, and resource type, joined with dots.

ITEM ON THE ROW	WHAT IT MEANS
In / Out / PAR times	Check-in time, check-out time (if any), and the time they were last PAR-confirmed.
Location line	The last known location, if one has been entered, shown with a marker.

A colored left edge tells you each row status at a glance: green means PAR-confirmed, red means checked in but not yet confirmed in this PAR, and a faded gray row means the person is checked out. The action buttons on each row are **PAR** (confirm this person), **Location** (record their last known location), and **Check Out** (see 21.10). A **Check All Out** button at the top of this tab checks everyone out at once for end-of-operation demobilization.

21.5 Conducting a Personnel Accountability Report (PAR)

A Personnel Accountability Report is a deliberate, time-stamped roll call confirming that every person is accounted for. Run one at every operational-period change and shift change, whenever a hazard escalates, and immediately whenever anyone cannot be located.

1. Open **Personnel Accountability** from the dashboard and choose the incident and period.
2. Click **Conduct PAR**. The time is stamped, the PAR badge reads "PAR in progress...", and the board switches to the ICS-211 list with unconfirmed people sorted to the top.
3. Contact each supervisor or crew boss by radio and have them confirm every person on their resource.
4. As each person is confirmed, click **PAR** on their row. The counts update immediately and the row turns green.
5. Check the **UNACCOUNTED** tab to see who is still outstanding (see 21.6). Work it down to zero.
6. When everyone is confirmed, the Unaccounted tile reads 0 and the PAR badge shows "All accounted for".
7. To begin the next roll call, click **Reset PAR** and confirm. Every confirmation is cleared and a new cycle begins.

21.6 The Unaccounted Tab

The UNACCOUNTED tab is the Safety Officer working list during a PAR. It shows only the people who are checked in but not yet confirmed in the current cycle, under a red banner telling you to contact their supervisor immediately and to escalate to the Incident Commander if you cannot make contact. Each row carries the same three actions -- **PAR -- Confirmed**, **Check Out**, and **Location** -- so you can clear a person without leaving the tab. When the list empties, it shows "All personnel accounted for".

WARN	If a person cannot be accounted for after you have exhausted radio contact, treat it as a missing-responder emergency: notify the IC, halt operations in the affected area, and start a search. Do not wait for the next scheduled PAR.

21.7 The T-Card Personnel Tab

This tab is resource-level accountability. It lists personnel grouped under the resource they are assigned to -- so instead of just "Jones is on this incident" you see "Jones is on Engine 12". Each resource group has a header showing the resource name, its assignment, and a running PAR count such as **3/4 PAR** -- how many of that crew are confirmed out of the total. Each person row shows name and callsign, ICS position, agency, and contact, plus a **PAR** button and a **Location** button. Confirming people here counts toward the same PAR cycle as the ICS-211 tab, so a large incident can be worked resource by resource.

WARN Adding someone to a T-card does NOT check them into the ICS-211, and checking someone in does NOT put them on a T-card -- the two lists are separate records. Keep both current, and use Cross-Reference (21.9) to catch anyone who is in only one of them.

21.8 Last Known Location

The last known location adds a layer of situational awareness to a PAR: when a supervisor confirms a crew, they can also report where that crew is. Click the **Location** button on any row -- on either the ICS-211 or the T-Card tab -- and the **UPDATE LAST KNOWN LOCATION** box opens. Type the location in the field (the placeholder suggests the style: "Division Alpha - Sector 3 - Base Camp") and click **Save**. The location is stamped with the time and then shows on that person's row with a marker. Click **Cancel** to close without saving.

21.9 Cross-Reference -- Closing the Accountability Gap

The CROSS-REFERENCE tab compares the ICS-211 check-in list against the T-card personnel rosters and flags two specific gaps. A clean cross-reference -- no flags either way -- means both lists are in sync.

CONDITION	WHAT IT MEANS	ACTION REQUIRED
Checked in -- not on any T-card	The person completed ICS-211 check-in but is not on any resource T-card. They may be floating staff, a late arrival, or the T-card was never updated.	Assign them to a resource and add them to its T-card, or confirm they are intentionally unassigned (for example, Safety Officer or PIO) and note their location.
On T-card -- no ICS-211 check-in	The person is on a resource T-card but has no ICS-211 entry. This is a **safety gap**: they are on the incident, possibly in a hazardous area, but not in the formal accountability system.	Direct them to complete ICS-211 check-in immediately. If they cannot be located, escalate to the IC.

NOTE Run the cross-reference at the start of each operational period and any time new personnel arrive. When it is clean, everyone on the incident is both in the formal check-in system and assigned to a resource.

21.10 Checking Personnel Out

Check-out matters as much as check-in. When a person leaves the incident for any reason, check them out so the board reflects who is actually present. People who leave without being checked out inflate the PAR count and can trigger an unnecessary search. To check one person out, click **Check Out** on their row and confirm. The check-out time is stamped, the row fades to gray, and they drop out of the active count -- their record is kept, not deleted. At the end of an operation, use the **Check All Out** button at the top of the ICS-211 tab to check out everyone still shown as present in one step, after confirming the count.

21.11 Troubleshooting

SYMPTOM	WHAT TO DO
The lists are empty and say "Select an incident above"	Choose an incident from the "Select incident..." menu in the control bar. Nothing loads until an incident is selected.
The ICS-211 tab says "No check-ins for this period"	People check in on the Incident Check-In / camera-scan pages, not here. Confirm they checked in, and that you have the right operational period selected.

SYMPTOM	WHAT TO DO
The Unaccounted tab says "Conduct a PAR first"	The Unaccounted list only fills once a PAR is running. Click **Conduct PAR** to start a roll call.
Cross-Reference flags someone whose name is spelled two ways	The match is by name. Fix the spelling so the ICS-211 entry and the T-card entry read the same, then Refresh -- the flag clears.
A checked-out person still shows in the list	That is expected -- checked-out records are retained (grayed out) for documentation. They no longer count toward Checked In or Unaccounted.
The counts look stale during a fast-moving incident	Click **Refresh**, or tick **Auto 60s** so the board reloads itself every minute.

22 ICS-213 General Message http://192.168.50.1/ics213.html

The ICS-213 General Message is the standard Incident Command System (ICS) form for short written messages that pass between sections, agencies, or individuals during an incident. It is the paper (or, here, on-screen) note that carries a request, an order, a status update, or a question from one person to another, with a clean record of who sent it, who it was for, when, and what came back. FieldCommand IMS gives you a fillable ICS-213 that auto-fills a sender name from the Federal Communications Commission (FCC) database, captures a drawn signature, produces a print-ready form, keeps a running log of the messages you have sent, and can hand a completed message to a Winlink radio-email client for delivery over the air.

TIP In one sentence: fill in the boxes, press Generate Form, then print it, save it to the log, or send it over Winlink.

22.1 Opening the Form and What You See

Open **ICS-213** from the dashboard or from the top navigation bar (the link reads **ICS-213**, sitting between **NTS Radiogram** and **ICS-214**). The page opens on the entry form, which is split into three titled cards you fill top to bottom, a row of action buttons, and a **Saved ICS-213 Messages** log at the bottom. A live Coordinated Universal Time (UTC) clock ticks in the top-right corner. The three cards are:

Incident & Routing - who the message is to and from, and the incident, number, date, time, and subject.

Message - the message text itself and the Reply Requested checkbox.

Approved / Received - the approving signature and the reply block the recipient fills in.

22.2 The Incident & Routing Card

This card is where the message is addressed. Every field is optional, but the more you fill in the more complete the finished form is.

FIELD	WHAT IT MEANS
Incident Name	The name of the incident this message belongs to, for example "Operation Winter Storm 2026". It prints in box 1 at the top of the form.
To (Name / Title)	The person or role the message is going to, such as "Operations Section Chief".
To (Position / Agency)	The position or agency of the recipient, such as "Franklin County EMA".
Message Number	A tracking number for this message, such as "ICS-213-001". Auto-fill can generate the next number for you (see 22.5).
From Callsign	Your amateur radio callsign. Type it and press Tab (or click out of the box) and the From name below fills in automatically from the FCC database. It forces itself to capital letters as you type. Leave it blank if you are not a licensed ham - the field is optional.
From (Name / Title)	Your name or role, such as "Communications Unit Leader". Fills itself from the callsign if you left it empty.
From (Position / Agency)	Your position or agency, such as "ARES / RACES".
Date	The message date, such as "Jun 7, 2026". Auto-fill can set this to today.

FIELD	WHAT IT MEANS
Time	The message time, such as "0930 UTC". Auto-fill can set this to now.
Subject	A one-line subject, such as "Communications Status Update".

NOTE The callsign lookup only works when the offline FCC database is installed on the server. If the name does not fill in, type it by hand - the message is not held up by a missing lookup.

22.3 The Message Card

The **Message Text** box is the body of the message - the actual thing you are telling the recipient. Write it clearly and concisely: say who needs to act, what action is needed, and by when. Below the message is one checkbox:

Reply Requested - tick this when the recipient must send an answer back. When ticked, the printed form marks the reply block with "REPLY REQUESTED" in red so it is obvious a reply is expected.

22.4 The Approved / Received Card

This card holds the approval (the sender side) and the reply (the recipient side). All of it is optional.

FIELD	WHAT IT MEANS
Approved By (Name)	The name of the person who approved sending the message, such as "Incident Commander". Prints in box 7.
Approved By (Title / Position)	That person's title or position, such as "Incident Command".
Approved By (Signature)	A signature pad you sign with a mouse, stylus, or finger. Use **Clear Signature** to erase and start over. Optional - leave it blank to sign the printed copy by hand.
Reply (if applicable)	The recipient's written reply. Usually filled in by whoever receives the message, not the sender.
Replied By (Name / Signature)	The name of the person who replied.
Reply Date / Time	When the reply was made.

TIP The signature is captured as a small image and prints on the form. If you sign, draw the whole signature in one continuous view before pressing Generate Form; a blank pad simply prints no signature.

22.5 Auto-fill and Generating the Form

Two buttons sit under the entry cards. **Auto-fill Date/Time** stamps the current UTC date and time into the Date and Time boxes and, if the Message Number is empty, fills in the next number in sequence (for example "ICS-213-004" when three are already saved). **Generate Form** builds the print-ready ICS-213 from everything you typed. To create a message:

1. Fill in the **Incident & Routing**, **Message**, and (if needed) **Approved / Received** cards.
2. Click **Auto-fill Date/Time** to stamp the date, time, and a message number, or type your own.
3. Sign the **Approved By** pad if a signature is wanted.
4. Click **Generate Form**. The entry cards hide and a clean, black-on-white ICS-213 appears, laid out in the standard numbered boxes, and the page scrolls to it.
5. If something is wrong, click **Edit** (the left-arrow button in the preview bar) to go back to the entry cards without losing anything.

22.6 Printing, Saving, and Exporting

Once the form is generated, a bar of buttons appears above it. Each does one thing with the finished message.

BUTTON	WHAT IT DOES
Print Form	Opens your browser's print dialog. Only the ICS-213 itself prints - the navigation bar, entry cards, and log are hidden on paper.
Save to Log	Stores the message in the Saved ICS-213 Messages log on this device and also sends a copy to the server so it becomes part of the incident record.
Export (.txt)	Downloads the message as a plain text file named after the message number, for archiving or pasting elsewhere.
Send via Winlink	Prepares the message as a Winlink text file (see 22.7).
Edit	Returns to the entry cards to make changes.

NOTE A new blank form is one click away at any time: the **New** button (the circular arrow, top-right of the page header) clears every field, wipes the signature, and returns you to an empty entry form.

22.7 Sending via Winlink

Winlink is amateur radio email - a way to move messages when there is no internet. FieldCommand does not connect to the radio itself; instead **Send via Winlink** writes the completed ICS-213 into a ready-to-send text file (named "winlink_" plus the message number) and downloads it. A short pop-up reminds you what to do next.

1. Generate the form, then click **Send via Winlink**.
2. A text file downloads with the To line, Subject line, and full message already filled in.
3. Open that file in your Winlink client - Pat (the Winlink software on the Pi, covered in Chapter 29) or Winlink Express on a laptop.
4. Put the recipient's Winlink address in the To line and send it over the radio from the Winlink client.

WARN FieldCommand only prepares the message; it never transmits. You still need a working Winlink session and a licensed operator to actually put it on the air.

22.8 Receiving a Winlink ICS-213

The flow also runs the other way. When an ICS-213 arrives over Winlink, the Winlink import tool (winlink-import.html) can push it straight into this page: the ICS-213 opens with every field - incident, to, from, number, date, time, subject, message, approval, and reply - already filled from the received message, and jumps directly to the print-ready view so you can read, print, or save it. You do not retype anything.

22.9 The Saved Messages Log

The **Saved ICS-213 Messages** table at the bottom of the page lists every message you have saved with **Save to Log**, newest first. It is your quick history of outgoing traffic on this device. The columns are:

COLUMN	WHAT IT SHOWS
#	The message number, such as "ICS-213-002".
Subject	The subject line of the message.

COLUMN	WHAT IT SHOWS
To	Who the message was addressed to.
From	Who sent it.
Date	The date and time it carried.
(remove)	The X button on the right of each row deletes that one saved message after a confirmation.

The log keeps up to fifty of your most recent messages on this device; the **Clear All** button beside the log heading empties the whole list after a confirmation. Because **Save to Log** also sends a copy to the server, a saved message is preserved in the incident record even if you later clear this local list.

22.10 Troubleshooting

SYMPTOM	WHAT TO DO
I typed my callsign but the From name did not fill in	The offline FCC database may not be installed, or the callsign was mistyped. Type the name in the From (Name / Title) box by hand - the lookup is only a convenience.
I clicked Print and got the whole web page	Print from the generated form, not the entry cards. Click Generate Form first; only then does the print output show just the ICS-213.
My signature did not print	The pad was empty when you generated the form, or you cleared it. Go back with Edit, sign the Approved By pad, then Generate Form again.
Send via Winlink did nothing on the radio	That is expected - it only prepares and downloads a file. Open that file in Pat or Winlink Express and send it from there with an active Winlink session.
A saved message vanished from the log	The local log holds only your fifty most recent, and Clear All empties it. A copy was still sent to the server when you saved it, so the incident record keeps it.
Auto-fill gave a message number I did not want	Just type over the Message Number box with your own value; auto-fill only fills it when the box is empty.

23 ICS-214 Activity Log & ICS-309 Communications Log

http://192.168.50.1/ics214.html

Two of the most-used records on any incident are the ICS-214 Activity Log and the ICS-309 Communications Log. The ICS-214 is a running diary: it records the notable activities and significant events for one unit or resource during an operational period, plus who was assigned and what equipment they used. The ICS-309 is a message log: it records the traffic that passed on a net or station - who called whom, when, and about what. FieldCommand gives you a clean fill-in-the-blanks screen for each, a print-ready federal form, and a way to save the finished record into the incident. The ICS-214 can also push its personnel straight into the Federal Emergency Management Agency (FEMA) labor-cost paperwork, so hours you already logged do not have to be retyped for reimbursement.

TIP In one sentence: the ICS-214 says what your unit did and who was there; the ICS-309 says what messages moved on the radio - and FieldCommand prints both on the official forms.

23.1 When to Use Each Log

The two logs answer different questions. Fill in whichever one - or both - matches the record you need to leave behind.

USE THIS	WHEN YOU NEED TO RECORD
ICS-214 Activity Log	What a unit or section actually did during a shift - the timeline of significant events, decisions, and actions, plus the people and gear assigned to that period.
ICS-309 Communications Log	The message traffic that moved on a net or at a station - each call, the from and to stations, the time, and the subject or message text.

Open either form from the top navigation bar, which carries links for **Dashboard**, **ICS-213**, **ICS-214**, and **ICS-309**. The clock at the right of the bar shows the current time in Coordinated Universal Time (UTC), which is the time base these logs use.

23.2 ICS-214: Incident and Unit Information

The ICS-214 screen is built as four numbered cards. The first card, **1. Incident & Unit Information**, sets the header of the form. Fill in what applies; blank fields simply print empty.

FIELD	WHAT IT MEANS
Incident Name	The name of the incident this log belongs to, such as "Operation Winter Storm 2026".
Operational Period Date From / To	The calendar dates the shift covers. The two boxes combine into the printed "Operational Period".
Time From	The clock time the operational period began, in 24-hour form (for example 0600).
Unit Name / Designator	The unit or resource this log is for, such as "ARES Communications Unit".
Unit Leader Name	The person in charge of the unit for this period, name and callsign if they have one.

FIELD	WHAT IT MEANS
Unit Leader Position	That leader's ICS role, such as "Communications Unit Leader (COML)".

TIP The **Auto-fill Dates** button (a clock icon) drops today's UTC date into the "Date From" box if it is empty and stamps the "Date / Time Prepared" field, so you do not have to look up the date by hand.

23.3 Personnel and Resources Assigned

The next two cards record who worked and what equipment they used - the information the ICS-214 needs, and the information FEMA later wants for cost recovery.

23.3.1 Personnel Assigned to This Period

Card **2. Personnel Assigned to This Period** is a small table. Click **+ Add Person** for each new row; click the red X at the end of a row to remove it. The screen starts you with three blank rows.

COLUMN	WHAT TO ENTER
Name	The person's name (and callsign if licensed).
ICS Position	Their assignment on the incident, such as "COML" or "NET".
Affiliation	The group they belong to, such as "ARES" or "EMA".

23.3.2 Resources Involved

Card **2b. Resources Involved** is optional. Use it to note the equipment, vehicles, and other resources the unit used this period - radios, generators, go-boxes, vehicles. Click **+ Add Resource** for each item.

COLUMN	WHAT TO ENTER
Resource	The item, such as "Generator", "Go-Box 1", or "Vehicle".
Type / Kind	The category, such as "Equipment" or "Vehicle".

NOTE If you leave the Resources card empty, it simply does not appear on the printed form - it prints only when at least one resource is filled in.

23.4 The Activity Log Itself

Card **3. Activity Log** is the heart of the form: a time-stamped list of the notable things that happened. Each entry has a time and a one-line description.

1. To add a blank line, click **+ Add Entry**, then type the time and the activity.
2. To add a line already stamped with the current UTC time, click **+ Add Timestamped Entry** (the clock icon). The cursor jumps to the description box so you can start typing at once. The **Add Entry Now** button at the top of the page does the same thing.
3. Type a short, plain description in the box - for example, "Established contact with EOC on primary net".
4. To delete a line, click the red X at the right of that row.

TIP Times are stamped in UTC and shown with a trailing "Z" (for example 1430Z). Keep every entry in the same time base so the timeline reads correctly later.

23.5 Signing and Generating the ICS-214

Card **4. Prepared By** captures who completed the form. Fill in **Prepared By (Name)**, **Position / Title**, and **Date / Time Prepared**. Below those is a **Signature** pad - sign it with a mouse, stylus, or finger, or click **Clear Signature** to start over. The signature is optional; leave it blank to sign the printed copy by hand.

1. When the form is complete, click **Generate Form** (the page icon). The screen switches to a black-and-white, print-ready ICS 214 that matches the official layout.
2. Review it. To go back and change anything, click **Edit** (the back arrow) to return to the fill-in cards.
3. Click **Print Form** (the printer icon) to send it to a printer or save it as a PDF through your browser's print dialog.
4. Click **Save** to store the ICS-214 in the incident record on the server so it is archived with the rest of the incident.

NOTE

Use **New** (the top-of-page circular-arrow button) to clear the whole form and start a fresh ICS-214. It wipes the fields, the tables, and the signature.

23.6 Export to FEMA Labor

On the print preview there is a **Export to FEMA Labor** button (a money icon). It takes the people you listed in the Personnel card and adds each of them to the FEMA Force Account Labor cost sheet for the active incident, carrying their name, position, affiliation, and the operational-period date. This is the bridge between day-to-day activity logging and the reimbursement paperwork - you do not retype the roster.

1. Make sure the incident you are working is selected on the dashboard, so the labor entries attach to the right incident.
2. Fill in at least one person in the Personnel card - the export refuses to run with an empty roster.
3. Click **Generate Form**, then **Export to FEMA Labor**.
4. A message confirms how many people were exported. Open the FEMA cost page to enter each person's hours and pay rate - only the names and roles cross over; the hours and dollars are entered there.

WARN

If no incident is selected, the people are still added to the labor sheet but without an incident link. The confirmation message warns you when that happens - select the incident first to avoid it.

23.7 ICS-309: Two Ways to Produce It

There are two ways to make an ICS-309 Communications Log, and they suit different situations.

METHOD	HOW IT WORKS
Automatic - from the Net Logger	On the Net Logger (netcontrol.html), the **ICS-309** button (a page icon, titled "Export ICS-309 for currently selected net") builds a Communications Log from the net you have open. It lists the message traffic first (number, time, from station, to station, handling, and message text), then the station check-in list, with the net name, frequency and mode, and net open and close times in the header. It opens a print view you can save or print. A net flagged as a drill prints with a DRILL / EXERCISE watermark.
Manual - the ICS-309 page	The ics309.html screen is a blank ICS-309 you fill in by hand, for traffic a Net Logger did not capture - a single station, an informal exchange, or a paper log you are transcribing.

TIP

If you ran the traffic on a logged net, always prefer the automatic export - it is faster and cannot miss a check-in. Use the manual page only for traffic the net never recorded.

23.8 Filling In the Manual ICS-309

The manual ICS-309 page opens with a **Log Header** card, then a **Log Entries** table. Fill in the header first:

FIELD	WHAT IT MEANS
Incident / Event Name	The incident or event this traffic belongs to.
Operational Period - From / To	The start and end of the shift the log covers, date and time.
Task No.	An optional task or assignment number.
Operator (Name / Callsign)	The operator who kept the log, such as "J. Smith W9XYZ".
Station / Net	The station or net position, such as "Net Control" or "EOC".
Page	The page count, such as "1 of 1". Left blank, it prints as "1 of 1".

23.9 ICS-309 Log Entries and Saving

Each row of the **Log Entries** table is one message. The rows are numbered for you. Add rows with the two buttons below the table.

COLUMN	WHAT TO ENTER
#	The line number, filled in automatically.
Date / Time	When the message passed, in "DD HHMMZ" form (day, then UTC time).
From	The callsign or station the message came from.
To	The callsign or station the message went to.
Subject / Message	A short description of the traffic, or the message text itself.

1. Click **+ Add Entry** for a blank line, or **+ Add (timestamp now)** to add a line already stamped with the current UTC date and time (the cursor jumps to the Subject box).
2. Remove any line with the red X at the right of that row.
3. Click **Generate Form** to see the print-ready ICS-309, then **Print** to print or save it. **Back to Edit** returns you to the table.
4. To file the log with an incident, pick the incident from the drop-down next to **Save to Incident**, then click that button. The log is stored on the server under that incident.

WARN If the server cannot be reached when you click **Save to Incident**, the log is saved locally on the device instead and you are told so. Save it again to the incident once the server is back, so it lands in the permanent archive.

23.10 Troubleshooting

SYMPTOM	WHAT TO DO
"Add at least one person before exporting to FEMA Labor."	The Personnel card is empty. Add at least one row with a name in the ICS-214, then run Export to FEMA Labor again.
FEMA Labor entries are not linked to my incident	No incident was selected when you exported. Select the incident on the dashboard first, then re-export; the entries will attach to it.

SYMPTOM	WHAT TO DO
The Resources section is missing from my printed ICS-214	It prints only when at least one resource is filled in. Add a row in card 2b, then Generate Form again.
My typed times look out of order on the log	Mixed time bases. Keep every entry in UTC (shown with a trailing "Z") - use the timestamp buttons rather than typing local time.
"Server unreachable - saved locally on this device." on the ICS-309	The log is safe on this device but not yet in the incident archive. When the server is reachable, open the form and click Save to Incident again.
The ICS-309 export from the Net Logger is grayed out or empty	No net is selected. Open the net you want on the Net Logger first, then click the ICS-309 button; it builds the log from the selected net.

24 Wide Area Network (WAN) Settings & Dual-Source Internet Configuration http://192.168.50.1/wan_settings.html

FieldCommand IMS is built to run with no internet at all, but when a Wide Area Network (WAN) connection is available it lights up the online-only extras: live weather radar, callsign and map lookups, and off-site backup. The WAN / Internet Settings page lets you tell the system about two internet sources at once - a **preferred** one that is tried first, and a **fallback** that takes over automatically if the preferred source goes down. No carrier names or hardware brands are wired into the system: either source can be a cellular modem, a phone hotspot, a satellite dish, or fixed home broadband. What matters is the role you give it, not the brand. This same page also sets which status cards appear on the dashboard and names the Universal Serial Bus (USB) drive used for automatic backups.

TIP In one sentence: fill in up to two internet sources, mark one Preferred and one Fallback, click Save Settings, and FieldCommand switches between them on its own.

24.1 Opening the Page and What You See

Open **WAN / Internet Settings** from the dashboard (it carries an **ADMIN** badge in the page header). The page is laid out top to bottom as two source cards, a swap button between them, a Dashboard Display box, a USB Backup Drive box, and the Save Settings row. The blue banner at the top restates the rule: the preferred source is tried first, the fallback is used automatically if the preferred one is down, and changes take effect within 30 seconds.

The top card is **Source A** and starts as Preferred (a green stripe down its left edge). The lower card is **Source B** and starts as Fallback (an amber stripe). Each card header shows a type icon, the display name you give it, a colored role badge, and an **Enabled** checkbox.

24.2 The Source Card Fields

Both cards carry the same fields. Fill them in for each internet source you have. You do not have to use both cards - a group with a single hotspot only fills in Source A.

FIELD	WHAT IT MEANS
Enabled (checkbox)	Turns the source on or off without erasing what you typed. A disabled card dims and its badge reads **Disabled**.
Display Name	The plain name shown on the dashboard status bar - for example "Cellular", "Satellite", or "Office Wi-Fi". The card header updates as you type.
Role	A drop-down: **Preferred - use this first** or **Fallback - use if preferred is down**. This is the setting that decides failover order.
Type	A drop-down that picks the icon and category. Choices are listed in 24.3.
Provider / Carrier (optional)	A free-text label for your own reference, such as "T-Mobile", "Starlink", or "Comcast". It has no effect on detection - leave it blank if you like.

FIELD	WHAT IT MEANS
Detection Method	How FieldCommand decides whether this source is up right now. The three methods are explained in 24.4.
Gateway IP to Ping	Appears only when the method is **Ping a gateway IP**. The address to ping.
Admin URL	Appears only when the method is **Modem admin page responds**. The web address of the modem or router admin page.

24.3 Source Type Options

The **Type** drop-down only changes the icon and the category label - it does not change how the source is detected. Pick the closest match.

OPTION	MEANING
Cellular (modem, router, dongle)	A dedicated cellular internet device - a modem, a cellular router, or a USB LTE dongle plugged into the Pi.
Phone / mobile hotspot	A smartphone or tablet sharing its data connection over Wi-Fi or USB.
Satellite (Starlink, HughesNet, ViaSat...)	A satellite internet dish and its indoor router.
Fixed ISP (cable, fiber, DSL, WISP)	Wired home or venue broadband - cable, fiber, Digital Subscriber Line (DSL), or a Wireless Internet Service Provider (WISP).
Other	Anything that does not fit the categories above.

24.4 Detection Methods

The detection method is how the WAN monitor tests whether a source is really working. When you choose a method, a help box under the card fills in with example addresses for common devices. Pick the method that fits your hardware.

METHOD	HOW IT WORKS	BEST FOR
Internet reachable - no device check	Simply tests whether the internet is reachable. Needs no IP address. If both sources use this method, Preferred wins.	Phone hotspots, a USB Long Term Evolution (LTE) dongle, or a single WAN source.
Ping a gateway IP	Pings an address that answers only when this path is active - usually the modem or hotspot gateway. Enter it in the Gateway IP field.	Any source with a known gateway, such as Starlink on 192.168.100.1.
Modem admin page responds	Checks that the modem or router admin web page answers. May also read the carrier name and signal strength. Enter the Admin URL.	Cellular modems and routers that have a local admin web page.

The help box lists real example addresses. Common ones: an Android hotspot gateway is usually **192.168.43.1**, an iPhone hotspot **172.20.10.1**, Starlink **192.168.100.1**, and HughesNet **192.168.0.1**. For a modem admin page, try **http://192.168.1.1** or **http://192.168.0.1**. Use your own device manual if it differs - do not type the examples blindly.

24.5 Setting Up Your Two Sources

1. On **Source A**, tick **Enabled**, type a **Display Name** (for example "Cellular"), and choose the **Type** that matches your device.
2. Set the **Role** to **Preferred - use this first** for the source you want used whenever it is available.
3. Choose a **Detection Method**. If you pick **Ping a gateway IP**, fill in the **Gateway IP to Ping**; if you pick **Modem admin page responds**, fill in the **Admin URL**. Otherwise no address is needed.
4. Repeat on **Source B** for your backup source, setting its **Role** to **Fallback - use if preferred is down**.
5. Click **Save Settings**. A green "Saved - changes active within 30s" message appears; the failover takes effect on the next monitor cycle.

> **NOTE** If you only have one internet source, fill in Source A, leave Source B disabled, and set Source A to the **Internet reachable - no device check** method. That gives you simple up/down status with no address to configure.

24.6 Swapping Preferred and Fallback

Between the two cards is the **Swap preferred and fallback roles between the two sources** button (marked with an up/down arrows icon). Click it to flip the two roles at once - whichever card was Preferred becomes Fallback and vice versa. The card stripes and badges update immediately so you can see the new order before you save. Use it when you want to manually favor, say, satellite over cellular for a while without touching any physical connections. The change is not stored until you click **Save Settings**.

24.7 Dashboard Display

The **Dashboard Display** box controls which status cards appear on the main dashboard. Tick only the cards you want operators to see.

CHECKBOX	WHAT IT DOES
Show Source A card on dashboard	Shows the Source A internet status card on the dashboard. On by default.
Show Source B card on dashboard	Shows the Source B (fallback) status card. Off by default - turn it on when you actually run a second source.
Show AMPRNet / 44Net status card	Shows the amateur-radio 44Net (AMPRNet) gateway status card. On by default.

24.8 USB Backup Drive

The **USB Backup Drive** box names the external drive that triggers an automatic backup when it is plugged in. Any USB drive formatted and labeled **FIELDCOMMAND** (all capitals) starts a backup on insertion - any brand, any size, in ext4 or exFAT format.

FIELD	WHAT IT MEANS
Drive Label (udev trigger)	The volume label the system watches for. Defaults to **FIELDCOMMAND**. The drive itself must carry this exact label for the automatic backup to fire.
Display Name	A friendly name shown in the app for the drive, such as "LaCie Rugged" or "WD Passport". For your reference only.

> **WARN** If you change the trigger label away from FIELDCOMMAND, you must also edit the file **/etc/udev/rules.d/99-fieldcommand-backup.rules** and run **sudo udevadm control --reload-rules** at the command line, or the drive will no longer be recognized. Most groups should leave the label as FIELDCOMMAND.

24.9 Saving and Applying Changes

At the bottom are two buttons. **Save Settings** writes the whole page - both sources, the dashboard choices, and the USB drive names - to the server. A status message confirms the save in green, or shows a red "Save failed" message if the server rejected it. **Reload** (the circular-arrow button) throws away unsaved edits and reloads the last saved values from the server, which is handy if you have changed things and want to start over. After a successful save, allow up to 60 seconds for the WAN monitor to pick up the new configuration and update the dashboard status bar.

24.10 Troubleshooting

SYMPTOM	WHAT TO DO
My changes did not seem to take effect right away	The monitor polls on a cycle. Wait up to 60 seconds after Save; the dashboard status bar updates on the next poll.
The Gateway IP field never appears	That field only shows when the Detection Method is "Ping a gateway IP". Choose that method and the field appears; choose "Modem admin page responds" for the Admin URL field instead.
A source shows down even though the internet works	The detection method may be wrong for that device. If it has no admin page or fixed gateway, switch it to "Internet reachable - no device check". If you ping a gateway, confirm the IP is correct for that device.
Both sources are set to Preferred (or both Fallback)	Roles are independent per card. Set one to Preferred and one to Fallback, or use the Swap button. If both use the internet-only method, Preferred wins.
I plugged in my USB drive but no backup started	The drive must be labeled exactly FIELDCOMMAND in all capitals and be ext4 or exFAT. Relabel the drive, or match the Drive Label field to the drive and update the udev rule as noted in 24.8.
Save failed with a red message	The page could not reach the platform server (port 5055). Confirm you are on 192.168.50.1, the ics-platform service is running, then click Reload and try Save again.

25 National Weather Service (NWS) Animated Radar

http://192.168.50.1/radar.html

The Next Generation Radar (NEXRAD) page is your live weather picture. It plays an animated loop of the last few hours of national radar over a dark map, so you can watch a line of storms move toward the incident area and brief command before it arrives. The page pulls fresh radar from an internet source, so it works only when the server has a Wide Area Network (WAN) connection - cellular, satellite, or site internet. When the internet is gone it keeps the last picture on screen and quietly tries to reconnect. You can open the radar from any of the three dashboard modes.

TIP In one sentence: press Play to watch a radar loop of the incoming weather - it needs internet, and it holds the last frame if the internet drops.

25.1 What You Need to Use It

Radar imagery is downloaded live, so this is one of the few FieldCommand pages that is **not** fully offline. The server must have a working WAN link at the moment you open the page. FieldCommand checks the WAN status for you before it tries to load anything, so you will get a clear message rather than a blank map if the internet is down. The map, the controls, and the timeline all live on one full-screen view; there is nothing to install and no login. The header reads **NEXRAD RADAR** (with a storm-cloud icon) and a reminder that it **Requires internet connection**.

25.2 How the Screen Is Laid Out

The whole page is one big radar map with small control strips floating over it. Knowing where each strip sits makes the rest of this chapter quick to follow.

AREA	WHAT IS THERE
Header (top)	The **NEXRAD RADAR** title, the station drop-down, the **Reflectivity** and **Velocity** view buttons, and a **Dashboard** link back to the home page.
Info bar (over the map, top)	A pulsing loading dot, a status line (for example "NEXRAD CONUS composite"), the frame count, and the three **Palette** color swatches on the right.
Legend (top-right corner)	The **Reflectivity (dBZ)** color scale that translates the on-screen colors into storm intensity.
Control bar (bottom)	The **Play / Pause** button, the step-back and step-forward buttons, the timeline scrubber, the frame time, and the **Speed** buttons.

25.3 The Playback Controls

The control bar along the bottom of the map runs the animation. It behaves like a video player.

CONTROL	WHAT IT DOES
Play / Pause	Starts or stops the loop. The button shows **Pause** while running and **Play** when stopped. The loop repeats from the start automatically.
Step back (left arrow)	Stops the loop and moves back one frame, so you can study one moment of a storm.

CONTROL	WHAT IT DOES
Step forward (right arrow)	Stops the loop and moves forward one frame.
Timeline scrubber	The slider across the middle. Drag it to jump to any point in the loaded loop; the oldest frame is on the left, the newest on the right.
Frame time	Shows the date and time of the frame you are looking at. The newest frame is highlighted in amber so you can tell "now" from history at a glance.
Speed (Slow / Med / Fast)	Sets how fast the loop plays. **Med** is the default; **Fast** is good for a quick trend, **Slow** for a careful look at storm rotation.

25.4 Reflectivity vs. Velocity

Two buttons in the header choose what the radar colors mean. Pick the one that answers the question you have.

VIEW	WHAT IT SHOWS
Reflectivity	The default. Shows how heavy the precipitation is - light rain through heavy rain, hail, and the strongest cores. This is the everyday "where is the rain and how bad is it" view. The dBZ legend applies to this view.
Velocity	Shows the motion of the precipitation toward or away from the radar, which is what forecasters use to spot rotation and possible severe weather. The reflectivity legend hides while this view is selected.

25.5 Reading the Reflectivity Legend

The color bar in the top-right corner is labeled **Reflectivity (dBZ)**. "dBZ" is the radar strength number - the bigger the number, the heavier the precipitation. You do not need to memorize numbers; read the colors.

COLOR	WHAT IT MEANS
Green	Light rain - drizzle to a steady light shower.
Yellow	Moderate rain - a noticeable rain, worth watching.
Red	Heavy rain - a strong cell; expect downpours and reduced visibility.
Purple / White	Extreme returns - the most intense cores, possible hail or a severe storm. Treat these as the priority to brief.

NOTE The legend numbers 5, 20, 35, 50, and 65+ are the dBZ scale. Anything in the red-to-purple end (roughly 50 dBZ and up) is the part of the storm to warn field teams about.

25.6 Color Palettes

The three round swatches on the info bar, next to the word **Palette**, change only the colors used to draw the radar - not the data. Use whichever is easiest to read on your screen and lighting.

PALETTE	BEST FOR
Default	The standard green-to-red scale most people recognize from television weather. The everyday choice.

PALETTE	BEST FOR
Dark	A muted blue-to-orange scheme that is easier on the eyes on a big screen in a dim operations room.
NOAA	The blue-to-magenta National Oceanic and Atmospheric Administration (NOAA) style, familiar to anyone who works from official NWS products.

25.7 Centering on a NEXRAD Station

The drop-down in the header (it starts on **Composite (National)**) is a quick way to jump the map to a part of the country. Choosing a station - for example **KLOT - Chicago, IL** - recenters and zooms the map on the coverage area of that radar. The imagery stays the national composite; this is a "look here" shortcut, not a switch to a single-site radar. Pick **Composite (National)** again to zoom back out to the default view.

TIP You can also just drag the map with your finger or mouse and scroll to zoom, exactly like any other map in FieldCommand. The station list is only there to save you the panning.

25.8 Watching a Storm Move In

The point of the loop is to see where the weather is going, not just where it is. A quick routine:

1. Open **NEXRAD Radar** from the dashboard. It centers on the incident area and begins playing automatically.
2. Use the station drop-down or drag the map so your incident area is in the middle.
3. Watch a full loop at **Med** speed to see the overall direction the storms are traveling.
4. When a heavy (red or purple) cell is heading your way, click **Pause**, then use the step-forward button to walk through the last few frames one at a time and judge how fast it is closing in.
5. Read the amber **frame time** to confirm how recent the newest picture is before you brief command.

25.9 How Fresh the Data Is

FieldCommand reloads new radar frames on its own about every five minutes, so a page left open on a wall display stays current without anyone touching it. The status line on the info bar tells you what is loaded (for example "NEXRAD CONUS composite" from the primary source, or "Using IEM NEXRAD WMS (fallback)" if the main source is briefly down). The frame count shows how many frames are in the current loop. Always trust the **frame time** over your assumptions: it is the timestamp of the actual image on screen.

25.10 When the Internet Is Down

Because radar comes from the internet, FieldCommand handles a lost WAN gracefully instead of showing a broken page. If no WAN link is present when you open the page, a full-screen overlay reads **RADAR UNAVAILABLE OFFLINE** with the note that radar requires an internet connection and a **Check WAN Status** button that jumps you to the WAN status page. If the internet drops **after** radar has already loaded, the last frames stay drawn on the map and the status line changes to something like "Showing last radar (from 2:45 PM) - reconnecting every 30 s". FieldCommand rechecks the connection every thirty seconds and resumes live loading the moment the internet returns - you do not have to reload the page.

WARN A stale radar picture can be dangerous if it is mistaken for live weather. When the status line says "Showing last radar", tell anyone reading the display that the loop is frozen until the internet is back.

25.11 Troubleshooting

SYMPTOM	WHAT TO DO
A full-screen "RADAR UNAVAILABLE OFFLINE" message appears	The server has no internet. Click **Check WAN Status** and restore the cellular, satellite, or site internet link, then return to the radar page.
The map shows storms but the time is old / says "Showing last radar"	The internet dropped after loading. The picture is frozen; FieldCommand reconnects every 30 seconds. Do not brief it as live until the time updates.
The loop is not moving	It may be paused. The button reads **Play** when stopped - click it. If it still will not play, only one frame may have loaded; wait for the next 5-minute refresh.
Status line reads "Using IEM NEXRAD WMS (fallback)"	The primary radar source was briefly unavailable, so FieldCommand switched to the backup. This is normal - the radar is still current; no action needed.
The colors are hard to read on the big screen	Try a different **Palette** swatch (Default, Dark, or NOAA). Dark is usually easiest in a dim operations room.
The map is centered on the wrong part of the country	Pick your area from the station drop-down, or drag the map and scroll to zoom. Choose **Composite (National)** to zoom back to the default view.

26 High Frequency (HF) Propagation Tool http://192.168.50.1/propagation.html

The High Frequency (HF) Propagation tool tells you, at a glance, which amateur radio bands are likely to work right now and which are dead. HF is the part of the radio spectrum that bounces signals off the upper atmosphere to reach across a county, a state, or the country when nothing local is left standing. Whether a band carries or fails changes hour by hour with the Sun. This page pulls the current solar numbers from an online space-weather source, turns them into plain "good / fair / poor" band ratings, and draws a set of reference charts so a net control operator or a Communications Unit Leader can pick a working frequency without guessing.

TIP In one sentence: open the page, read the color of each band, and steer your HF traffic to the green ones.

26.1 What This Tool Shows and What It Needs

The page has one job: convert space-weather data into a usable band plan. It fetches live solar indices from HamQSL (the well-known N0NBH feed at hamqsl.com) every 15 minutes and, from those numbers, estimates how each band is behaving by day and by night. It is a **read-only reference** — there is nothing to fill in, save, or submit. Nothing here is stored with the incident; it is a live look at the sky.

This tool is one of the few features in FieldCommand IMS that **works better with internet.** The solar numbers come from the internet. When the deployment has a Wide Area Network (WAN) connection, the page shows real, current data. When the deployment is fully offline, the page cannot fetch the Sun's numbers and instead shows a model-based estimate and a plain "Offline" notice. The estimate is still useful as a rough guide, but it is not live.

26.2 Reading the Page: Header and Refresh Bar

At the very top is the page header. On the left is a **Home** link back to the dashboard. In the center is the title **HF PROPAGATION**. On the right is a status line that reads **Fetching data...** while the page loads, then changes to **Updated** with a timestamp when live data arrives, or **Offline - internet unavailable** when it cannot reach the source.

Just below is the **refresh bar**. It carries a one-line reminder that solar data updates every 15 minutes when internet is available and that band conditions are model-estimated from the solar indices. On the right of that bar is a **Refresh** button (a circular-arrow icon). The page refreshes itself automatically every 15 minutes; press **Refresh** only when you want the newest numbers immediately.

26.3 The Solar Indices Strip

The first block of content, under the **Solar Indices** heading, is a strip of seven cards. Each shows one space-weather number with its full name underneath. These are the raw inputs; everything else on the page is derived from them.

CARD	WHAT IT MEANS
SFI (Solar Flux)	Solar Flux Index. The overall strength of the Sun's radio energy. Higher is better for HF. Above 130 the high bands (20m and up) open; below 70 you are limited to the low bands. The number turns green when high, amber mid, red low.

CARD	WHAT IT MEANS
SN (Sunspot #)	The sunspot count. More sunspots generally means stronger high-band propagation. It moves with the SFI.
A-Index (Geomagnetic)	A daily measure of how disturbed the Earth's magnetic field has been. Low is calm and good; a high A-index means bands have been rough. Green under 10, amber to 30, red above.
K-Index (3-hr Planetary)	The most immediately useful number. A right-now, 0-to-9 measure of geomagnetic activity updated every three hours. 0-2 is excellent, 3-4 is fair, 5 or more signals a storm that degrades HF. Green, amber, or red accordingly.
X-Ray (Flux Class)	The current solar X-ray level, shown as a letter class (A, B, C, M, or X). A class M or X flare shown in red or amber can cause a sudden radio blackout on the daylight side of the Earth.
Proton (Event)	Whether a solar proton event is in progress. "None" (green) is normal; a flagged event (red) can knock out paths that cross the polar regions.
Signal (Noise Level)	The reported background radio noise level. Higher noise makes weak signals harder to copy.

NOTE When the page is offline, all seven cards read **N/A** instead of a number. That is your cue that the ratings below are the built-in estimate, not live data.

26.4 The K-Index Scale

Below the strip is a panel titled **K-Index Scale (Geomagnetic Activity)**. It draws ten cells labeled K0 through K9. The cells up to the current K value are filled in with color that runs from green at the calm end to deep red at the stormy end, and the current level is outlined in white. A one-line legend under the cells spells out the bands: **K0-K2** Quiet (good HF), **K3-K4** Unsettled, **K5-K7** Storm (degraded HF), and **K8-K9** Severe storm (HF blackout possible). This is the quickest single check on the page: if the white-outlined cell is in the green, HF is worth trying.

26.5 Band Conditions (Day/Night Model)

The **Band Conditions (Day/Night Model)** section is a grid of cards, one per band group. Each card shows the band name, its frequency span, and two colored bars - a **Day** bar and a **Night** bar - each labeled with a word rating. The bar's length and color both encode the rating, so you can read the card from across a room.

BAND GROUP	FREQUENCY SPAN AND TYPICAL EMCOMM USE
80m-40m	3.5 to 7.3 MHz. The regional and national workhorses. 80m carries local and regional traffic at night; 40m is the backbone band, reliable day and night out to a few hundred miles and beyond.
30m-20m	10.1 to 14.35 MHz. Longer-haul daytime bands. 20m reaches across the country and farther when the Sun is up.
17m-15m	18.1 to 21.45 MHz. High bands that open only when solar flux is strong; good for long distances by day, usually closed at night.
10m	28 to 29.7 MHz. Opens only under high solar flux. When it is open it is excellent; most of the time it is closed.

The word ratings and their colors mean:

RATING	WHAT IT TELLS YOU
Good (green)	The band is open and reliable. Use it.
Fair (amber)	The band is workable but marginal; expect fading and weaker signals.
Poor (orange-red)	The band is mostly unusable; try only if nothing else works.
Closed or a dash (gray)	The band is effectively dead for that time of day.

26.6 MUF / LUF Estimates

In the left panel of the two-column row is **MUF / LUF Estimates**, labeled for the deployment's reference location (shown as Columbus OH, grid EN90, in the shipped build). Maximum Usable Frequency (MUF) is the highest frequency that will still bounce back to you for a given distance; Lowest Usable Frequency (LUF) is the lowest that will get through. You want to work between the two. The table has one row per path distance.

COLUMN	WHAT IT MEANS
Path	The distance category - Local (0 to 500 km), Regional (500 to 2000 km), DX (2000 to 5000 km), or Long Path (over 5000 km).
LUF (MHz)	The lowest frequency that will complete that path. Below this, signals are absorbed and never return.
MUF (MHz)	The highest frequency that will still bounce back for that path. Above this, signals punch through and are lost to space.
Best Band	The recommended amateur band for that path right now, picked from the estimated MUF.
Reliability	A rough percentage confidence, colored green (strong), amber (marginal), or red (weak).

A small note under the table reminds you these are estimates from a simplified ionospheric model - confirm with propagation beacons or an actual on-air call before you rely on a path for critical traffic.

26.7 Ionosphere Layers Diagram

The right panel, **Ionosphere Layers**, is a drawn diagram, not a control. It shows the atmospheric layers that reflect HF signals, stacked by altitude above the Earth's surface, with a sketched skip path arcing off the top layer. It is here to explain **why** the bands behave as they do.

LAYER	ROLE IN HF PROPAGATION
F2 (~300 km)	The main reflecting layer for long-distance HF, roughly 14 to 30 MHz. It is what makes coast-to-coast contacts possible.
F1 (~200 km)	A daytime-only layer reflecting roughly 10 to 20 MHz.
E (~110 km)	A daytime layer; also the source of short-lived "sporadic-E" openings.
D (~70 km)	The daytime absorber. It soaks up the low bands during the day, which is why 80m works at night but not at noon.

26.8 24-Hour Band Activity Guide

Below the two panels is the **24-Hour Band Activity Guide**, a timeline chart. The rows are the major bands (80m, 40m, 20m, 15m, 10m); the horizontal axis is the hour of the day in Coordinated Universal Time (UTC), 00 through 24. A colored block marks the hours each band is typically open, and a white dashed vertical line marked **NOW** shows the current UTC time so you can read straight down it to see what is open this minute. The chart scrolls sideways on a narrow screen. As with the other estimates, the openings shift with season and solar activity - the guide is typical, not guaranteed.

26.9 EmComm Propagation Quick Reference

The last panel, **EmComm Propagation Quick Reference**, is a printed cheat sheet that does not change with the data. On the left it lists each band from 160m through 10m with a plain day-use and night-use rating. On the right it lists the space-weather conditions (SFI ranges, K-index thresholds, X-ray flares, proton events, gray line) and what each one does to HF. A closing note states the emergency-communications priorities: 40m is the backbone for regional-to-national paths day and night, 80m covers local and regional at night, and the 60m channels support interoperability with served agencies.

26.10 Using It During an Incident

A net control operator or Communications Unit Leader can plan an HF net in under a minute:

1. Open **HF Propagation** from the dashboard and confirm the top-right status reads **Updated** with a recent time. If it reads **Offline**, treat the ratings as a rough model estimate only.
2. Glance at the **K-Index Scale**. If the current cell is in the green (K0-K2), HF is worth using; if it is red (K5 or higher), expect degraded or failed paths.
3. Read the **Band Conditions** cards for the distance you need to cover, using the Day or Night bar that matches the current time.
4. Cross-check the **MUF / LUF Estimates** row for your path distance and note the **Best Band** it recommends.
5. Pick the band your radio and license privileges support that is rated best, set up the net on it, and confirm with an actual on-air check before committing traffic.

WARN Amateur HF transmission requires a properly licensed operator with privileges on the band and mode in use. This tool tells you which bands are open; it does not grant anyone the authority to transmit on them.

26.11 Troubleshooting

SYMPTOM	WHAT TO DO
Every card reads N/A and the top says "Offline"	The deployment has no internet, so the live solar feed cannot load. The band ratings shown are a built-in model estimate. Connect a WAN if you need live data, or use the estimate as a rough guide.
The status stays on "Fetching data..." and never updates	The source (hamqsl.com) may be slow or unreachable. Wait a moment, then press the Refresh button. If it still hangs, the internet path is down - treat it as offline.
The numbers look old	The page refreshes on its own every 15 minutes. To force the newest data now, press the Refresh button in the refresh bar.
All bands show Poor even though it is daytime	Check the K-Index. A geomagnetic storm (K5 or higher) forces the model to rate bands Poor regardless of the hour. Wait for the storm to pass or try 40m/80m.
The 24-hour chart is cut off on the side	On a narrow screen the timeline scrolls. Swipe or drag it sideways, or view the page on a wider display, to see the full 00-to-24 UTC span.
The recommended band is one my radio cannot use	The tool assumes full HF access. Choose the next-best band your equipment and license privileges actually support from the Band Conditions cards.

27 Winlink Radio Email http://192.168.50.1/winlink-import.html

Winlink is the worldwide amateur radio email network. It carries email-style messages, including standard Incident Command System (ICS) forms, over radio when the internet is down. FieldCommand IMS does not send or receive Winlink itself - you run a separate Winlink program (Winlink Express on a Windows laptop, or Pat on the Raspberry Pi) connected to a radio. What FieldCommand gives you is the **Winlink Form Import** page: it takes the ICS form that arrived (or that you sent) in Winlink and files it into the permanent incident record on the server, so the traffic that came in over the air becomes part of the same archive as everything else the incident produced. This chapter is about that import page and how to feed it.

TIP

In one sentence: get the ICS form out of your Winlink program as a file or as copied text, drop it on the Winlink Form Import page, check the fields, tag it to the incident, and save it into the record.

27.1 What This Page Is For

Open **Winlink Form Import** from the dashboard. The banner at the top reads **WINLINK FORM IMPORT**, with the line "Bring Winlink Express ICS form data into the incident record on the server." The page is a simple three-step column: **Step 1** - provide the form data, **Step 2** - review and correct the extracted data, and **Step 3** - tag to incident and file. Steps 2 and 3 stay hidden until you have parsed a form in Step 1, so at first you only see Step 1.

The page understands three ICS forms it can fully re-render: the **ICS-213 General Message**, the **ICS-214 Activity Log**, and the **ICS-309 Communications Log**. Any other Winlink form is still captured and archived - you just cannot open it back into a printable ICS page.

27.2 Getting the Form Out of Winlink

Before you touch FieldCommand, you need the form data out of your Winlink program. You have two ways to hand it over, and both work for messages you received and messages you sent.

1. In Winlink Express, open the message that carries the ICS form.
2. Either **save the attached file** - it is named like RMS_Express_Form_*.xml; right-click the attachment and choose **Save** - **or** select and **copy the message text**.
3. Keep that file handy (or keep the text on the clipboard) for the next step.

NOTE

The saved **XML** file is the better choice when you have it: it carries the form field-by-field, so FieldCommand fills every box cleanly. Copied message text still works, but the page has to guess the fields from the layout, so check the result more carefully.

27.3 Step 1 - Provide the Form Data

Step 1 has a dashed **drop box** that reads "Drop an RMS_Express_Form_*.xml file here, or click to browse," a line that says "- or paste below -," and a large paste box under it.

1. To use the saved file: drag it onto the drop box, or click the drop box to open a file picker and choose it. The page reads the file and parses it right away.
2. To use copied text instead: click in the paste box (its placeholder reads "Paste the form XML or the Winlink message text here...") and paste, then click **Parse Form Data**.

3. If you picked the wrong thing, click **Clear** to empty the box and start over.

The moment a form is recognized, Step 2 and Step 3 appear below.

27.4 Step 2 - Review and Correct the Data

Step 2 is where you make sure the machine read the form correctly before it goes into the permanent record. At the top is a colored **detection banner** that tells you what kind of form the page thinks it is:

BANNER COLOR	WHAT IT MEANS
Green (recognized)	The page identified the form - "Detected an ICS-213 General Message form," or the ICS-214 or ICS-309 equivalent. The fields below are filled in and ready to check.
Amber (not recognized)	The form type could not be identified automatically. The data is still captured and can be archived, but re-rendering back into a printable ICS page works only for the ICS-213, ICS-214, and ICS-309.
Red (nothing found)	No form data was found at all. You probably pasted the wrong thing - go back to Step 1 with the RMS_Express_Form XML or the actual form message.

Under the banner is a grid of the form fields, each already filled in with what the page extracted. Every box is editable - correct anything that came across wrong or blank, exactly as you would fix a typo, before you file it. The long fields (the message body, activities, reply, and log) are shown as multi-line boxes.

27.5 The Three Recognized Forms and Their Fields

Which fields appear in the review grid depends on the form the page detected. The tables below list them with a plain-language meaning.

27.5.1 ICS-213 General Message

FIELD	WHAT IT MEANS
Incident	The incident this message belongs to.
To (Name) / To (Position)	Who the message is addressed to, and their ICS role.
From (Name) / From (Position)	Who sent it, and their ICS role.
Message #	The message number the sender assigned.
Date / Time	When the message was written.
Subject	The one-line subject of the message.
Message Text	The body - the actual message.
Approved By / Approver Position	Who released the message, and their role.
Reply / Replied By / Reply Date/Time	The reply half of the ICS-213, if the form carried one.

27.5.2 ICS-214 Activity Log

FIELD	WHAT IT MEANS
Incident	The incident this log belongs to.

FIELD	WHAT IT MEANS
Unit Name	The unit or position the log covers.
Unit Leader / Leader Position	Who led the unit, and their title.
Op Period From / To	The operational period the log covers.
Time From / Time To	The start and end clock times within that period.
Prepared By / Prep Position	Who filled the log out, and their role.
Activities	The chronological list of what happened.
Personnel	The people assigned to the unit.

27.5.3 ICS-309 Communications Log

FIELD	WHAT IT MEANS
Incident	The incident this radio log belongs to.
Op Period From / To	The operational period the log covers.
Task No.	The task or assignment number, if used.
Operator	The radio operator who kept the log.
Station / Net	The station or net name the log came from.
Page	The page number.
Log Entries (raw)	The block of log lines. When you open this into the ICS-309 page, FieldCommand splits it into date/time, from, to, and subject columns automatically.

27.6 Fields Not Automatically Mapped

Winlink forms vary by version, so some fields may not line up with a known ICS box. When that happens, an amber panel titled **Fields not automatically mapped** appears under the grid, listing each leftover field and its value. These are **not lost** - the page keeps them and stores them with the archived record. Read the list in case one of them really belongs in a box above; if so, copy the value up into the correct field by hand before you file. Pure Winlink plumbing (version stamps and the like) is filtered out and never shown here.

27.7 Step 3 - Tag to Incident and File

Step 3 decides where the form lands and what you do with it. It has two drop-down menus and two action buttons.

CONTROL	WHAT IT DOES
Incident	Choose which incident this form is filed under. The list is pulled live from the ICS platform. If there is no active incident, it files to the general archive instead.
Direction	Mark the traffic as **Received (incoming traffic)** or **Sent (outgoing traffic)**, so the record shows which way the message went.
Open in ICS form (print)	Opens the form back into its matching ICS page (ICS-213, ICS-214, or ICS-309) in a new tab, filled in and ready to print. Available for those three forms only.

CONTROL	WHAT IT DOES
Archive to incident	Saves the reviewed form - fields, unmapped extras, direction, and the original raw text - into the permanent incident record on the server.

After you click **Archive to incident**, a line under the buttons confirms it was filed and names the form type and direction. Archiving is the step that makes the message a durable part of the incident record - do it even when you also open the form to print, because printing alone does not save anything to the server.

TIP

The two buttons do different jobs. **Open in ICS form** gives you a printable page. **Archive to incident** writes it into the record. For traffic you want to keep - which, on an incident, is all of it - always archive.

27.8 If the Server Cannot Be Reached

If you click **Archive to incident** and the server is unreachable, the page does not throw the form away. It saves a local copy on the device you are using and shows an amber line telling you the server was unreachable and to re-archive when it is back. When the server returns, open the form again and archive it so the permanent record on the server is complete - the local copy is only a safety net on that one device, not the archive.

27.9 Winlink Client Options (Background)

FieldCommand imports the form; a separate Winlink program does the sending and receiving over radio. The common choices, for reference:

CLIENT	RUNS ON	TYPICAL PATHS
Winlink Express	Windows laptop	VARA HF, VARA FM, Telnet, Pactor
Pat	Raspberry Pi / Linux	VARA FM (via Wine), Telnet, packet (AX.25)

Whichever client you use, the import step is the same: get the ICS form out as a file or copied text, then bring it into this page.

27.10 Troubleshooting

SYMPTOM	WHAT TO DO
Red banner - "Could not find any form data"	You pasted the wrong text. Go back to Winlink, save the RMS_Express_Form_*.xml attachment (or copy the actual form message), and try Step 1 again.
Amber banner - form type not recognized	The form is not one of the three the page re-renders. You can still Archive it - the data is captured - but the "Open in ICS form" button will not work for it.
Fields came across blank or wrong	Use the saved XML file rather than copied message text; if you only have text, fix each box by hand in Step 2 before archiving.
An amber "Fields not automatically mapped" panel appeared	That is normal for some form versions. The data is preserved with the archive; copy anything that belongs in a real field up into the grid before you file.
Incident menu shows "no active incident"	The ICS platform has no incident running, or is unreachable. The form files to the general archive; start or open an incident first if you want it tagged there.
Amber line - "Server unreachable - saved a local copy"	Nothing is lost, but it is only on this device. When the server is back, open the form again and click Archive to incident so the server record is complete.

28 Amateur Packet Radio Network (AMPRNet) / 44Net Gateway http://192.168.50.1/amprgate.html

WARN

This chapter applies only to deployments that include an amateur radio group as the lead or a key partner. If your deployment is operated entirely by a public safety agency, municipality, or served organization without a licensed amateur radio group involved, the AMPRNet gateway does not apply to you -- skip this chapter. FieldCommand IMS runs fully and completely without AMPRNet. All tools, ICS forms, net loggers, FEMA documentation, personnel accountability, and every other feature operate on EMCOMM-NET with no dependency on AMPRNet whatsoever.

The AMPRNet / 44Net Gateway page is a live status board for a completely separate piece of hardware: a second, dedicated gateway Raspberry Pi at address 192.168.50.2 that carries amateur radio internet traffic. AMPRNet -- the Amateur Packet Radio Network -- is the global amateur radio Internet Protocol (IP) network operating on the 44.0.0.0/8 address block permanently assigned by the Internet Assigned Numbers Authority (IANA) to Amateur Radio Digital Communications (ARDC) for amateur radio use. The ARDC gateway (**amprgw.ampr.org:51820**) provides a WireGuard-encrypted tunnel into that network over the internet. When the gateway Pi is configured and the tunnel is up, every device on EMCOMM-NET can reach AMPRNet resources -- Winlink gateways, APRS Internet Service (APRS-IS) servers, and other amateur stations worldwide that live on 44.x.x.x addresses. This page is where you watch that tunnel, read its traffic counters, see who is connected, and (only from the gateway Pi keyboard) bring the tunnel up or down.

TIP

In one sentence: this page tells you at a glance whether the amateur radio internet gateway is up, how much data it is moving, and who is connected -- and it is the only place, from the gateway Pi itself, to control that tunnel.

The gateway Pi is completely isolated from the primary FieldCommand server. If it goes down or is never deployed, nothing else on EMCOMM-NET is affected -- the dashboard, ICS forms, and every other tool keep working normally.

28.1 Who Should Deploy AMPRNet?

SITUATION	DEPLOY AMPRNET?
ARES/RACES group leads or co-leads the deployment	Yes -- Winlink via the AMPRNet path, APRS-IS, and inter-node data. Licensed operators handle the registration under their callsign.
Amateur radio club operates the EOC comms section	Yes -- same benefits. A club callsign (for example, W9XYZ) is preferable for organizational deployments.
Licensed amateurs are supporting partners with the served agency	Consider -- only if the licensed operators take full ownership of the AMPRNet registration and its ongoing maintenance.
Public safety agency only -- no amateur radio group involved	Not applicable -- an FCC amateur license is required for registration. AMPRNet is a Part 97 resource; it cannot be registered or operated by a non-licensed agency.

28.2 The Amateur Radio Group Must Lead This

AMPRNet IP addresses are assigned to a licensed amateur callsign and governed by Part 97 of the Federal Communications Commission (FCC) rules. The registration process, the ongoing maintenance, and all operational use must be led by the amateur radio group -- not the served agency, not the EOC information technology (IT) department, and not the municipality. The practical steps are:

STEP	WHO
Designate a licensed technical lead (any FCC license class)	ARES EC or club President
Contact the regional AMPRNet coordinator before registering	Technical lead
Register at portal.ampr.org using the club or personal callsign	Technical lead
Request a /29 subnet (6 usable IPs) or a /28 (14 IPs)	Technical lead -- allow 2 to 6 weeks for approval
Download the WireGuard config and set up the gateway Pi	Technical lead -- see Installation Guide Step 11
Maintain the portal account and license going forward	Technical lead plus a designated backup

28.3 What AMPRNet Enables

CAPABILITY	DESCRIPTION
Winlink via AMPRNet path	Reach Winlink Radio Message Server (RMS) gateways at 44.x.x.x addresses without using the commercial internet -- keeps message handling within the amateur network.
APRS-IS via AMPRNet	APRS-IS servers are reachable on 44.x.x.x. Direwolf can use the AMPRNet path instead of the public internet.
Inter-node FieldCommand	Two FieldCommand deployments, each with a 44Net gateway, can share net log data and resource status over AMPRNet -- no commercial internet required.
Global amateur station reach	Any amateur station worldwide with a 44.x.x.x address is directly reachable from any EMCOMM-NET device.
Permanent static IPs	Your 44.x.x.x block is yours permanently -- fixed addresses that never change regardless of internet provider or location.

NOTE	**Part 97 applies to all AMPRNet traffic:** no encryption of message content (the WireGuard tunnel encryption of the transport layer is permitted), no commercial traffic, and station identification is required. All use must comply with Part 97 rules.

28.4 Opening the Page and Reading the Hero Banner

Open the gateway page from the dashboard, or type http://192.168.50.1/amprgate.html into any browser on EMCOMM-NET. The header reads **AMPRNet / 44Net Gateway**; the **Dashboard** link at the top right returns you to the main menu. The page fetches status the moment it loads and again every 30 seconds on its own -- the line **Last polled** at the top right tells you when the figures were last refreshed.

The large colored banner near the top -- the hero banner -- is the one thing to read first. Its color and words tell you the tunnel state at a glance:

BANNER	WHAT IT MEANS
Green -- TUNNEL UP, AMPRNet Connected	The WireGuard tunnel is established. The banner also shows your assigned AMPRNet address. AMPRNet resources are reachable from EMCOMM-NET.
Red -- TUNNEL DOWN	The gateway Pi is responding but the tunnel is not established. AMPRNet is not reachable. Bring it up from the gateway Pi keyboard (28.8).
Gray -- Gateway Unreachable	The gateway Pi at 192.168.50.2 is not responding at all. Check that it is powered on and connected. The banner reads "not responding -- is it powered on?"

28.5 The Status Cards

Below the banner is a grid of small cards, each a single live figure read from the gateway Pi. A dash (-) means the value is not available yet -- usually because the tunnel is down or the gateway is unreachable.

CARD	WHAT IT SHOWS
AMPRNet Address	The 44.x.x.x/29 address assigned to your gateway. Blank until the tunnel comes up and an address is assigned.
Last Handshake	When the WireGuard peer was last in contact. A recent time means the tunnel is alive and passing keep-alives.
Data Received	Total bytes received through the AMPRNet tunnel, shown in B, KB, MB, or GB.
Data Sent	Total bytes sent through the AMPRNet tunnel.
Gateway CPU Temp	The gateway Pi processor temperature in degrees Celsius -- a health check on the gateway hardware.
Gateway Memory	Memory used versus total on the gateway Pi, in megabytes (MB).
IP Forwarding	Shows Enabled (green) or DISABLED (red). This must be Enabled for the gateway to route 44Net traffic; DISABLED in red means routing is broken even if the tunnel is up.
Gateway Uptime	How long the gateway Pi has been running since its last reboot.

WARN

If **IP Forwarding** shows **DISABLED** in red while the tunnel is UP, devices on EMCOMM-NET still will not reach 44Net. Forwarding is set on the gateway Pi during Installation Guide Step 11; have the technical lead check it there.

28.6 Active Routes and Connected Peers

Two tables in the middle of the page show the plumbing behind the tunnel.

28.6.1 Active Routes

The **Active Routes** table lists the network routes the gateway is advertising. A green dot and **Active** means the route is in place; a red dot and **Not routed** means it is missing. If the table reads "No 44Net routes found," the tunnel is most likely down.

COLUMN	WHAT IT MEANS
NETWORK	The destination network being routed, such as the 44.0.0.0/8 block.
VIA	The interface or next hop the traffic travels through.

COLUMN	WHAT IT MEANS
STATUS	Green dot with Active, or red dot with Not routed.

28.6.2 Connected Peers

The **Connected Peers** table lists the WireGuard peers. When the tunnel is down or no peers are configured, it reads "No peers configured."

COLUMN	WHAT IT MEANS
PEER	The peer public key that identifies the far end.
ENDPOINT	The internet address and port of the peer.
ALLOWED IPs	The IP ranges this peer is allowed to carry.
LAST HANDSHAKE	When this peer was last in contact.
STATUS	Green dot with Connected, or red dot with Idle.

28.7 The Access Log and Part 97 Access Control

The **Access Log** section shows the recent gateway logins and station identifications, kept for Part 97 record-keeping. The gateway enforces access control: only a licensed amateur operator with a valid FCC callsign may authenticate, and each callsign is validated against the offline FCC database. Every access attempt is recorded -- callsign, IP address, timestamp, and result -- and a login session lasts 8 hours before it must be renewed. When there is nothing to show, the section reads "No access-log entries yet."

28.8 Tunnel Control (Gateway Pi Only)

The **Tunnel Control** section holds the buttons that start and stop the tunnel. By deliberate design, these buttons only work when this page is open in the Chromium browser on the gateway Pi itself -- never from an operator laptop or phone on EMCOMM-NET. A valid callsign login is also required. This keeps a single, physically present, licensed operator in control of the amateur radio link.

BUTTON	WHAT IT DOES
Bring Tunnel UP	Establishes the WireGuard tunnel to AMPRNet.
Bring Tunnel DOWN	Tears the tunnel down and stops carrying 44Net traffic.
Restart Tunnel	Brings the tunnel down and back up -- the usual first fix.
Refresh Status	Re-reads the gateway now instead of waiting for the 30-second poll.
Open Gateway Dashboard	Opens the gateway Pi own status page directly at http://192.168.50.2:9000 in a new tab.

To actually control the tunnel:

1. Sit down at the **gateway Pi** keyboard (the second Pi, at 192.168.50.2).
2. Open the **Chromium** browser on that Pi.
3. Go to **http://localhost:9001** and log in with your amateur radio callsign.
4. Use **Bring Tunnel UP**, **Bring Tunnel DOWN**, or **Restart Tunnel** as needed. The message line under the buttons reports the result.

NOTE If you press a control button from an operator laptop or phone, the page tells you the control requires physical access to the gateway Pi keyboard, or asks you to log in with your callsign first. This is expected -- it is not a fault. Only the read-only status view (port 9000) is available from other EMCOMM-NET devices.

28.9 About This Gateway

The **About This Gateway** table at the bottom of the page is a quick reference card of the fixed facts about your gateway:

ITEM	VALUE
Gateway Pi IP	192.168.50.2
Gateway status API	http://192.168.50.2:9000/api/status
AMPRNet block	Your assigned 44.x.x.x/29 (fills in once the tunnel is up)
WireGuard endpoint	amprgw.ampr.org:51820
What it routes	All 44.0.0.0/8 traffic for every device on EMCOMM-NET
AMPRNet portal	https://portal.ampr.org

NOTE Full AMPRNet setup procedures -- WireGuard configuration, IP forwarding, route advertisement, and the verification checklist -- are in Installation Guide Step 11. Portal registration: portal.ampr.org. WireGuard gateway endpoint: amprgw.ampr.org:51820.

28.10 Troubleshooting

SYMPTOM	WHAT TO DO
The banner is gray and says "Gateway Unreachable / not responding."	The gateway Pi at 192.168.50.2 is off or disconnected. Check that it is powered on and on the same network, then press Refresh Status. Remember the gateway is a second, separate Pi from the main FieldCommand server.
The banner is red -- TUNNEL DOWN -- but the cards show gateway temp and memory.	The gateway Pi is up but the tunnel is not established. At the gateway Pi keyboard, open http://localhost:9001, log in with your callsign, and use Restart Tunnel.
IP Forwarding shows DISABLED in red even though the tunnel is UP.	Routing is off, so EMCOMM-NET devices still cannot reach 44Net. Forwarding is set on the gateway Pi in Installation Guide Step 11 -- have the technical lead re-check it.
I press a control button from my laptop and nothing happens.	Tunnel control only works from the Chromium browser on the gateway Pi itself (port 9001, localhost). This is by design. Use the gateway Pi keyboard, or just watch the read-only status from your laptop.
The control buttons say "Login required -- enter your callsign first."	You are on the gateway Pi but not logged in. Log in at http://localhost:9001 with a valid FCC amateur callsign; sessions last 8 hours and then need renewing.
Active Routes reads "No 44Net routes found" and Connected Peers is empty.	These both point to a tunnel that is down or never brought up. Bring the tunnel up from the gateway Pi, then press Refresh Status; the routes and peers should populate.

29 JS8Call — High Frequency (HF) Digital Messaging

http://192.168.50.1/

JS8Call is a weak-signal High Frequency (HF) digital mode for keyboard-to-keyboard messaging and store-and-forward relay. When the local infrastructure is gone and even Very High Frequency (VHF) repeaters are down, JS8Call can still carry short text messages hundreds or thousands of miles on HF, using signals so weak a human ear could not hear them. It does not run inside FieldCommand. It runs on a separate Windows laptop that is wired to the incident HF transceiver and joined to the same EMCOMM-NET network, and FieldCommand gives you a one-tap link to reach it. This chapter explains what JS8Call does, how the laptop fits into your station, how to point the dashboard card at that laptop, and how to fold the messages it receives into the permanent incident record.

TIP In one sentence: JS8Call is your long-haul text lifeline on HF radio, run from a Windows laptop next to the radio, and the FieldCommand dashboard has a card that opens it.

29.1 What JS8Call Does for Emergency Communications (EMCOMM)

JS8Call was built for exactly the kind of long-distance, no-infrastructure messaging an incident needs when everything else fails. Its main capabilities are below.

CAPABILITY	WHAT IT GIVES YOU
Keyboard messaging	Real-time, chat-style text between stations. You type a message, the radio sends it as tones, and the receiving station reads it on screen.
Store and forward	A message can be held at a relay station and passed along automatically when the destination station is finally heard, so you reach places you cannot hear directly.
Heartbeat beacons	The software sends a short automatic beacon every few minutes that announces your station and shows who can hear whom, which reveals live propagation.
Group messages	A message addressed to a group name such as @EMCOMM, @ARES, or @RACES reaches every station monitoring that group at once.
Extreme sensitivity	Copies signals down to roughly -24 decibels signal-to-noise ratio (SNR) — far weaker than a Single Sideband (SSB) voice signal you could understand by ear.

29.2 How the JS8Call Laptop Fits Into Your Station

JS8Call is not part of the Raspberry Pi that runs FieldCommand. It lives on its own Windows laptop for one practical reason: the laptop is what is physically wired to the HF radio through a Universal Serial Bus (USB) sound-and-control interface. A typical setup is an IC-7300 transceiver connected to the laptop with a Digirig or a similar USB audio interface, with JS8Call installed on the laptop. The laptop then joins the same EMCOMM-NET Wi-Fi as everything else, so operators at other FieldCommand screens can reach it over the network.

PIECE	ROLE
HF transceiver	The radio that actually transmits and receives on the HF bands (for example an IC-7300).

PIECE	ROLE
USB audio interface	The Digirig (or similar) that carries receive audio, transmit audio, and push-to-talk keying between the radio and the laptop.
Windows laptop	Runs the JS8Call program itself. This is the machine whose network address you give to the dashboard card.
EMCOMM-NET network	The shared Wi-Fi. Both the laptop and every FieldCommand screen sit on it, which is what lets the dashboard reach the laptop.

WARN

JS8Call transmits on the amateur radio bands, so it may be operated only by a properly licensed operator with privileges on the bands and modes in use. A group with no licensed operator leaves the callsign blank and does not use this feature.

29.3 The JS8Call Card on the Dashboard

On the FieldCommand dashboard, switch to **Amateur Radio** mode. Among the operator cards you will see a purple **JS8Call** card with a radio-antenna icon. It is a shortcut, not the program — tapping it opens the JS8Call laptop in a new browser tab. The card shows the parts below.

CARD ELEMENT	WHAT IT SHOWS
Name	JS8Call — the label of the card.
Description	"HF digital messaging via keyboard" — a one-line reminder of what it is.
Port label	Reads **Port 2442** until you configure the laptop address. Once set, it shows the laptop address and port, such as 192.168.50.2:2442.
Status line	A small line under the port. Before setup it warns "Windows laptop — tap to configure IP". After setup it shows "Windows:" and the address you entered.

NOTE

The Amateur Radio mode button is grayed out until a station callsign is set in Setup. If you do not see the JS8Call card, add a callsign first (see the Setup chapter), then return to the dashboard and choose Amateur Radio mode.

29.4 Telling the Card Where the Laptop Is

The card cannot guess which laptop on the network is running JS8Call, so the first time you use it you tell it the laptop network address. You do this once per screen; the address is remembered in that browser.

1. On the JS8Call laptop, find its network address: click the Start menu, type **cmd**, press Enter, type **ipconfig**, and press Enter. Read the **IPv4 Address** line — it looks like 192.168.50.2.
2. On the FieldCommand dashboard, in Amateur Radio mode, tap the **JS8Call** card.
3. A small box asks for the address of the Windows laptop running JS8Call. Type the IPv4 Address you just read (for example 192.168.50.2) and click OK.
4. The card now remembers that address: its port label changes to that address with :2442 after it, and the status line reads "Windows:" and the address. JS8Call also opens in a new browser tab.

TIP

To clear a wrong address, tap the card again and leave the box empty, then click OK. The card returns to showing "Port 2442" and the configure-IP warning, ready for you to enter the correct address.

29.5 Opening JS8Call from the Dashboard

Once the address is set, tapping the card opens the JS8Call laptop at port 2442 in a new browser tab. Port 2442 is the address JS8Call listens on. Note that the keyboard operating — typing and reading messages — is done at the laptop itself, on the JS8Call program window. The dashboard card is the quick way to jump to that machine and to keep its address handy for everyone at the incident.

> **NOTE** If the new tab does not open, your browser may have blocked the pop-up. Allow pop-ups for the FieldCommand address (192.168.50.1) and tap the card again, or simply walk to the laptop and use JS8Call there.

29.6 Sending a Message on HF Digital

Operating happens in the JS8Call program on the laptop. The general flow of a single message is the same every time.

1. Make sure the radio is on the agreed HF frequency and band for your net, and that JS8Call shows signals scrolling in its waterfall display (the moving picture of activity across the channel).
2. In the message box, type the callsign of the station you are calling, then your text. Keep it short — HF digital is slow, so every extra word costs time on the air.
3. Send the message. The radio keys up and plays the tones; a progress indicator shows the message going out.
4. Watch the receive area for the other station to answer. Their reply appears as text as it decodes.
5. For a message to a whole group, address it to the group name (for example @EMCOMM) instead of a single callsign, so every monitoring station in that group receives it.

29.7 Directed Calls and Group Names

JS8Call lets you aim a message at one station or at a named group. The common emergency group names are below; your net may agree on others.

ADDRESS	WHO RECEIVES IT
A single callsign	Only the station with that callsign is being called, though others can still see the exchange.
@EMCOMM	Every station monitoring the general emergency-communications group.
@ARES	Stations monitoring the Amateur Radio Emergency Service group.
@RACES	Stations monitoring the Radio Amateur Civil Emergency Service group.
@ALLCALL	A general call to any station hearing you — used to find out who is on frequency.

29.8 Bringing JS8Call Traffic Into the Incident Record

Incident data is permanent, so a message that arrives by HF digital should not live only on the laptop. Today the bridge between JS8Call and the incident record is done by hand, and that is deliberate: an operator reads the message and decides where it belongs.

1. Read the incoming message in JS8Call on the laptop.
2. If it is formal traffic — a request, a report, a message with a sender and an addressee — open the **ICS-213 Message** page on the dashboard and type the message into a new ICS-213 General Message form, so it is saved and can be re-printed with the rest of the incident paperwork.
3. If it is routine net traffic — a check-in, a status, a short exchange — enter it in the Net Control Logger instead, so it lands in the running log for the operational period.

4. Note the time and the sending station exactly as JS8Call shows them, so the record matches what was on the air.

> **NOTE** A future FieldCommand release is planned to ingest JS8Call messages into the incident log directly. Until then, the manual copy step above is the correct practice — it keeps a person in control of what becomes an official record.

29.9 Good Operating Practice on HF Digital

HF digital rewards patience and short messages. A few habits keep a net running cleanly.

Keep it short. The mode is slow. Say only what is needed; abbreviate where your net has agreed abbreviations.

Listen before you transmit. Watch the waterfall and the decode area so you do not send on top of another station.

Let heartbeats do their job. The automatic beacons show who can hear whom; use that picture to pick a relay station when you cannot reach a destination directly.

Log as you go. Copy anything that matters into the ICS-213 page or the Net Control Logger while it is fresh, not at the end of the shift.

29.10 Troubleshooting

SYMPTOM	WHAT TO DO
The JS8Call card is missing from the dashboard	You are not in Amateur Radio mode, or no callsign is set. Choose Amateur Radio mode; if that button is grayed out, add a station callsign in Setup first.
Tapping the card opens nothing	The browser blocked the new tab. Allow pop-ups for 192.168.50.1 and tap again, or operate at the laptop directly.
The card still says "Port 2442" and warns to configure the IP	No laptop address is saved on this screen yet. Tap the card and enter the laptop IPv4 Address (find it with ipconfig on the laptop).
The new tab opens but shows an error or blank page	The address is wrong, JS8Call is not running, or the laptop is off the network. Confirm JS8Call is open on the laptop and re-check the address with ipconfig.
No signals scroll in the JS8Call waterfall	The radio audio is not reaching the laptop. Check the USB audio interface cabling and that JS8Call is set to the correct sound device and radio.
A message came in but is not in the incident record	JS8Call does not log to the incident automatically yet. Copy the message into the ICS-213 page (formal traffic) or the Net Control Logger (net traffic) by hand.

30 National Traffic System (NTS) Radiogram Generator

http://192.168.50.1/nts.html

The National Traffic System (NTS) Radiogram Generator builds a properly formatted American Radio Relay League (ARRL) radiogram - the standard form for passing formal written messages by radio. When telephones, cell service, and the internet are down, a radiogram is how a short, exact message about a person's safety or an urgent request moves across the country, operator to operator, until it reaches the person it is for. This page fills in the message number, precedence, check (word count), and address for you, shows a clean printable form, and keeps a running log of every radiogram you have handled so nothing is lost.

TIP In one sentence: type the message, click Generate Radiogram, and you get a correctly formatted, printable NTS form with the word count filled in for you.

30.1 What a Radiogram Is and When to Use One

A radiogram carries one short message in a fixed format that every traffic handler in the country understands. Because the format never changes, an operator who receives your message can relay it word for word to the next station without confusion. Radiograms are used most heavily right after a disaster, when families outside the area want to know that relatives are safe (Health and Welfare traffic) and when officials need to move an urgent request that has no other path out.

The message text is kept short on purpose - 25 words or fewer. Short messages relay quickly and accurately, and a whole net full of them clears faster. The generator counts your words as you type and warns you the moment you go over 25.

30.2 The Preamble (Header)

The **Preamble** card is the message header - the bookkeeping that travels ahead of the actual words. Fields marked with a star (*) are required. The **Check** field is filled in for you and cannot be typed into.

FIELD	WHAT IT MEANS
Message Number *	The sequential number of this message from your station, such as 001. Click Auto-fill Msg # to take the next number after your last saved one.
Precedence *	How urgent the message is - Routine, Welfare, Priority, or Emergency (see the table in 30.3). Choosing one shows a plain-language reminder of what it is for.
Handling Instructions	An optional service instruction (the HX codes) telling the delivering station how to handle the message - see 30.4. Leave it on None if there is no special instruction.
HX Supplement (if any)	The extra detail some handling codes need, such as a number of hours or a person's name. Leave blank when the chosen code needs nothing extra.
Station of Origin *	The callsign of the station that first put the message into the system. Typed in capital letters automatically.
Check (word count) *	The number of words in the message text. Filled in and updated for you as you type; you cannot edit it by hand.

FIELD	WHAT IT MEANS
Place of Origin *	The city and state the message started from, such as COLUMBUS OH.
Date / Time Filed *	When the message was filed, such as 0930 JUN 7. Click Auto-fill Date/Time to insert the current time in Coordinated Universal Time (UTC).

30.3 Precedence - How Urgent the Message Is

Precedence tells every operator who touches the message how quickly it must move and which messages to pass first when a net is busy. The generator shows the matching reminder under the drop-down as soon as you pick one.

PRECEDENCE	WHAT IT IS FOR
R - Routine	Normal message traffic with no urgency. The everyday default.
W - Welfare	A message about the health and welfare of a person, or an inquiry from a family member. Common after a disaster.
P - Priority	Important traffic with some urgency - for example official or government messages that should not wait behind routine traffic.
E - Emergency	Any message with life-or-death urgency, a disaster declaration, or official government communications. Handled ahead of everything else.

30.4 Handling Instructions (HX Codes)

The optional **Handling Instructions** drop-down carries the standard ARRL service codes. They tell the station that finally delivers the message what the sender wants done. Pick one only when it applies; most messages leave this on None.

CODE	WHAT IT ASKS THE DELIVERING STATION TO DO
HXA	Collect on delivery (a phone-call charge limit may apply).
HXB	Cancel the message if it is not delivered within a set number of hours (put the hours in HX Supplement).
HXC	Report the date and time of delivery back to the station of origin.
HXD	Report the identity of the station the message came from, and the delivery route, back to the originating station.
HXE	Ask the addressee whether there is a reply, and report it back.
HXF	Hold the message for arrival of the addressee (put the date in HX Supplement).
HXG	Deliver by mail or landline toll call. If that is not possible, cancel the message and report it back to the origin.

NOTE When a code needs a detail - hours for HXB, a name or date for HXF - type that detail into the HX Supplement box next to the drop-down.

30.5 Address, Message Text, and Signature

Below the preamble are three more cards that hold who the message is for, what it says, and who sent it.

30.5.1 Address

FIELD	WHAT IT MEANS
Addressee Name *	The person the message is for, such as John Smith. Required.
Phone Number	The delivery phone number, if known - used by the station that finally delivers the message.
Street Address	The street address for delivery.
City	The delivery city.
State / Zip	The delivery state and ZIP code, such as OH 43215.

30.5.2 Message Text

Type the message itself into the **Message Text** box. Use plain words and avoid punctuation - spell out the word X where you would put a period (for example, ARRIVED SAFELY X EVERYONE FINE X LOVE). Under the box a live **Word count** shows your total against the 25-word maximum; if you go over, the number turns red and the words over limit are flagged.

30.5.3 Signature

FIELD	WHAT IT MEANS
Sender Name / Callsign	Who the message is from, such as Jane Smith / W8ABC. If left blank, the Station of Origin callsign is used.
Phone / Email	A contact number or email for the sender, if wanted.

30.6 Generating and Printing the Radiogram

The buttons at the bottom of the form build the finished radiogram. The generator will not proceed until the required fields are filled and the check is 25 words or fewer.

1. Fill in at least the required fields: **Message Number**, **Station of Origin**, **Addressee Name**, and **Message Text**.
2. Use **Auto-fill Date/Time** and **Auto-fill Msg #** if you want the current UTC time and the next message number filled in for you.
3. Click **Generate Radiogram**. The form is replaced by a clean, black-on-white radiogram preview showing the preamble, the address, the message text in capitals, the signature, and blank received/sent tracking lines.
4. Click **Print Radiogram** to print the form. Only the radiogram itself prints - the editing controls and the saved log are hidden on paper.
5. To make a change, click **Edit** (the left-arrow button) to return to the form with your entries intact.

WARN If a required field is missing, or the check is over 25 words, a message tells you exactly what to fix. Shorten the text - spell out X for periods and drop non-essential words - until the check is 25 or fewer, then try again.

30.7 Preview Actions - Save, Copy, and Send to the Net

Once the preview is showing, a row of action buttons appears beneath it:

BUTTON	WHAT IT DOES
Print Radiogram	Opens your device print dialog and prints just the radiogram.

BUTTON	WHAT IT DOES
Save to Log	Stores this radiogram in the Saved Radiograms table below and also sends a copy to the server forms record so the message is kept as part of the incident.
Copy Text	Copies the full message - preamble, address, text, and signature - to the clipboard as plain text, ready to paste into another log or a digital mode.
Send to Net Log	A reminder to record this message as a traffic entry in Net Control (open Net Control and use the Traffic Log tab).
Edit (left arrow)	Returns to the form so you can change any field.

30.8 The Saved Radiograms Log

At the bottom of the page the **Saved Radiograms** table lists every radiogram you have saved, newest first (the log keeps up to the most recent 50). Each row shows the columns below. Use **Load Saved** at the top of the page to jump straight to this table.

COLUMN	WHAT IT SHOWS
#	The message number.
Prec	The precedence, shown as a colored badge (R, W, P, or E).
To	The addressee name.
Origin	The station of origin callsign.
Date Filed	The date and time the message was filed.
Check	The word count.
Load / X	Load reopens the saved radiogram in the form; the X button deletes that one row after you confirm.

The **Clear All** button on the Saved Radiograms header removes every saved radiogram after you confirm. The **New Form** button at the top of the page clears the current form and preview so you can start a fresh message without touching the log.

NOTE Saving keeps a copy both on this device and on the FieldCommand server, so a radiogram you save is part of the permanent incident record even if this browser is later cleared.

30.9 Troubleshooting

SYMPTOM	WHAT TO DO
Generate Radiogram does nothing and a message pops up	A required field is empty. Fill in Message Number, Station of Origin, Addressee Name, and Message Text, then click Generate again.
The word count is red and it will not generate	The message text is over the 25-word limit. Spell out X for periods and remove non-essential words until the count is 25 or fewer.
The Check field will not let me type in it	That is intended. The check is the word count and is filled in for you as you type the message; edit the message text to change it.
The date/time or message number is blank	Click Auto-fill Date/Time for the current UTC time, and Auto-fill Msg # for the next number after your last saved radiogram. You can also type either by hand.

SYMPTOM	WHAT TO DO
Printing shows the whole page, not just the form	Click Generate Radiogram first so the preview is on screen, then Print. Only the radiogram preview is set to print; the form and log are hidden on paper.
A saved radiogram disappeared from the log	The log keeps the 50 most recent entries and older ones roll off. Radiograms you saved are also stored on the server as part of the incident record.

31 Repeater Database http://192.168.50.1/repeaters.html

The Repeater Database is your offline directory of the VHF and UHF voice repeaters in and around your operating area. A repeater is an automated relay station on a hilltop or tower that listens on one frequency and re-transmits on another, greatly extending the range of a hand-held or mobile radio. This page lets you load a full county-or-region export from RepeaterBook, search and filter it in seconds, plot every machine on a map, and push the ones you will use straight into the Channel Library so they flow into your ICS-205 communications plan. Once the data is loaded it lives in this browser, so it keeps working with no internet at all.

TIP

In one sentence: load a RepeaterBook export, filter to the repeaters you want, and one click sends any of them into your communications plan.

31.1 How the Screen Is Laid Out

The page has five working areas, top to bottom. Knowing where each one lives makes the rest of this chapter quick to follow.

AREA	WHAT IT IS
Source tabs	A row of three tabs just under the title - **Offline File**, **Server API**, and **Demo Data** - that pick where the repeater data comes from. Each tab shows a small count of how many repeaters that source holds.
Toolbar	The search box, the filter drop-downs, and the action buttons (Add Repeater, Map View, CSV, Print). It stays hidden until data is loaded.
Table	The main list - one row per repeater, sortable by any column.
Detail panel	A slide-in panel on the right that shows everything about the one repeater you clicked, with Copy, Program, and Channel Library buttons.
Status bar	The thin strip pinned to the bottom of the window. On the left it shows how many repeaters match your filters out of the total; on the right it shows the current sort.

31.2 Loading Repeater Data

The three source tabs are three different ways to get repeater data onto the page. For almost every deployment the **Offline File** tab is the right one - it needs only a free RepeaterBook account and no internet at the Pi.

SOURCE TAB	HOW TO USE IT	WHAT IT NEEDS
Offline File (recommended)	Log in at repeaterbook.com, search your county or area, and click Export then CSV. Drag the downloaded file onto the drop zone, or click the drop zone (or the "Load file" box) to browse for it.	A free repeaterbook.com account. No API token - any logged-in user can download a CSV. RepeaterBook CSV, RepeaterBook JSON, and a FieldCommand repeaters.json file are all accepted.

SOURCE TAB	HOW TO USE IT	WHAT IT NEEDS
Server API (optional)	Switching to this tab calls the FieldCommand server, which serves whatever fetch_repeaters.py last downloaded. Place your token in /opt/fieldcommand/data/repeaterbook_token.txt, then run python3 scripts/fetch_repeaters.py on the Pi.	An approved RepeaterBook API token - a separate application and approval, NOT the same as an account login. Apply at repeaterbook.com/api/token_request.php and allow several weeks.
Demo Data	Shows a handful of clearly-labeled SAMPLE placeholder entries so you can see how the page looks and behaves before real data is loaded.	Nothing. These are not real repeaters and must never be used on the air.

1. Click the **Offline File** tab (it is selected by default).
2. On your computer, open **repeaterbook.com**, sign in, search your area, and download the **Export - CSV** file.
3. Drag that file onto the dashed drop zone, or click the drop zone to browse and pick it.
4. A progress box reads the file and reports "Loaded N repeaters." The toolbar and table appear, and a green banner names the file and the import time.

NOTE The RepeaterBook API token is not the same as a RepeaterBook login. It takes a separate application and staff approval. The CSV export works with any free account and is the method to use in the field. An imported file is remembered in this browser, so it is still there after a reload - but it is stored per-browser, not on the server, so each operator device imports its own copy (or use the Server API tab to share one).

31.3 Searching, Filtering, and Sorting

Once data is loaded the toolbar appears. A big RepeaterBook export can hold hundreds of machines, so the toolbar is how you cut it down to the few you care about. Every control narrows the table instantly as you use it, and they stack - set several at once to zero in.

CONTROL	WHAT IT DOES
Search	Free-text box. Matches callsign, city, state, frequency, tone, sponsor, notes, and county all at once.
Band	Limits to one amateur band: 10m, 6m, 2m, 1.25m, 70cm, 33cm, or 23cm (the band is worked out from the output frequency).
Mode	Limits to one mode: FM / Analog, D-STAR, C4FM / Fusion, DMR, or P25.
Status	All, On-Air, or Off-Air - hide machines RepeaterBook lists as off the air.
State	All States, or one state. The list is built automatically from the data you loaded, so a two-state export offers both.
Use	All, Open, Closed, or Private - how the repeater owner allows access.
EmComm	All, ARES, SKYWARN, or RACES - show only repeaters flagged for that emergency-communications program.
Sort by distance	Off, Nearest first, or Farthest first. Turning it on reveals a Dist column of miles from your station location and orders the table by it.

To sort by any other column, click that column heading; click it again to reverse the direction. The status bar shows which column is sorting and which way.

31.4 Reading the Table

Each row is one repeater. The columns, left to right:

COLUMN	WHAT IT SHOWS
Output	The output frequency in megahertz - what you set your radio to receive on. This is the number people mean by "the frequency."
Input	The input frequency - what your radio transmits on. Dashed if not known.
Callsign	The repeater callsign (usually the trustee or club callsign).
Tone	The access tone. A green badge is the transmit tone (CTCSS / PL); a fainter "in:" badge appears when the input tone differs.
Mode	A color badge for the mode - FM, D-STAR, C4FM, DMR, or P25.
City	The nearest city or landmark.
State	The state.
EmComm	Emergency-program badges: ARES, RACES, SKY (SKYWARN), CAN (CANWARN).
Links	Internet-linking badges: EL (EchoLink), AS (AllStar), WX (Wires-X). Hover a badge to see the node number.
Use	Open, Closed, or Private access.
Status	A green dot for On-Air, a red dot for Off-Air.
Dist	Miles from your station location. Hidden until you turn on "Sort by distance."

31.5 The Detail Panel

Click any row and a panel slides in from the right with the full record for that repeater. The large green number at the top is the output frequency, with the callsign and the band, city, and state under it, then a row of mode and program badges. Below that is a labeled list:

ROW	WHAT IT MEANS
Input	Transmit frequency in megahertz.
CTCSS (out)	The tone your radio must send to open the repeater ("None / CSQ" means no tone needed).
CTCSS (in)	The tone the repeater sends back, if any.
Offset	The difference between input and output, worked out for you (for example -0.600 MHz on 2 meters).
Digital Code	The color code or digital ID for a digital repeater.
County	The county the repeater is in.
Distance	Miles from your station location.
Status	On-Air or the status RepeaterBook lists.
Use	Open, Closed, or Private.
Trustee/Sponsor	The person or club that runs the repeater.

ROW	WHAT IT MEANS
Last Updated	When the RepeaterBook record was last changed.
Website	A link to the sponsor site, when one is listed.

Any notes on the repeater appear in a shaded box under the list. Four buttons sit at the bottom of the panel:

Copy - copies a one-line summary (callsign, frequencies, tone, mode, location) to the clipboard for pasting into a log or message.

Program - opens a plain "Programming Summary" (RX, TX, offset, transmit and receive tones, mode, and a suggested name) to copy into a radio or a programming cable tool such as CHIRP.

Print (the printer icon) - sends the current view to the printer.

+ Channel Lib - adds this repeater to the Channel Library so it is available to drop into the ICS-205 communications plan (see 31.6).

31.6 Sending a Repeater to the Channel Library

The Repeater Database is a reference list; the Channel Library (Chapter 32) is the working set of channels that feeds your ICS-205. The **+ Channel Lib** button is the bridge between them. It saves the repeater to the server, so unlike an imported file it is shared with every operator device.

1. Find the repeater you want and click its row to open the detail panel (or open the Map View and click its pin).
2. Click **+ Channel Lib**. The button briefly reads "Added" in green.
3. The channel now appears in the Channel Library, ready to insert into the ICS-205 communications plan.

> **NOTE** The saved channel is filed as an **Interop** channel when the repeater is flagged ARES or RACES, and as an **Amateur** channel otherwise, so it lands in a sensible group in the Channel Library. You can change that later in the Channel Library itself.

31.7 Adding a Repeater by Hand

For a machine that is not in RepeaterBook - a private club repeater, a temporary cross-band unit, or a single local repeater when you do not want a whole export - use manual entry. Click **Add Repeater** in the toolbar (or **Add a Repeater Manually** on the empty drop-zone screen). A small form opens with these fields:

FIELD	WHAT TO ENTER
Callsign	The repeater callsign (typed in capitals).
Output Freq (MHz) *	Required. The frequency you listen on, such as 146.940.
Input Freq (MHz)	The frequency you transmit on. Leave blank for a simplex machine or to work it out later.
Tone / PL (Hz)	The access tone, such as 103.5. Leave blank for none.
Mode	FM, D-STAR, C4FM, DMR, P25, or NXDN.
City	The nearest city, such as Woodstock.
State	The two-letter state, such as IL.
Latitude / Longitude	Decimal coordinates, such as 42.32 and -88.38. Needed for the repeater to appear on the map.

FIELD	WHAT TO ENTER
Notes	Anything useful - "EOC primary, wide-area coverage."
ARES / RACES / SKYWARN	Tick the box for each emergency program the repeater supports.

1. Fill in at least the **Output Freq -** it is the only required field.
2. Add the input frequency, tone, mode, location, and any flags you know.
3. Click **Add to List**. The repeater joins the table immediately.

> **WARN** A hand-added repeater is added to the list you are viewing but is not saved on its own. To keep it, click **CSV** in the toolbar to export the list (hand entries included), or open its detail panel and click **+ Channel Lib** to save it to the server as a channel.

31.8 Map View

The **Map View** button in the toolbar opens a full-screen map of every repeater in the current filtered view that has coordinates. RepeaterBook exports include latitude and longitude for each repeater, so this usually means all of them. The button shows the plottable count in parentheses, for example **Map View (247)**.

MAP FEATURE	WHAT IT DOES
Pin color by mode	Green FM, blue D-STAR, orange C4FM / Fusion, purple DMR, red P25. A repeater flagged ARES or RACES is drawn amber, overriding the mode color. The legend across the top names each color.
Filter sync	The map plots exactly the repeaters visible in the table. Set your band, mode, state, or EmComm filters first, then open the map to see only those.
Pin popup	Click a pin for the frequency, callsign, offset, mode, tone, location, county, sponsor, badges, and notes.
Add to Channel Library	Each popup has a "+ Add to Channel Library" button that imports that repeater as a channel without leaving the map.

1. Load repeater data and, if you want, apply filters to narrow the view.
2. Click **Map View** in the toolbar. The map opens zoomed to fit every plotted pin.
3. Click any pin to read its details or add it to the Channel Library.
4. Press **Escape** or click **Close** to return to the table.

> **NOTE** Only repeaters with coordinates are plotted. A machine with no latitude and longitude still appears in the table but not on the map. The map draws its background from an online map service, so the streets fill in only when the Pi has internet; the pins themselves plot with or without a connection.

31.9 Repeater Overlays on the Tactical and Resource Maps

The same repeater data can be laid over the other two big maps in FieldCommand. Both read from the FieldCommand server, so import repeater data here first for the overlay to have anything to show.

MAP	HOW TO TURN IT ON	WHY
Tactical Map (tactical.html)	Click the **Repeaters** button in the map button bar. The first click loads and plots the repeaters; later clicks toggle the layer.	See which repeaters cover the areas where APRS stations are active.

MAP	HOW TO TURN IT ON	WHY
Resource Map (resource_map.html)	Click the **Repeaters** button in the toolbar to toggle the layer on or off.	Check repeater coverage over staging areas, shelters, and deployed resources.

31.10 Exporting and Printing

Two toolbar buttons get the list off the screen. **CSV** downloads the currently filtered list as a spreadsheet file (repeaters_fieldcommand.csv), which is the way to save hand-added entries or hand a channel plan to another operator. **Print** sends the current view to the printer for a paper cheat sheet to keep at the radio.

31.11 Troubleshooting

SYMPTOM	WHAT TO DO
The import failed with an error	The file is probably not a RepeaterBook export. Re-download it from repeaterbook.com as an Export - CSV, or use a RepeaterBook JSON file. The page accepts .csv, .json, and .txt.
The Server API tab shows "unreachable" or is empty	That tab needs an approved RepeaterBook API token and fetch_repeaters.py to have been run on the Pi. Use the Offline File tab instead - download a CSV and drag it in.
Map View is grayed out or says "No repeaters with coordinates"	The loaded repeaters have no latitude and longitude, or your filters hid them all. A RepeaterBook CSV includes coordinates; a hand-added repeater plots only if you filled in Latitude and Longitude.
My hand-added repeater vanished after I reloaded	Manual entries are added to the list, not saved on their own. Export the list to CSV, or click + Channel Lib to save it to the server as a channel.
Another operator device does not see my imported repeaters	An imported file is stored in that one browser. Either import the CSV on each device, or run fetch_repeaters.py and use the Server API tab so all devices share one server copy.
Distances look wrong or are all blank	Distance is measured from the station location built into the page, and needs each repeater to have coordinates. Turn on "Sort by distance" to reveal the Dist column; blanks mean that repeater has no coordinates.

32 Channel Library http://192.168.50.1/channel_library.html

The Channel Library is your agency's saved list of radio channels. Each entry holds one channel - its name, receive and transmit frequencies, tone, mode, the job it does, and which Division or Group uses it. The value of the library is not just storage: every channel you save here appears in the Channel Library picker on the ICS-205 Communications Plan, so building a communications plan becomes point and click instead of typing frequencies by hand while the clock is running. Enter a channel once, correctly, and reuse it on every incident.

TIP In one sentence: save your local repeaters, interop channels, and tactical frequencies here once, and they drop straight into the ICS-205 on every incident.

32.1 Universal Channels vs. Your Agency Channels

FieldCommand already knows the channels that are the same everywhere. Universal channels - National Simplex, Automatic Packet Reporting System (APRS), the National Interoperability Field Operations Guide (NIFOG) channels, and the standard Mutual Aid channels - are always available automatically and do not need to be entered. The Channel Library is for the channels that are specific to your jurisdiction: your local repeaters, your county interop and tactical frequencies, and your served agency talk paths. Only add your agency-specific channels here.

Channels that FieldCommand shipped with (rather than ones you typed) carry a small **seed** badge next to the name in the table, so you can tell a built-in starting channel from one your agency added.

32.2 What a Channel Record Holds

Every channel is described by the same set of fields. These are the fields you fill in when adding or editing a channel, and the same fields shown as columns in the main table.

FIELD	WHAT IT MEANS
Channel Name	Required. The full, readable name - for example "McHenry County Command". This is what appears on the ICS-205.
Alpha Tag	A short code-plug label, up to ten characters, such as "MHCO CMD" - the kind of tag that fits a radio display. Optional.
RX Freq	Required. The receive frequency in megahertz (MHz) - what the radio listens on, for example 155.3400.
TX Freq	The transmit frequency in MHz - what the radio sends on. Leave it blank for a simplex channel; the library then shows "= RX" to mean transmit equals receive.
PL/DCS	The sub-audible tone the channel needs, if any - a Private Line (PL / CTCSS) tone like "100.0 Hz" or a Digital Coded Squelch (DCS) code like "D023N". Leave blank for none.
Mode	The radio mode - see the reference table in 32.6.
Function	The job the channel does - Command, Tactical, Medical, and so on (see 32.6).
Division	The Division or Group that uses this channel, such as "Div A", "Grp SAR", or "All". Optional.

FIELD	WHAT IT MEANS
Notes	Free text - repeater location, coverage area, or special instructions.

32.3 Reading the Channel Table

The main screen is one wide table with a row per channel. The columns are, left to right: **Channel Name** (with the seed badge where it applies), **Alpha Tag**, **RX Freq** shown in green, **TX Freq** shown in amber when it differs from RX or as "= RX" when it is the same, **PL/DCS**, **Mode**, **Function** shown as a colored badge, **Division**, and **Notes**. The last column holds two small buttons on every row: a pencil to edit the channel and an X to delete it. A count at the right of the filter bar tells you how many channels are currently listed.

32.4 Searching and Filtering

The filter bar sits just above the table:

- **Search box** - type any text (a name, a frequency, a function) and the table narrows as you type; it matches anywhere in the channel record.
- **All Functions drop-down** - pick one function (Command, Tactical, Interop, Medical, Data, Amateur, Calling, Mutual Aid, or Air) to show only channels with that job. Choose "All Functions" to clear it.

The two filters work together, so you can, for example, search for a repeater name and limit the result to Tactical channels at the same time.

32.5 Adding or Editing a Channel

1. Click **+ Add Channel** at the top right to create a new one, or click the pencil button on a row to change an existing one. The Add / Edit Channel window opens.
2. Fill in the fields. **Channel Name** and **RX Frequency (MHz)** are required (marked with an asterisk); everything else is optional.
3. Leave **TX Frequency** blank for a simplex channel - the library treats it as the same as RX.
4. Choose the **Function** and **Mode** from their drop-downs, and set the **Division / Group Assignment** if this channel belongs to a specific part of the org.
5. Click **SAVE CHANNEL**. The window closes and the table refreshes with your change. Click **Cancel** (or outside the window) to discard.

> **NOTE** A new channel defaults to Function "Tactical" and Mode "FM" so you can save a plain simplex frequency with only a name and RX. If you leave the required name or RX blank, the app warns you and does not save.

The editor lays the fields out in a two-column grid. They are exactly the fields listed in 32.2, plus the required-field markers:

EDITOR FIELD	NOTES
Channel Name *	Required. Full readable name.
Alpha Tag	Up to ten characters; shown in a fixed-width font.
Function	Command, Tactical, Interop, Medical, Data, Amateur, Calling, Mutual Aid, Air, or Other.
RX Frequency (MHz) *	Required receive frequency.
TX Frequency (MHz)	Leave blank if same as RX.
PL Tone / DCS	For example "100.0 Hz" or "D023N".

EDITOR FIELD	NOTES
Mode	FM, NFM, P25, DMR, D-STAR, C4FM, AM, SSB, or Analog.
Division / Group Assignment	For example "Div A", "Grp SAR", or "All".
Notes	Repeater location, coverage area, special instructions.

TIP Deleting a channel asks for confirmation and then hides it - it is a soft delete (the record is marked inactive, not erased), so an accidental delete does the least possible harm.

32.6 Reference - Function and Mode Options

The drop-down choices for a channel, in plain language.

32.6.1 Function

OPTION	MEANING
Command	The command net - command staff and unit leaders coordinate here.
Tactical	A working channel for a specific job or team on the ground.
Interop	An interoperability channel shared with other agencies.
Medical	Medical coordination and patient traffic.
Data	A data channel - packet, APRS, or another digital link.
Amateur	An amateur (ham) radio channel or repeater.
Calling	A calling / national simplex channel used to make first contact.
Mutual Aid	A standard mutual aid channel for cross-agency response.
Air	An air-to-ground or aircraft coordination channel.
Other	Any purpose that does not fit the categories above.

32.6.2 Mode

MODE	MEANING
FM	Frequency Modulation - the standard analog voice mode on VHF/UHF.
NFM	Narrow FM - FM at reduced deviation, where narrow channel spacing is required.
P25	Project 25 - a public-safety digital voice standard.
DMR	Digital Mobile Radio - a common commercial digital voice mode.
D-STAR	An amateur digital voice and data mode.
C4FM	Continuous 4-level FM - the amateur System Fusion digital mode.
AM	Amplitude Modulation - for example, aircraft-band use.
SSB	Single Sideband - a common voice mode on the HF bands.
Analog	A generic analog channel when the exact mode is not specified.

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32.7 Import and Export (CSV)

The whole library moves as a Comma-Separated Values (CSV) file - a plain spreadsheet file - so you can back it up, edit it in a spreadsheet program, or share it with another FieldCommand server.

- **Export CSV** - downloads every channel as one file named **channel_library.csv**. Keep it as a backup or hand it to another agency.
- **Import CSV** - opens a file picker; choose a CSV file and its channels are added. The columns are the same field names used above (name, alpha_tag, rx_freq, tx_freq, pl_tone, mode, function, division, notes). A row missing a name or an RX frequency is skipped, and TX defaults to RX when blank.

WARN Import adds channels; it does not replace your library. If you re-import a file you already have, expect duplicate rows. Export first if you want a clean backup before experimenting.

32.8 Importing from RadioReference

If you have a RadioReference Premium subscription, you can pull your county's channels straight in instead of typing them. Click **Import from RadioReference** in the agency-configuration notice to open the import panel.

Enter your own **RR Username** and **RR Password** (they are used only for that one request and are never stored on the Pi), then either type your **County ID** directly or type a **ZIP Code** and click **Lookup ZIP** to find it. The **Filter by Tag** drop-down narrows what you fetch to one category:

TAG	WHAT IT PULLS
All frequencies	Everything RadioReference lists for the county.
Fire Dispatch / Fire-Tac	Fire main dispatch and fire tactical channels.
EMS Dispatch / EMS-Tac	Emergency Medical Services dispatch and tactical channels.
Law Dispatch / Law-Tac	Law enforcement dispatch and tactical channels.
Interop	Interoperability channels shared across agencies.
Emergency Ops	Emergency Operations Center and related channels.
Amateur	Amateur (ham) repeaters and frequencies.

1. Enter your RadioReference username and password.
2. Type a ZIP code and click **Lookup ZIP**; the County ID fills in and the status line shows the city it found. (Or type the County ID yourself.)
3. Optionally choose a category in **Filter by Tag**, then click **Fetch Channels**.
4. Review the results list. Tick the channels you want (the top checkbox selects all), then click **Import Selected** - or click **Import All** to take every result. Imported channels are tagged in their Notes as a RadioReference import.

NOTE RadioReference import needs the internet - it is one of the features that lights up only when the Pi has a Wide Area Network (WAN) connection. Offline, type or CSV-import your channels instead.

32.9 How Channels Reach the ICS-205

The Channel Library is not a stand-alone list - it is the source for the communications plan. When you build an **ICS-205 Communications Plan** on an incident, the form has a Channel Library picker that lists everything saved here (plus the universal channels). Pick a channel and its name, frequencies, tone, mode, and function drop into the ICS-205 row for you. That is why it is worth getting each channel right once in the library: the effort pays off on every incident, and every operator reads the same correct frequency instead of a hand-copied one.

32.10 Troubleshooting

SYMPTOM	WHAT TO DO
The table says "Cannot reach server"	The ICS platform service (port 5055) is not responding. Check that the Pi is on and the ics-platform service is running, then reload the page.
A channel I expected is not listed	Check the filter bar - a search term or a Function filter may be hiding it. Clear the search box and set the drop-down back to "All Functions".
I do not see the common simplex / NIFOG channels	Those are universal and are added automatically on the ICS-205 - they are not stored in this library, so they will not appear in this table. Only agency-specific channels live here.
TX shows "= RX" and I need a repeater offset	That channel has no separate transmit frequency. Edit it and enter the correct TX Frequency in MHz; the table will then show the transmit value in amber.
RadioReference lookup returns an error	Confirm your Premium subscription username and password, that the Pi has internet (WAN) right now, and that the ZIP or County ID is correct. The status line under the buttons shows the exact message.
I deleted a channel by mistake	Delete is a soft delete, so the record still exists but is inactive. Re-add the channel (or re-import from your last CSV export) to bring it back.

33 Hospital Proximity & Facilities Directory http://192.168.50.1/hospitals.html

When someone is hurt, the first question is always the same: where do we take them, and how fast can they get there? The Hospital Proximity page answers it. You keep a list of the trauma centers, burn centers, and community emergency departments in your response area, type in where the incident is, and FieldCommand instantly ranks every hospital by air miles, helicopter flight time, driving distance, and drive time -- and flags which ones have a helipad, a burn unit, or a pediatric, stroke, or cardiac capability. It works entirely offline once the hospitals are entered, so it keeps working when the internet does not. The companion Facilities Directory does the same job for every other operational site: Emergency Operations Centers (EOCs), shelters, staging areas, supply depots, and command posts.

TIP

In one sentence: type in the incident location, and the page sorts your hospital list nearest-first with air time, drive time, and capabilities shown on every card.

33.1 How the Screen Is Laid Out

The Hospital Proximity page (hospitals.html) is one scrolling screen with four working areas stacked from top to bottom. Knowing where each lives makes the rest of this chapter quick to follow.

AREA	WHAT IT IS FOR
Top bar	The blue MEDICAL header. On the right are the buttons that act on the whole list: **+ Add Hospital**, **Import CSV**, **Export CSV**, **Import from CMS**, and the **Dashboard** link back home.
Incident Location bar	The pale band with **Latitude** and **Longitude** boxes. This is where you tell the page where the incident is so it can measure distances.
Air Transport bar	The **Helicopter Type** picker. It sets the aircraft speed used to work out flight time to each hospital.
Sort bar and list	The row of **Sort by** buttons, the search box and filter checkboxes, and below them the list of hospital cards -- one card per hospital.

33.2 Adding and Editing a Hospital

Click **+ Add Hospital** (top right) to open the Add Hospital form, or click the **Edit** button on any card to change one you already have. The form is one panel with the fields below. Only the name is required; fill in as much of the rest as you know. Latitude and longitude are what make the distance ranking work, so add them whenever you can.

FIELD	WHAT IT MEANS
Hospital Name *	Required. The official facility name, for example "Northwestern Medicine McHenry Hospital".
Address	The street address. Used for reference and to build the address line on the card.
City / State / County	Location details. State is a two-letter code (for example IL); the card shows the county with the word "County" after it.

FIELD	WHAT IT MEANS
Switchboard Phone	The main hospital number. It becomes a tap-to-call link on the card.
Emergency Dept (ED) Phone	The direct emergency department line, if you have one. It becomes a separate tap-to-call link marked ED. Optional.
Latitude / Longitude	The hospital coordinates in decimal degrees (for example 42.3089 and -88.4356). Required for the distance, air time, and drive time to be calculated.
Trauma Level	A drop-down: None / Unknown, Level I, Level II, Level III, or Level IV. Level I is the highest trauma capability.
Burn Center	Tick if the hospital has a dedicated burn unit. Adds a Burn Center badge and enables the burn-first sort.
Helipad	Tick if the hospital can receive a medical helicopter. Ticked by default on a new hospital. Adds a Helipad badge.
Peds Trauma	Tick if the hospital has pediatric (child) trauma capability. Adds a Peds Trauma badge.
Stroke Center / Cardiac Center	Tick if the hospital is a designated stroke or cardiac center. Each adds its own badge to the card.
Notes	A free-text line for anything else -- diversion status, special capabilities, or current capacity.

Click **SAVE** to store the hospital on the server. When you are editing an existing hospital a **Delete** button also appears; it asks you to confirm before removing the record. **Cancel** closes the form without saving.

NOTE A hospital with no latitude and longitude still shows in the list, but it always sorts to the bottom and shows no distance block -- the page cannot measure a hospital it does not have coordinates for. Add them by editing the hospital.

33.3 Bringing In Hospitals in Bulk

Typing hospitals one at a time is fine for a handful, but there are faster ways to load a whole region. Three buttons in the top bar handle bulk data.

33.3.1 Import CSV

Click **Import CSV** and choose a comma-separated (spreadsheet) file. The page matches your column headings automatically -- it looks for headings that contain words like name, address, city, state, county, phone, lat, lon, trauma, burn, heli, peds, stroke, cardiac, and notes, so a spreadsheet exported from almost any registry usually loads without editing. The file must have a **name** column; any row without a name is skipped. When it finishes, a message tells you how many hospitals were imported and how many failed.

33.3.2 Export CSV

Click **Export CSV** to download your entire hospital list as a spreadsheet file named hospitals.csv. This is your backup and your sharing format -- hand the file to another FieldCommand server and it imports straight back in with Import CSV.

33.3.3 Import from CMS

Click **Import from CMS** to open a panel that pulls Medicare-certified hospitals from the public federal Centers for Medicare & Medicaid Services (CMS) dataset. Enter the two-letter **State** (required) and optionally a **County**, tick **Acute Care** and/or **Critical Access**, and click **Search CMS**. Results appear in a table with a tick box on each row; use **Import All** or tick the ones you want and use **Import Selected**.

> **WARN** The CMS dataset gives you names, addresses, and phone numbers, but it does NOT include coordinates or trauma level. Imported hospitals arrive with blank latitude, longitude, and trauma level -- you must edit each one to add them before distance ranking will work. This search also needs a working internet connection; the rest of the page does not.

33.4 Setting the Incident Location

Distances are measured from one point: the incident location. Until you set it, the list is unsorted and no distance blocks appear. Set it in the Incident Location bar by any of three ways.

1. Type the **Latitude** and **Longitude** into the two boxes directly -- for example 42.3089 and -88.4356. The list re-sorts the moment you finish typing.
2. Or click **From General Info** to pull the coordinates from the currently selected incident's General Info form. If that form has no coordinates yet, the page tells you to enter them there first.
3. Or click **Use GPS** to take the coordinates from the device you are holding. Handy when you are standing at the scene. Your browser will ask permission to share location.

When coordinates are set, a small confirmation line shows the numbers the page is measuring from, every card gains its distance block, and each card shows a rank number (#1, #2, and so on) in whatever order you are sorting by.

33.5 Air Transport -- the Helicopter Selector

Helicopter flight time depends on how fast the aircraft flies, so the Air Transport bar lets you pick the aircraft. The **Helicopter Type** drop-down lists common medical helicopters with their cruise speeds; the standard Bell 407 at 155 miles per hour is selected by default. Choose **Custom speed...** to type any speed between 80 and 250 miles per hour if your local air-medical service flies something else.

HELICOPTER TYPE	CRUISE SPEED
Air Methods / Metro (Bell 206)	145 mph
Helicopter EMS Midsize (EC135)	150 mph
Air Methods Standard (Bell 407) -- default	155 mph
Agusta AW109	165 mph
Sikorsky S-76	170 mph
MedStar / AirLife (EC145)	180 mph
Custom speed...	Any value you type, 80 to 250 mph

> **NOTE** The Helo Time on each card is not pure flight time. FieldCommand adds a realistic 8 minutes for scene arrival and liftoff plus 5 minutes for the hospital approach, so the number you see is a usable door-to-door estimate rather than a best-case straight line.

33.6 Sorting, Searching, and Filtering

The Sort bar controls the order of the list. Click a **Sort by** button and the list re-orders at once; the active button is highlighted. Hospitals with no coordinates always fall to the bottom of a distance sort because there is nothing to measure.

SORT BUTTON	ORDERS THE LIST BY
Air Miles	Straight-line distance from the incident, nearest first. The default.
Air Time	Estimated helicopter time (flight plus the scene and approach minutes), soonest first.
Ground Miles	Estimated road distance -- about 1.3 times the air miles, a standard road multiplier.
Drive Time	Estimated driving time, figured at an average 45 miles per hour, soonest first.
Trauma Level	Highest trauma capability first (Level I, then II, III, IV), with nearer hospitals first inside each level.
Burn Center First	Burn centers at the top, then ordered by trauma level.

On the right of the Sort bar are three narrowing tools. The **Search hospitals** box hides any card that does not contain the text you type -- a name, a city, a capability. The **Helipad only** checkbox hides hospitals with no helipad. The **Trauma only** checkbox hides hospitals with no trauma-level designation. A count line above the list shows how many hospitals match and which sort is active.

33.7 Reading a Hospital Card

Each hospital is one card. A colored stripe down the left edge shows its trauma level at a glance -- red for Level I, amber for Level II, blue for Level III, plain for the rest. The card carries these parts.

PART OF THE CARD	WHAT IT SHOWS
Name and rank	The hospital name, with its rank number (for example #1) when a sort is active.
Edit and phone buttons	An **Edit** button, plus tap-to-call links: a green switchboard link and, if entered, a red **ED** link for the emergency department.
Address line	The street, city, state, and county.
Capability badges	Small pills for each capability the hospital has: Trauma level, Burn Center, Helipad, Peds Trauma, Stroke, Cardiac, and ICU.
Distance block	A four-part strip -- Air Miles, Helo Time, Ground Mi, and Drive Est -- shown only when the incident location and the hospital both have coordinates.
Notes	Any note you entered, shown at the bottom of the card.

33.8 The Facilities Directory

The Facilities Directory (facilities.html) is the sister page for every operational site that is not a hospital. Open it from the dashboard. It keeps EOCs, shelters, staging areas, supply depots, command posts, and anything else in one searchable directory, each shown as a colored card with a status dot -- green for active, amber for standby, gray for inactive. Because it records radio frequencies and an on-site callsign for each site, it doubles as a facility communications plan.

A toolbar at the top lets you **Search** by name, address, or notes and filter by **Type** (EOC, Hospital, Shelter, Staging Area, Supply Depot, Command Post, Other) or by **Status**. The **CSV** button downloads the directory and **Print** sends it to your printer. Click **+ Add Facility** to open the form; click any card to see its full detail, where **Edit**, **Delete**, and **Copy Address** buttons wait.

FACILITY FIELD	WHAT IT MEANS
Facility Name *	Required. The site name, for example "Franklin County EOC".
Type *	What kind of site it is -- EOC, Hospital, Shelter, Staging Area, Supply Depot, Command Post, or Other. Sets the card color.
Operational Status	Active, Standby, or Inactive -- the colored status dot.
Address / City / State / ZIP	The location, used for reference and Copy Address.
Latitude / Longitude	Decimal coordinates for the site.
Primary / Secondary Phone	Up to two phone numbers for the site.
Primary / Secondary Frequency	Radio frequencies in megahertz that reach this site.
CTCSS Tone	The sub-audible tone the site repeater or channel uses, if any.
Capacity	How many people or how much the site holds, for example "250 persons".
Contact Person / Ham Callsign on Site	Who to reach at the site and the amateur radio callsign operating there.
Generator / ADA Accessible	Backup power and accessibility status -- Yes, No, Partial, or Unknown.
Notes	Parking, access codes, gate instructions, and anything else.

TIP Facility records, like hospitals, are stored on the server, so every operator on EMCOMM-NET sees the same directory and it survives a restart.

33.9 Troubleshooting

SYMPTOM	WHAT TO DO
The list is not sorting by distance	No incident location is set. Type latitude and longitude into the Incident Location bar, or click From General Info or Use GPS.
A hospital shows no distance block and sits at the bottom	That hospital has no coordinates. Click Edit and fill in its Latitude and Longitude, then Save.
I imported from CMS but nothing sorts by distance	CMS data has no coordinates or trauma level. Edit each imported hospital to add latitude, longitude, and trauma level.
Search CMS says it failed or timed out	The CMS search needs the internet. Check your connection, or add the hospitals by hand or by CSV instead -- the rest of the page works offline.
My CSV imported fewer hospitals than expected	Rows with no name are skipped, and the file must have a name column. Open the file and confirm every hospital has a value in the name column.
Cannot reach server appears in the list area	The core service on port 5050 is not answering. Confirm you are on 192.168.50.1, then check the fcc-lookup service on the Health page.
My changes vanished after I reloaded	They should not -- hospitals and facilities are saved on the server. Confirm the Save button was clicked (not Cancel) and that you are on the right server.

34 Reference Tools — Grid, Cheat Sheets, Resources, Print Center

FieldCommand IMS gathers the small, everyday helper tools an operator reaches for again and again into one set of reference pages. None of them run an incident by themselves; they support the work you do everywhere else in the app. The Grid Square Calculator turns coordinates into a Maidenhead grid and back. The Radio Cheat Sheets give you the phonetic alphabet, Q-codes, prowords, band plans, and signal-report scales on one screen. The National Incident Management System (NIMS) Resource Typing Library holds the standard resource definitions that feed your T-cards. The Incident Command System (ICS) Position Checklists spell out the duties of every ICS job. The Print Center puts every form, reference card, and log one click from your printer. Every one of these tools works with the internet turned completely off.

TIP

In one sentence: this chapter covers the reference and print helpers — grid math, radio cheat sheets, the resource-type library, position checklists, and the Print Center — that support the rest of FieldCommand and run fully offline.

34.1 Grid Square Calculator

Open the Grid Square Calculator from the dashboard. Maidenhead grid squares are the short location codes hams use — for example EN52 or EN52ab. This page converts a grid to latitude and longitude, converts coordinates to a grid, and measures the distance and bearing between any two grids. The screen is divided into panels.

PANEL	WHAT IT DOES
Lat / Lon to Grid Square	Type a decimal Latitude and Longitude, then click Calculate Grid. The big amber readout shows the 6-character grid, with the 4-character grid below it, and a detail list of Field, Square, Subsquare, decimal, and degrees-minutes-seconds (DMS) lines.
Use My Location (GPS)	A button in that same panel. It asks your device for its location and fills the Latitude and Longitude for you, then calculates. Needs a device with location services turned on.
Grid Square to Lat / Lon	Type a Maidenhead locator (4 or 6 characters) and click Decode Grid. The readout shows the center latitude and longitude of that square in both decimal and DMS.
Distance and Bearing Between Two Grids	Type Grid 1 and Grid 2 and click Go. You get the great-circle distance in kilometers and miles, plus the true bearing each way with a compass label.
Grid Square Map (North America)	A drawn map you can click or tap to look up a grid. Your current result is highlighted with an amber box and a green center dot.

Grid input boxes force upper case as you type and accept 4, 6, or 8 characters. The Maidenhead Grid Reference panel at the bottom explains the precision levels: a Field (2 letters) is roughly 20 degrees by 10 degrees, a Square (add 2 digits) is 2 by 1 degrees, and a Subsquare (add 2 lowercase letters) narrows it to about 5 by 2.5 arc-minutes.

NOTE

The GPS button and the map only need your device — no internet. If Use My Location does nothing, your browser blocked location access; type the coordinates by hand instead.

34.2 Radio Cheat Sheets

The Cheat Sheets page is a tabbed quick-reference card for common amateur radio and ICS procedures. A row of tabs across the top switches between eight cards; the Print All button in the header prints every card at once, each on its own page, so you can laminate a full set.

TAB	WHAT IS ON IT
Phonetic Alphabet	The full NATO phonetic alphabet A through Z with pronunciation, plus the ITU spoken numbers zero through nine.
Q-Codes	Two columns — common HF and amateur Q-codes (QRM, QSY, QTH, and so on) and the emergency-communications net Q-codes (QNI, QND, QTC, and more).
Prowords	A table of procedure words such as SAY AGAIN, ROGER, WILCO, and BREAK, each with its meaning and an example of how it is used on the air.
Band Plan	The 2-meter and 70-centimeter band segments, a table of HF emergency frequencies, and a Special Service Bands table (MURS, FRS, GMRS, marine, aviation).
NTS Precedence	The National Traffic System message precedence levels — EMERGENCY, PRIORITY, WELFARE, ROUTINE — plus the ICS triage priority colors.
ICS Structure	A drawn ICS organization chart and a table of the key ICS forms (ICS-201 through ICS-309) with the section that owns each and its purpose.
CTCSS / DCS	The full 42-tone CTCSS (Continuous Tone-Coded Squelch System) table and a table of common Digital Coded Squelch (DCS) codes.
Signal Reports	The RST (Readability, Strength, Tone) scale, APCO P25 signal quality, FM voice-report shorthand, and a dBm-to-watts reference.

Resource typing is deliberately NOT one of these cards — it is a large, searchable library of its own, described next in 34.3.

34.3 NIMS Resource Typing Library

The Resource Typing Library holds the standard NIMS resource-type definitions — the kinds of teams, vehicles, and equipment you order and track on an incident. These are the same definitions that populate the T-card board, so keeping this library accurate keeps your resource ordering NIMS-compliant. The definitions are drawn from the Federal Emergency Management Agency (FEMA) Resource Typing Library Tool.

Resources appear as collapsed cards grouped by category (Fire, Search and Rescue, Medical, and so on). Click any card to expand its full definition. The header and filter bar give you these controls:

CONTROL	WHAT IT DOES
Search box	Type any part of a name, capability, or description to filter the list as you type.
Category buttons	Colored pills below the search box — click one to show only that category, or All to show everything.
Expand All / Collapse All	Open or close every card at once.
+ Add Custom Type	Opens the editor so you can add a resource type your organization uses that is not in the standard set.

An expanded card can show up to five labeled sections: What It Is, Minimum Standards / Specifications, Capability Summary, When to Order This Resource, and Common Confusion / Watch Points. When you add or edit a type, the same fields appear in the editor window.

34.3.1 Adding or Editing a Resource Type

1. Click **+ Add Custom Type** (or the pencil icon on a custom card you made earlier). The editor window opens.
2. Fill in **Kind / Name** — this is the only required field, for example "Swiftwater Rescue Team".
3. Set the **Category** from the drop-down and, if it applies, the **Type Level** (Type I, Type II, or N/A) and **Min Personnel**.
4. Add any of the longer fields you want: Full Description, Capability Summary, Metrics / Specifications, What It Is, Minimum Standards, When to Order, and Common Confusion.
5. Click **Save Resource Type**. The card is saved on the FieldCommand server and becomes available on the T-card board immediately.

> **NOTE** Standard shipped types show no pencil icon and cannot be edited or deleted — only types you add carry a CUSTOM tag and the pencil. This protects the NIMS baseline from accidental changes.

34.4 ICS Position Checklists

The Position Checklists page is a job-duty book for every ICS position. Pick a position from the left sidebar and the main area fills with that job's duties, the forms it must produce, the meetings it attends, and who it reports to and supervises. Each duty is a checkbox you can tick off as you go and then print for the position binder.

The sidebar groups positions by section — Command, Operations, Planning, and the rest — and has a search box at the top. Positions include the Incident Commander (IC), Deputy IC, Safety Officer (SO), Public Information Officer (PIO), Liaison Officer (LNO), Operations Section Chief (OSC), Branch Director, Division/Group Supervisor, Strike Team / Task Force Leader, Planning Section Chief (PSC), Resources Unit Leader (RESL), Situation Unit Leader (SITL), Staging Area Manager, Technical Specialist, and the air-operations supervisors, among others.

Each position screen is laid out the same way:

PART OF THE SCREEN	WHAT IT SHOWS
Position header card	The job title, who it Reports To, and who it Supervises, with a plain-language description of the role.
Fallback box	A red note explaining who absorbs the duties when this position is left unfilled — common on smaller incidents.
Activation checklist	The tasks to do when you first take the position.
Operational checklist	The recurring duties for each operational period. Small colored tags mark items tied to a Form, a Briefing, Coordination, Safety, or Documentation.
Deactivation checklist	The tasks to close out and hand off the position.
Deliverables	The ICS forms this position must produce, each with a short description and how often it is due.
Meetings	The incident meetings and briefings this position attends.

Click any checklist line to mark it done — it dims and a green check fills the box, and a progress bar tracks how far you are. The **Print This Position** button in the header prints just the position you are viewing, and **Reset Checks** clears all the ticks so the checklist is fresh for the next operational period or the next person.

34.5 Print Center

The Print Center gathers every printable form, reference card, and log into one page of cards, grouped into sections. Each card has an Open (or Open Form) button that takes you to the live tool, and most also have a Preview button that loads the document into a preview pane on the same page so you can print it without leaving the Print Center.

SECTION	CARDS IT HOLDS
ICS / NTS Forms	ICS-213 General Message, ICS-214 Activity Log, ICS-309 Communications Log, NTS Radiogram, and the Pre-Flight Checklist.
Reference Cards	Phonetic Alphabet, Q-Codes & Prowords, ICS Structure & Forms, and CTCSS / DCS & Signal Reports — each opens the matching Cheat Sheet card (34.2).
Operations	Net Control Log (ICS-309), Starcom Net Log (ICS-309), Roster / Member Directory, and the Resource Board — each jumps to that live tool to print its current data.

When a Preview pane is open it has its own Print button (which prints just the framed document) and a Close button. Printing always uses your own device's browser print dialog, so you can send the page to any printer that device can reach, or choose Save as PDF. There is no server-side print spooling to set up.

34.6 Incident Cover Sheet Generator

At the bottom of the Print Center is a Cover Sheet Generator that builds a clean title page for an incident packet. Fill in the boxes, click **Generate Cover Sheet** to see the formatted page, then click **Print**.

FIELD	WHAT TO ENTER
Incident Name	The incident title, for example "Winter Storm Alpha".
Incident Number	Your local tracking number, such as "2026-001".
Date / Time	Pre-filled with the current date and time; edit if you need a different stamp.
Incident Commander	The IC's name and, if a ham, callsign.
Agency	The agency or group running the incident.
Operational Period	The period number and hours, e.g. "1 — 0800-2000".
Location / Jurisdiction	Where the incident is — city, county, or address.
Net Frequency	The primary net frequency and tone, e.g. "146.940 MHz / 100.0 Hz".
Situation Summary	A short paragraph describing the incident and current status.

The **Clear** button empties the form so you can start a fresh cover sheet. Like everything else here, the cover sheet prints through your device's own browser print dialog.

34.7 Troubleshooting

SYMPTOM	WHAT TO DO
Use My Location does nothing on the Grid page	Your browser blocked location access, or the device has no location service. Allow location for this site, or just type the Latitude and Longitude by hand and click Calculate Grid.

SYMPTOM	WHAT TO DO
Decode Grid says the grid is invalid	Enter at least the first two letters, then the digits and letters in order (for example EN52 or EN52ab). The box forces upper case; spaces are not allowed.
The Resource Typing Library will not load	It reads from the FieldCommand server. Confirm you are connected to EMCOMM-NET and on the server at 192.168.50.1, then reload the page.
A resource type has no pencil (edit) icon	That is a standard shipped type and is protected. To change it, add your own Custom Type with the values you want using + Add Custom Type.
A checklist I filled in came back blank	The Reset Checks button clears every tick for that position. Ticks are also per-device — another operator's device keeps its own set.
Print or Preview shows a blank or tiny page	Wait a moment for the document to finish loading in the frame, then use the Print button inside the preview. In the print dialog, set paper to Letter and scale to Fit or 100%.

35 Network Hardware — Routers, Switch, and Coverage Extension

FieldCommand IMS lives on its own private network called EMCOMM-NET (short for emergency communications). Everything an operator does happens over this network: a phone, tablet, or laptop joins the EMCOMM-NET Wi-Fi, opens a browser to http://192.168.50.1, and reaches every tool. What makes that network exist is a small, deliberate set of gear — a portable Wi-Fi router, a wired network switch, and optional mesh nodes that stretch the signal further. This chapter names each box, tells you what it does, gives the exact port and address layout that keeps the system predictable, and explains how to add coverage when one room is not enough. Neither Raspberry Pi ever broadcasts the Wi-Fi — the router does that job, which frees the servers to run the incident.

TIP

In one sentence: the router makes the EMCOMM-NET Wi-Fi and hands out addresses, the switch wires the servers and workstations together, and extra routers act as mesh nodes to spread the same network across a larger site.

35.1 The Three Pieces of EMCOMM-NET

EMCOMM-NET is self-contained. It does not need the internet, a phone company, or any outside service to exist — as long as the router and switch have power, the network is up and everyone on it can reach the FieldCommand server, even in a parking lot in the middle of nowhere. Three kinds of hardware build it:

PIECE	WHAT IT DOES
Wi-Fi router	The ASUS RT-BE58 Go. It broadcasts the EMCOMM-NET wireless network, hands every joining device an address automatically, and (when a WAN source is plugged in) routes the optional internet connection. This is the one box you must have.
Network switch	The UniFi Switch Lite 16 PoE. A box of Ethernet sockets that wires the two servers, the operator workstations, a laptop, and a printer together at full speed, with power on some ports.
AiMesh nodes	Extra RT-BE58 Go routers — identical to the primary — added to repeat the same EMCOMM-NET signal into more rooms or outdoor areas. Optional; add them only when the space is larger than one router covers.

35.2 Recommended Router — ASUS RT-BE58 Go

The ASUS RT-BE58 Go is the recommended primary router for FieldCommand IMS. It is a compact, travel-sized Wi-Fi 7 router built for mobile deployments — it runs on USB-C power, supports AiMesh for seamless coverage extension, and can take a cellular or satellite internet source on its Wide Area Network (WAN) uplink port. It is available from most electronics retailers for roughly $100 to $130. In the FieldCommand kit it plays two roles at once: the access point (AP) that makes the Wi-Fi signal, and the Dynamic Host Configuration Protocol (DHCP) server that hands out addresses.

SPEC	RT-BE58 Go
Wi-Fi standard	Wi-Fi 7 (802.11be) — backward compatible with all older devices
Bands	2.4 GHz + 5 GHz dual-band

SPEC	RT-BE58 Go
WAN uplink	2.5 gigabit port — takes an optional cellular or satellite internet source
Power	USB-C, about 18 watts — runs from a USB-C adapter or a battery bank
AiMesh support	Yes — extends EMCOMM-NET with additional identical RT-BE58 Go units
Typical range	About 2,500 square feet indoors; larger outdoors
Recommended qty	3 (1 primary + 2 mesh nodes) for the standard kit

NOTE The router is set up once, at the bench, during the build — you do not touch it during an activation. Operators only ever join the Wi-Fi it broadcasts.

35.3 The Fixed Addresses on EMCOMM-NET

A handful of addresses are fixed on purpose so that fault-finding is easy and the address operators type never collides with anything. Internet Protocol (IP) addresses are just each device number on the network. Learn these four and the network holds no surprises:

ADDRESS	WHAT IT IS
192.168.50.1	The FieldCommand application server (the main Raspberry Pi 5). This is the address everyone opens in a browser — every tool lives here.
192.168.50.2	The 44Net / Amateur Packet Radio Network (AMPRNet) gateway — the optional second Pi. Only amateur-radio groups use it; it is kept separate so it can never interfere with the main server.
192.168.50.254	The ASUS router itself — its own admin page. Kept deliberately apart from the server so 192.168.50.1 stays clean. Reach it at http://192.168.50.254.
192.168.50.100 to .200	The pool of addresses the router hands out automatically to phones, tablets, and laptops as they join EMCOMM-NET. The operator does nothing.

TIP If a device shows an address that is not in the 192.168.50.x range, it did not get one from the router. Turn its Wi-Fi off and on to rejoin EMCOMM-NET and pick up a fresh address, and remove any manually-set IP so it uses automatic (DHCP).

35.4 UniFi Switch Lite 16 PoE — The Wiring Hub

The Ubiquiti UniFi Switch Lite 16 PoE is the central wiring hub. It has 16 gigabit Ethernet ports: 8 of them supply Power over Ethernet (PoE) — sending electricity down the same network cable that carries data, so a camera or an extra access point needs no separate power brick — and 8 are ordinary data-only ports. It also has 2 high-speed fiber uplink slots a basic build never needs. Wiring the ports in this exact order keeps every address predictable and makes fault-finding easy:

PORT	WHAT PLUGS IN
Port 1	ASUS RT-BE58 Go router (LAN uplink) — the wired link to the router.
Port 2	FieldCommand Pi 5 (the main server) — fixed 192.168.50.1.
Port 3	44Net gateway Pi 5 — fixed 192.168.50.2. Optional; only if you run 44Net.
Port 4	Windows laptop (Winlink / JS8Call station). Can join by Wi-Fi instead.

PORT	WHAT PLUGS IN
Port 5	Shared color multifunction printer (MFP). Can join by Wi-Fi instead.
Ports 6-9	Operator workstations — up to four Raspberry Pi 500 units, one per port.
Port 10	Satellite dish (optional) — only if you use a satellite upstream source.
Ports 11-12	AiMesh nodes — the wired backhaul for coverage extension (see 35.5).
Ports 13-16	Spare — room to grow; plug in anything else you bring.

The UniFi management interface shows each port live — link status, speed, and PoE power draw — which makes it quick to see at a glance whether a cable is seated and a device is talking. The included 45-watt power supply runs the whole switch and budgets power across the eight PoE ports.

35.5 Extending Coverage with AiMesh

One RT-BE58 Go covers a single room easily. For a multi-room emergency operations center (EOC), a whole building, or an outdoor staging area, add one or two more of the same routers as AiMesh nodes. AiMesh is ASUS's built-in mesh feature: because every node broadcasts the same EMCOMM-NET name (its Service Set Identifier, or SSID) and the same password, a phone or laptop moves from one node to the next automatically — no reconnecting and no re-entering the password. Nodes connect back to the primary through the switch with a real Ethernet cable, which is far faster and more reliable than letting them talk over Wi-Fi.

1. Run a CAT 6 patch cable from a spare UniFi switch port (11 or 12) to the node's LAN port. This wired link between routers is called the backhaul.
2. Power the node on with its USB-C supply. Give it about a minute to boot.
3. Open the primary router's admin page at **http://192.168.50.254** and sign in (or use the ASUS Router app).
4. Go to the **AiMesh** page and click **Add AiMesh Node**. The new node appears automatically once it is found.
5. Click **Connect**. The node joins the mesh and begins broadcasting EMCOMM-NET within about 60 seconds; the primary pushes it the name, password, and settings.
6. Place the node where coverage is needed, overlapping the primary by 20 to 30 percent, and confirm a phone still shows full signal at the farthest point.

35.6 How Many Nodes Do I Need?

Match the number of routers to the size and shape of the space, then walk the far corners with a phone on EMCOMM-NET to confirm the signal holds. You do not have to deploy all three routers — bring what the space needs and leave the rest in the case.

YOUR SPACE	RECOMMENDED SETUP
Single-room EOC (under 2,500 sq ft)	1 primary router only — no extension needed.
Multi-room EOC or large shelter (2,500-7,500 sq ft)	1 primary + 1 node, wired backhaul via switch Port 11.
Large building or campus (7,500-20,000 sq ft)	1 primary + 2 nodes, wired backhaul via switch Ports 11 and 12.
Outdoor SAR staging area	1 primary at the command post + 1-2 nodes at field positions, each on battery power.

NOTE Add a node only when a phone starts dropping bars at the edge of your coverage. A small activation in one room runs perfectly on the single primary router.

35.7 Keeping the Internet Side Separate

EMCOMM-NET runs fine with no internet at all — that is the whole point. If you do want an upstream connection for the features that use it (weather radar, propagation data, some lookups), the internet source plugs into the router's WAN uplink port, not the switch. The primary source is a cellular modem, with a satellite link as an automatic fallback. The router keeps the internet side and the operator side apart and bridges traffic across only when a WAN source is up. That separation is why unplugging the internet never disturbs the operators: they are on a different side of the router entirely, still reaching 192.168.50.1 as normal.

WARN Never wire an internet source into the UniFi switch. Internet always enters through the router's WAN port so the router can protect the local network. See the WAN Source Configuration page for switching and monitoring the upstream sources.

35.8 Power Considerations

For field deployments without shore power, FieldCommand IMS can run from two Astron RS-35M-AP regulated linear power supplies (one per Pi cluster) fed from a portable generator, or from a high-capacity lithium iron phosphate (LiFePO4) battery with a pure-sine inverter. The complete system — both Pis, the router, and the switch — draws roughly 80 to 120 watts under typical load. For battery-only operation, a 100 amp-hour 12-volt LiFePO4 battery gives about 8 to 10 hours of runtime. The router itself runs from any USB-C source, so a small power bank keeps EMCOMM-NET alive on its own even while you swap the main supply.

35.9 Common Questions

QUESTION	ANSWER
Does the Raspberry Pi make the Wi-Fi?	No. The ASUS router makes EMCOMM-NET. Both Pis are servers on the network; if the router loses power, the Wi-Fi goes down even though the Pis are still running.
I joined EMCOMM-NET but the page will not load.	Confirm you typed the full address including http:// (for example http://192.168.50.1) and that you are on EMCOMM-NET, not a nearby Wi-Fi. Check that the router and the application-server Pi both have power.
Do the mesh nodes have to be cabled?	It is strongly recommended. A wired backhaul (a cable from switch Port 11 or 12 to the node) is faster and far more reliable and does not eat into operator Wi-Fi speed. Nodes can link over Wi-Fi in a pinch, but cable them when you can.
Can I rename EMCOMM-NET to my group name?	Yes. EMCOMM-NET is only the default network name. Change it on the router (or in the FieldCommand Setup screen). It is never tied to a callsign or a group.
A node will not join the AiMesh.	Pair it the router maker's way — through the router's AiMesh page or app, not through FieldCommand. Then confirm the node hands out addresses in the 192.168.50.x range before you rely on it.
Do I have to buy all three routers and four workstations?	No. One primary router and one Pi cover the core system; nodes and workstations are recommended, not required. Any phone, tablet, or laptop on EMCOMM-NET is a full station.

36 Appendix — Quick Reference & Administration

This appendix is the fold-out map at the back of the manual. It gathers, in one place, the facts you reach for again and again once FieldCommand IMS is running: the web address of every page, the network ports and background services that keep the system alive, the handful of routine administration tasks that keep it healthy, and a plain-language list of every abbreviation used in this book. Nothing here is new material — it is a fast lookup for things explained in full in the earlier chapters, arranged so a person who opens straight to this page still finds what they need.

TIP

In one sentence: keep this appendix open in a second browser tab during an activation, and you will rarely need to hunt through the rest of the manual.

36.1 How to Use This Appendix

Every device on the incident reaches FieldCommand IMS the same way. Join the **EMCOMM-NET** Wi-Fi network, open any web browser (Chrome, Safari, Edge, Firefox — any of them work), and type the server address into the address bar. The server address is **192.168.50.1**. There is no app to install, no account to create, and no password to the dashboard. The page tables below list every address, but you only ever need to remember the one number: 192.168.50.1.

WARN

If you type 192.168.50.1 and nothing loads, you are almost always on the wrong Wi-Fi network. Check that your phone or laptop is connected to EMCOMM-NET and not to a cellular data connection or a different Wi-Fi, then reload the page.

36.2 All Pages — Web Address (URL) Reference

Below is every page in FieldCommand IMS with its web address (Uniform Resource Locator, or URL) and what it does. To open any of them, type the address shown into the browser. For example, to open the main dashboard, type **192.168.50.1/index.html**. You can also reach every page from the buttons on the main dashboard, so you rarely need to type these by hand.

WEB ADDRESS	PAGE
192.168.50.1/index.html	Main Dashboard
192.168.50.1/incident.html	Incident Management / Command Section
192.168.50.1/incident_mgmt.html	Incident Archive, Restore, Delete, Exercise / Scenario Reset
192.168.50.1/event_templates.html	Pre-Planned Event Templates
192.168.50.1/resources.html	Resource Board (flat list)
192.168.50.1/ics/operations.html	T-Card Resource Board (drag-and-drop)
192.168.50.1/resource_map.html	GPS-Tracked Resource Map
192.168.50.1/resource_types.html	National Incident Management System (NIMS) Resource Typing Library
192.168.50.1/checkin.html	Manual Check-In (ICS-211)

WEB ADDRESS	PAGE
192.168.50.1/scan_checkin.html	Quick Response (QR) code / Barcode Scan Check-In
192.168.50.1/roster.html	Member Roster and QR Code Generator
192.168.50.1/netcontrol.html	Amateur Radio Net Control Logger
192.168.50.1/starcom.html	Public Safety Net Logger
192.168.50.1/observer.html	Observer Mode — Read-Only Net View
192.168.50.1/deadmans.html	Dead Man's Switch
192.168.50.1/iap.html	Incident Action Plan (IAP) Assembly — Planning Section
192.168.50.1/iap_compile.html	IAP One-Click Portable Document Format (PDF) Compilation
192.168.50.1/ics-form.html	Incident Command System (ICS) Form Suite — all forms
192.168.50.1/ics213.html	ICS-213 General Message
192.168.50.1/ics214.html	ICS-214 Activity Log
192.168.50.1/ics309.html	ICS-309 Communications Log
192.168.50.1/fema_costs.html	Federal Emergency Management Agency (FEMA) Public Assistance (PA) Cost Documentation
192.168.50.1/fema_rates.html	FEMA Equipment Rate Schedule
192.168.50.1/cost_dashboard.html	Real-Time Cost Dashboard
192.168.50.1/wan_settings.html	Wide Area Network (WAN) Source Configuration
192.168.50.1/wan-status.html	WAN Status Detail Page
192.168.50.1/radar.html	Animated Next Generation Radar (NEXRAD) Radar
192.168.50.1/propagation.html	High Frequency (HF) Propagation Data
192.168.50.1/tactical.html	Tactical Automatic Packet Reporting System (APRS) Map
192.168.50.1/resmap.html	Public Safety Resource Map
192.168.50.1/callsign.html	Federal Communications Commission (FCC) Callsign Lookup
192.168.50.1/amprgate.html	Amateur Packet Radio Network (AMPRNet) (44Net) Gateway Status
192.168.50.1/nts.html	National Traffic System (NTS) Radiogram Generator
192.168.50.1/winlink-import.html	Winlink Form Import
192.168.50.1/briefing_204a.html	ICS-204A Briefing Sheet
192.168.50.1/hospitals.html	Hospital Proximity Directory
192.168.50.1/facilities.html	Facilities Directory
192.168.50.1/repeaters.html	Repeater Database

WEB ADDRESS	PAGE
192.168.50.1/channel_library.html	Channel Library
192.168.50.1/cheatsheets.html	Radio / ICS Cheat Sheets
192.168.50.1/grid.html	Grid Square Calculator
192.168.50.1/position_checklists.html	ICS Position Checklists
192.168.50.1/meetings.html	Meeting Scheduler
192.168.50.1/printcenter.html	Print Center
192.168.50.1/caltopo_import.html	CalTopo / SARTopo GeoJSON Import
192.168.50.1/preflight.html	Preflight Deployment Checklist
192.168.50.1/refs.html	Reference Library
192.168.50.1/setup.html	Organization Setup
192.168.50.1/general_info.html	General Info / ICS-201

36.3 Network Ports — For the System Administrator

You do not need this table to use FieldCommand IMS. It is here for the person who maintains the server. Behind the scenes, the dashboard is served by the web server (nginx) on the standard web port, and several small Application Programming Interface (API) programs each handle one job on their own network port. If a feature stops working, this table tells the administrator which program to look at.

PORT	PROGRAM	WHAT IT HANDLES
80	nginx	Serves every HTML page and static file on the dashboard
5050	fcc_lookup_server.py	Core API — FCC callsign lookup, nets and roster, hospitals, repeaters, resource types, facilities, GPS and dead-man switch, WAN configuration
5051	health_monitor.py	System health — processor, memory, and disk use, service states, and the connectivity roll-up
5055	ics_platform_server.py	ICS platform — incidents, ICS forms, T-cards, check-ins, the IAP, and FEMA cost documentation
5056	reference_server.py	Offline reference library (renders the PDF documents)
8083	tile_server.py	Offline map tiles (MBTiles), used by every map when there is no internet
9000	amprgate_status.py	44Net status — read-only, on the gateway Pi at 192.168.50.2
9001	amprgate_status.py	44Net tunnel control — reachable only on the gateway Pi itself

NOTE The ICS platform lives on port **5055**. The health monitor is **5051**. These two are easy to confuse; they are different programs doing different jobs.

36.4 Background Services

Each program above runs as a background service that starts by itself when the server powers on and restarts on its own if it ever stops. The administrator manages them by name. This is the list of service names and what each one keeps running.

SERVICE NAME	WHAT IT KEEPS RUNNING
ics-platform.service	ICS platform server (port 5055)
fcc-lookup.service	FCC lookup and configuration server (port 5050)
health-monitor.service	System health monitor (port 5051)
fieldcommand-refs.service	Offline reference library server (port 5056)
fieldcommand-tiles.service	Offline map tile server (port 8083)
deadmans.service	Per-net dead-man switch monitor
wan-monitor.service	WAN source monitoring and failover
aprs-rf.service	RF APRS receive by way of Direwolf or a Terminal Node Controller (TNC)
amprgate-poll.service	44Net tunnel keepalive and reconnect (gateway Pi)
backup.service / backup.timer	Nightly SQLite backup to the external Universal Serial Bus (USB) drive

36.5 Everyday Administration Tasks

Almost everything in FieldCommand IMS is done from the browser, so day-to-day there is nothing to administer. The short list below covers the rare occasions when someone needs to touch the server itself. For the exact commands, see the FieldCommand Installation Guide; the steps here describe what to do in plain terms.

36.5.1 Checking That Everything Is Healthy

1. Open the main dashboard at **192.168.50.1**.
2. Look at the **System Health** panel. Green indicators mean the processor, memory, disk, and every background service are in good shape.
3. For more detail, open **Preflight Deployment Checklist** (192.168.50.1/preflight.html), which walks through each service and reports any that need attention before you rely on the system in the field.

36.5.2 Restarting the Server

A clean restart fixes most transient problems. Power the Raspberry Pi 5 server off and back on, wait about two minutes for it to finish starting, then reload the dashboard. Every background service starts automatically, so nothing needs to be launched by hand. Because all incident data is saved to disk, a restart never loses any of your work.

> **TIP** If a single feature is misbehaving but the rest of the dashboard is fine, the administrator can restart just that one background service by name (from the table in 36.4) instead of restarting the whole server.

36.6 Backups and Keeping Data Safe

Incident data in FieldCommand IMS is permanent — the system is built to save everything an incident produces (nets, logs, ICS forms, T-cards, the IAP, cost records, resources, roster snapshots, and attachments) and never treat any of it as throwaway. That data lives in a database on the server and is copied to an external USB drive automatically.

WHAT	HOW IT IS PROTECTED
Live data	Written to the server database as you work, using the standard SQLite storage engine. It survives a restart or a power loss.
Nightly backup	The backup service copies the whole database to the external USB drive every night on a timer, with no action needed from you.
Archived incidents	When an incident closes, you archive it from the Incident Archive page (192.168.50.1/incident_mgmt.html); the archive is kept and can be restored later.

WARN Confirm the external backup drive is plugged in and recognized before an activation. The nightly backup can only run if the drive is connected. Keeping a second copy of the drive off-site is good practice for records you must retain.

36.7 Two Networks and Two Maps — Quick Reference

Two pairs of things in FieldCommand IMS are commonly confused. This is the one-glance reminder of which is which.

36.7.1 The Two Servers

FieldCommand IMS runs on two Raspberry Pi 5 units. The **application server** is the one you reach at 192.168.50.1 — it serves the whole dashboard. The **44Net gateway** is a separate Pi at 192.168.50.2 that handles the Amateur Packet Radio Network (AMPRNet, also called 44Net) tunnel. Everyday users only ever talk to 192.168.50.1.

36.7.2 The Two Maps

MAP	WHAT FEEDS IT
Tactical APRS Map (tactical.html)	Live amateur-radio APRS heard off the air. Direwolf decodes the packets, a feed program serves them, and the map shows the stations automatically. This is the big-screen situational-awareness display.
Public Safety Resource Map (resmap.html)	Objects you place by hand — "+ Add Unit" and "Draw Zone". It is not fed by radio and never connects to APRS. It is the public-safety side of the picture.

NOTE The GPS-Tracked Resource Map (resource_map.html) is a third, separate map that tracks resources by their reported location; do not confuse it with either of the two above.

36.8 When the Internet Is and Is Not Available

FieldCommand IMS is offline-first: the full tool set works with zero internet. A few features simply add live data when a Wide Area Network (WAN) connection is present and quietly step aside when it is not. Knowing which is which prevents "is it broken?" questions in the field.

FEATURE	NEEDS INTERNET?
Incidents, ICS forms, T-cards, IAP, cost documentation	No — always available
Net loggers, roster, check-in, resource boards	No — always available
Offline maps and the reference library	No — served locally from the Pi

FEATURE	NEEDS INTERNET?
Animated NEXRAD radar	Yes — appears when a WAN is connected
HF propagation data and some online lookups	Yes — appears when a WAN is connected

The WAN Source Configuration page (192.168.50.1/wan_settings.html) is where the administrator selects and monitors the internet source; the WAN Status page (192.168.50.1/wan-status.html) shows whether one is currently connected.

36.9 Abbreviations and Acronyms

Every abbreviation is spelled out in full the first time it appears in each chapter. This table repeats the common ones in one place, for a reader who opens the manual to the middle.

SHORT FORM	FULL MEANING
APRS	Automatic Packet Reporting System
AMPRNet	Amateur Packet Radio Network (the 44Net address range)
API	Application Programming Interface
CTCSS	Continuous Tone-Coded Squelch System (a sub-audible tone)
EMCOMM	Emergency Communications
EOC	Emergency Operations Center
FCC	Federal Communications Commission
FEMA	Federal Emergency Management Agency
HF	High Frequency
IAP	Incident Action Plan
ICS	Incident Command System
NEXRAD	Next Generation Radar
NIMS	National Incident Management System
NTS	National Traffic System
NWS	National Weather Service
PA	Public Assistance (the FEMA reimbursement program)
PDF	Portable Document Format
QR	Quick Response (code)
SAR	Search and Rescue
TNC	Terminal Node Controller
UHF	Ultra High Frequency

SHORT FORM	FULL MEANING
URL	Uniform Resource Locator (a web address)
USB	Universal Serial Bus
VHF	Very High Frequency
WAN	Wide Area Network (an internet connection)

36.10 Copyright and License

36.11 Common Questions

QUESTION	ANSWER
What is the one address I need to remember?	192.168.50.1 — type it into any browser after joining the EMCOMM-NET Wi-Fi.
Do I need to install anything or log in?	No. Any smartphone, tablet, or laptop reaches the whole dashboard through a web browser with no app and no account.
A page will not load. What do I check first?	Confirm your device is on the EMCOMM-NET Wi-Fi (not cellular or another network), then reload. This fixes the large majority of "cannot connect" reports.
Will I lose my work if the power drops?	No. Incident data is saved to disk as you work and backed up nightly to the external USB drive; a restart never loses saved work.
Why is the radar blank?	NEXRAD radar needs an internet (WAN) connection. With no WAN, it is expected to be blank while every offline tool keeps working.
Which map is fed by radio?	Only the Tactical APRS Map (tactical.html). The Public Safety Resource Map (resmap.html) is hand-entered and never touches radio.